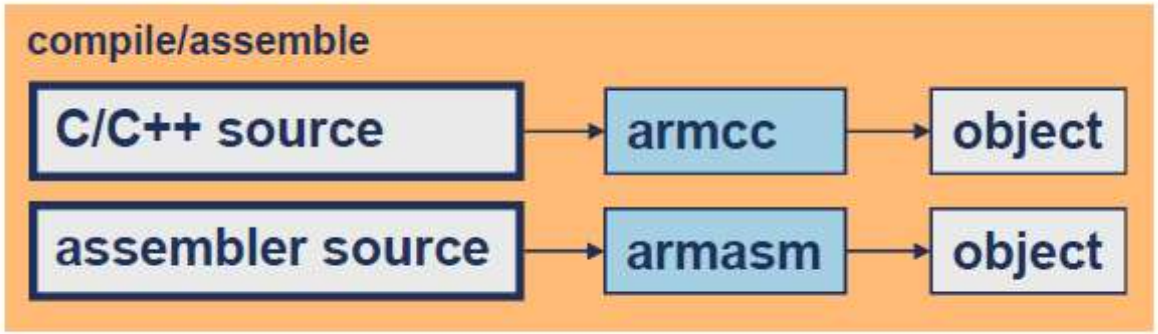
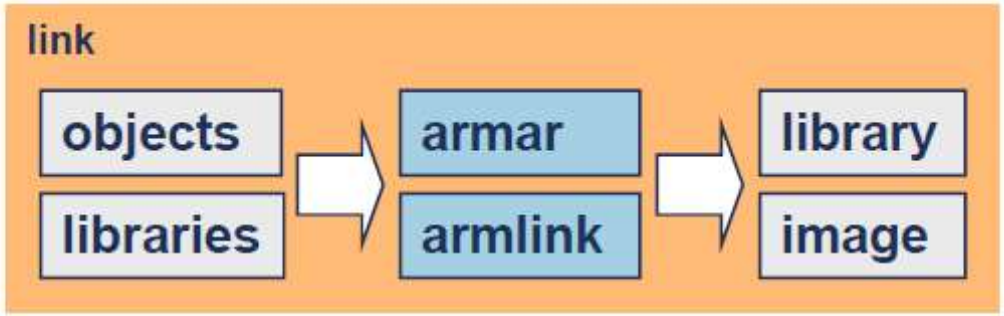
# Lecture 5:

# Project Creation
# From Scratch

# Keil MDK: ARM Compiler

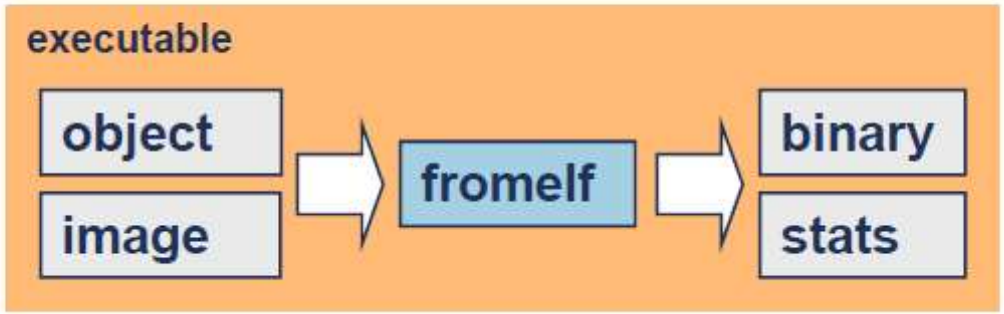
- Compiler
- Assembler
- Linker
- Format Converter
- Libraries
- Librarian




- These are all command line tools
  - Easy to script

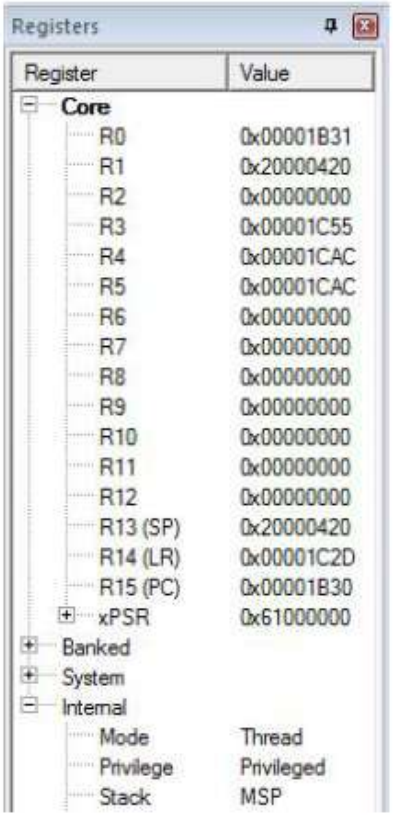
- All tools emit useful statistics
  - e.g. code size, data size, symbol table, call graph, image memory map etc.
  - See the documentation

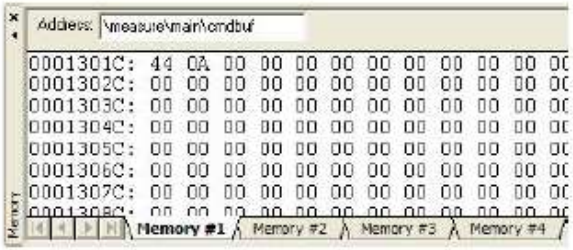


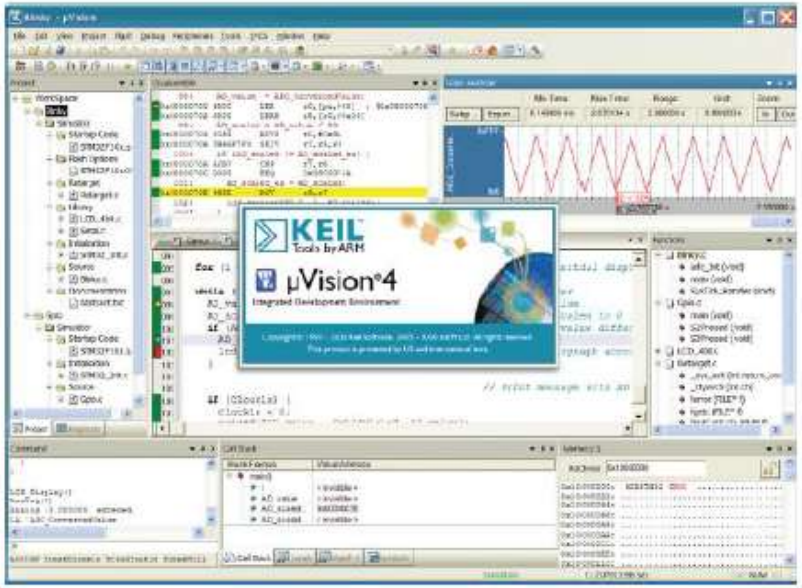
# Keil MDK: µVision Debugger

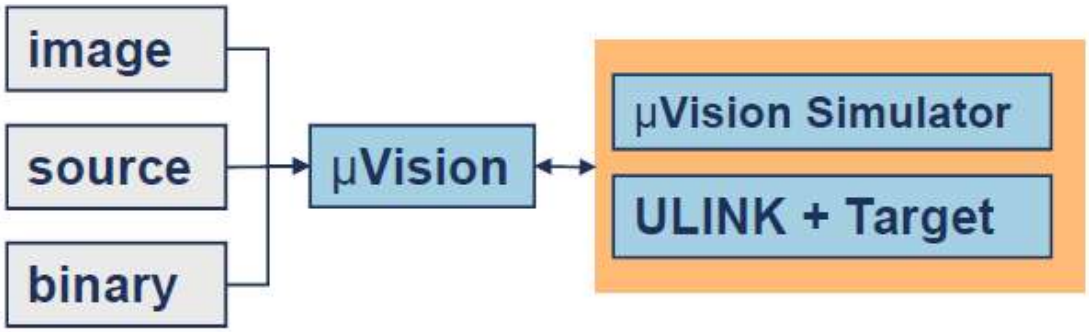
- **Source and disassembly level debugger**
- **Connects to:**
  - uVision Simulator
  - Hardware targets, via ULINK
- **Analysis Tools:**
  - Profiler, Logic and Performance Analyzers, Code Coverage
- **GCC images supported**



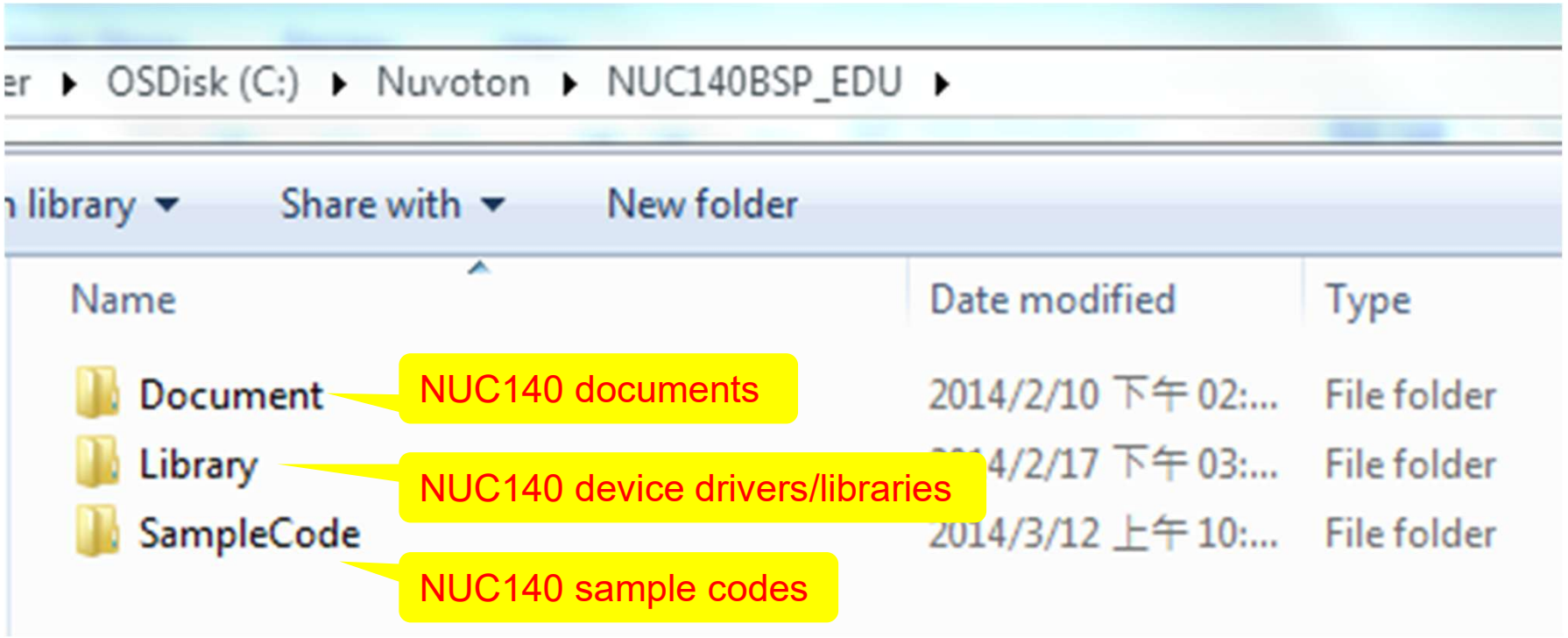


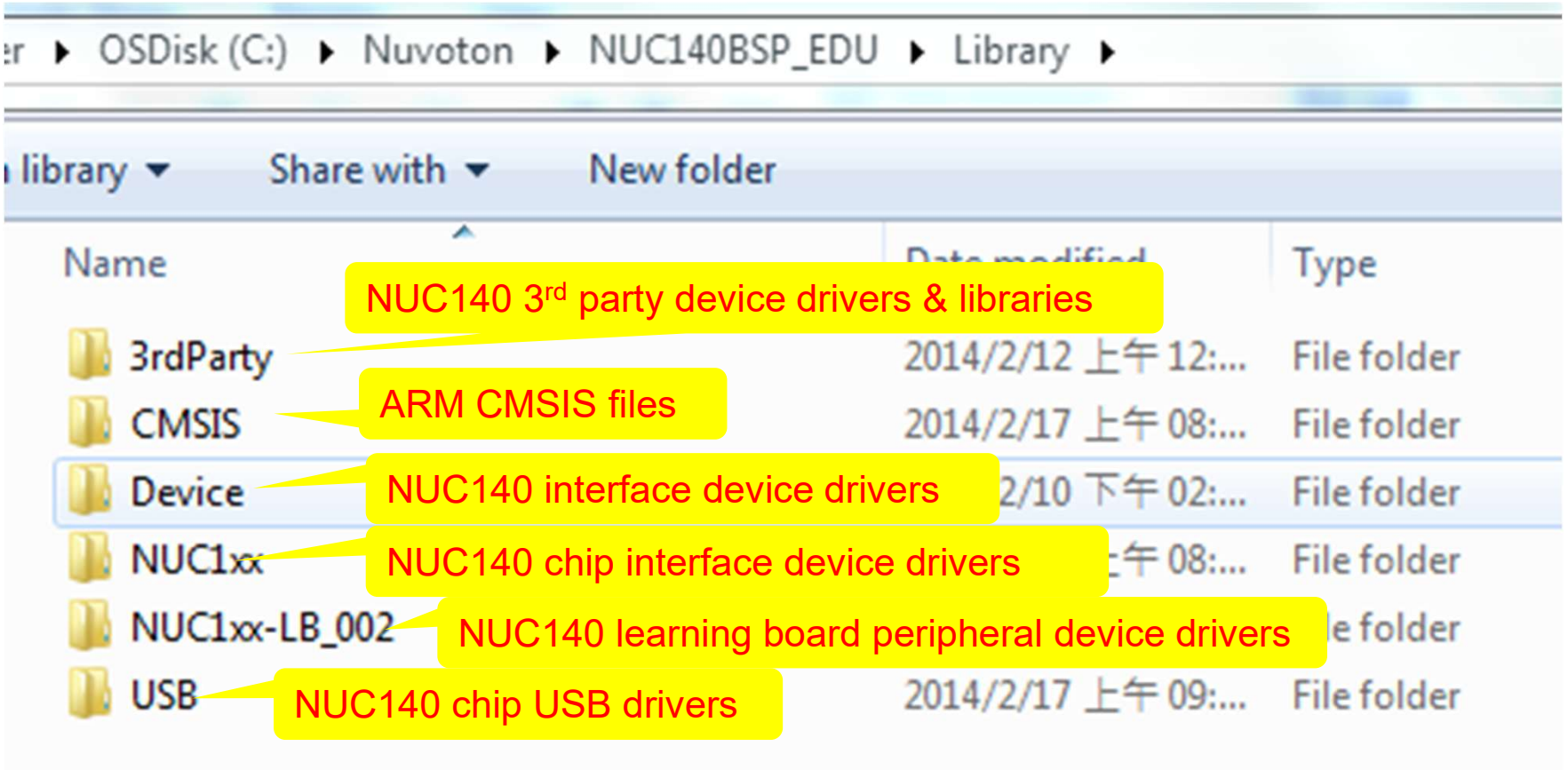



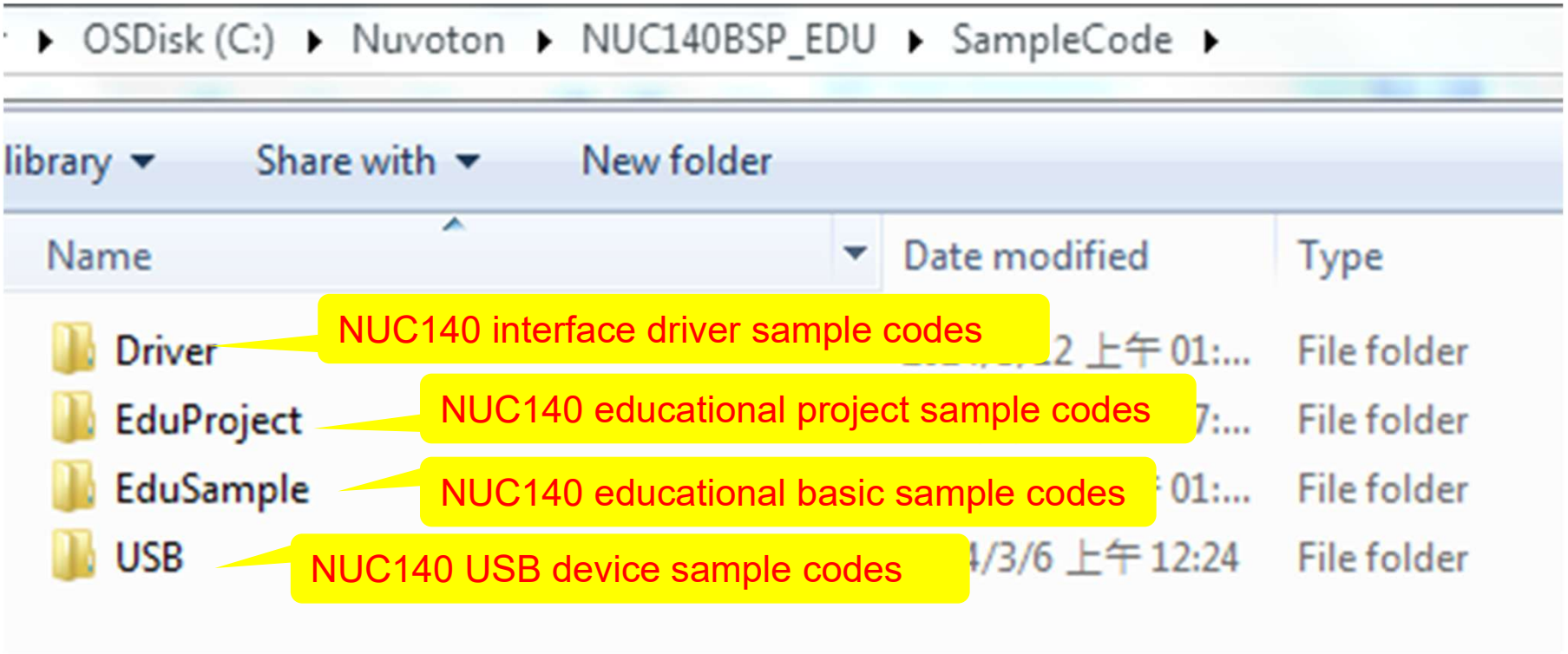
# \Nuvoton\NUC140BSP_EDU\



# \Nuvoton\NUC140BSP_EDU\Library

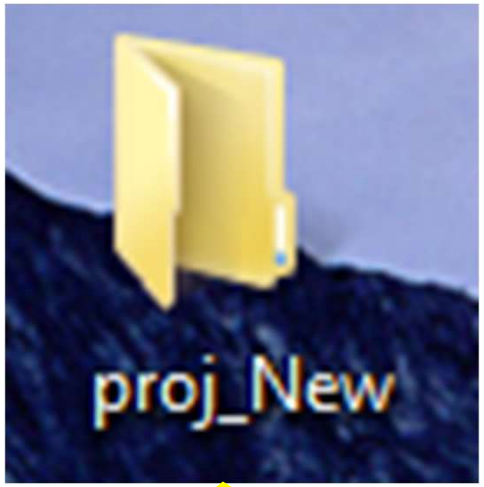


# \Nuvoton\NUC140BSP_EDU\SampleCode



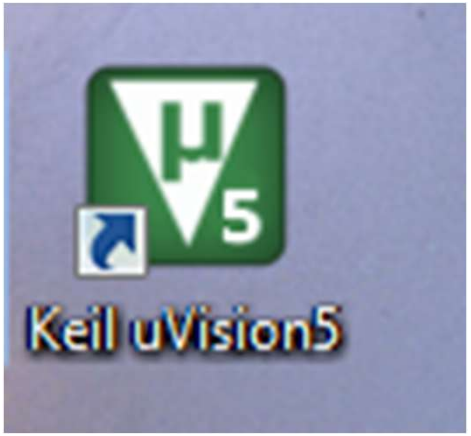
# NuMicro MCU New Project from scratch

# Create a new folder



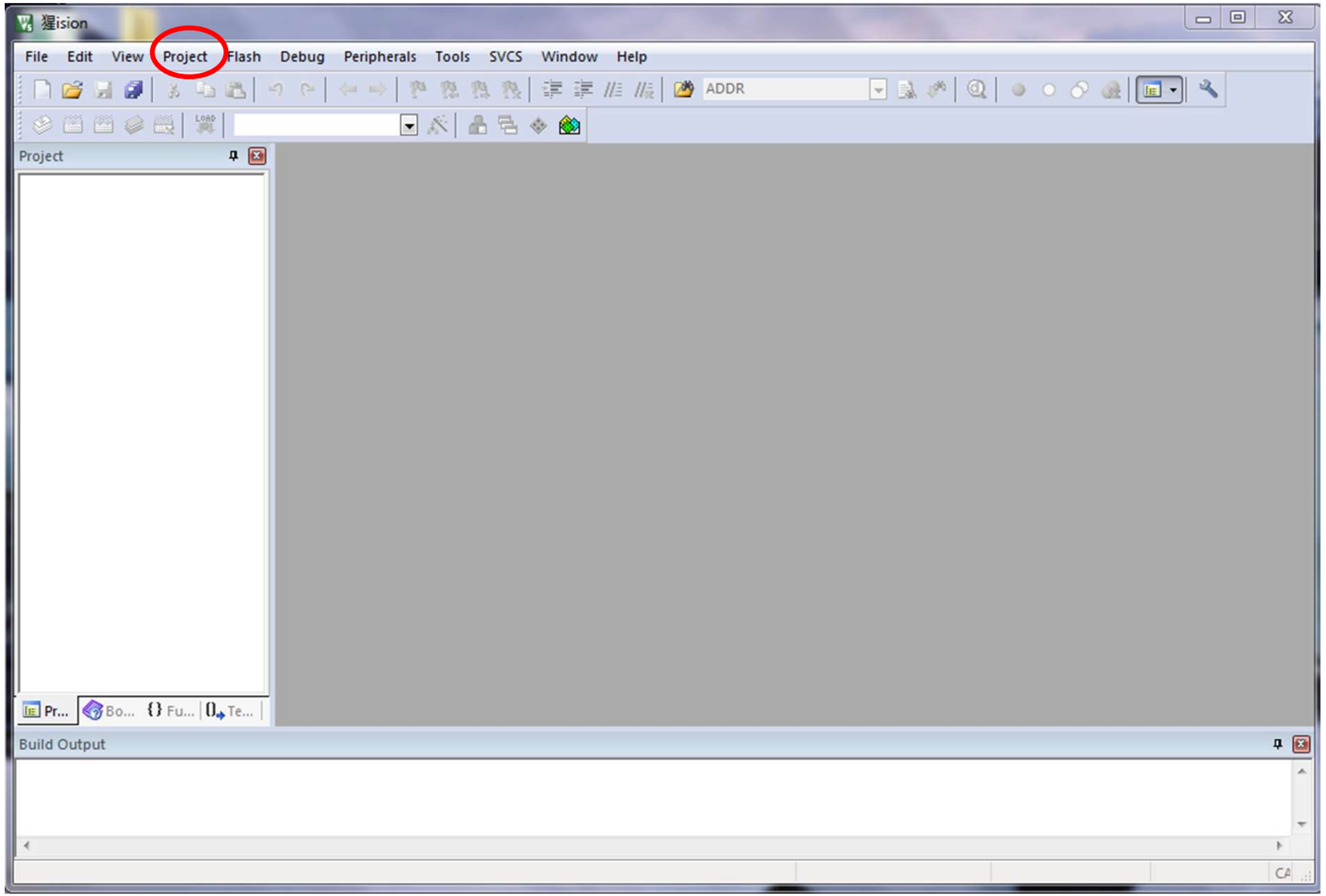

Create a new folder on desktop

# Start Keil MDK

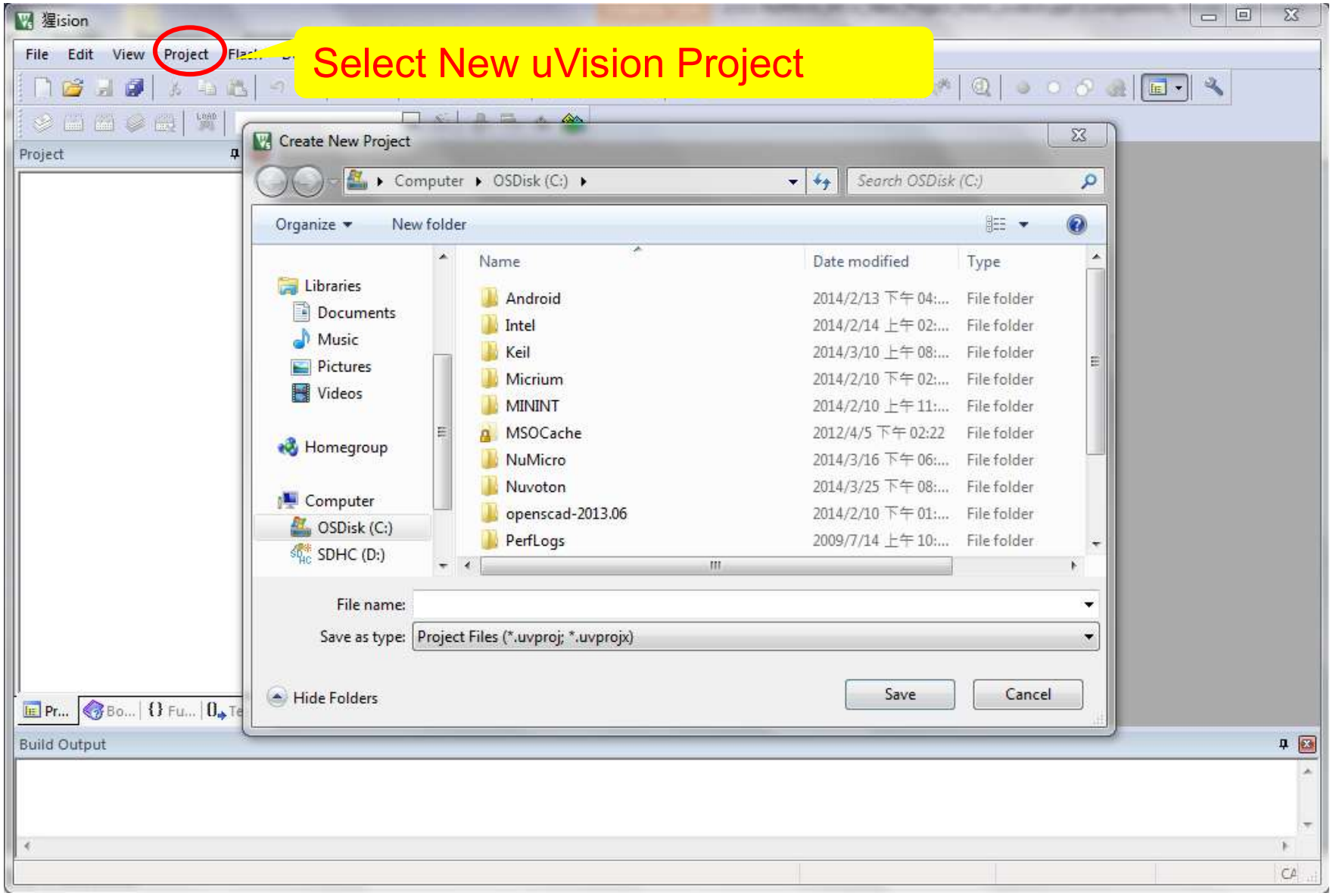



double click to start Keil MDK

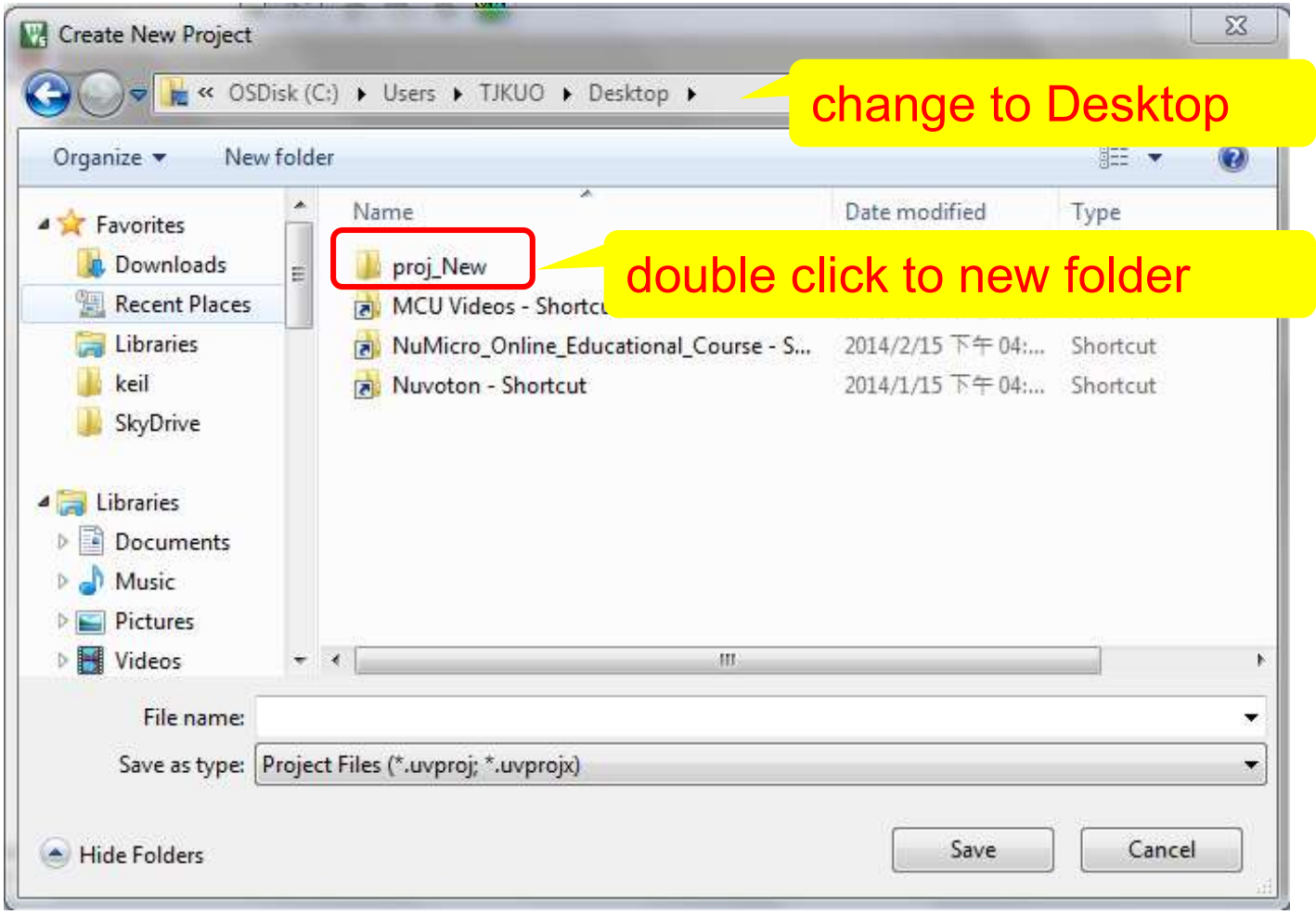
# New Keil Project



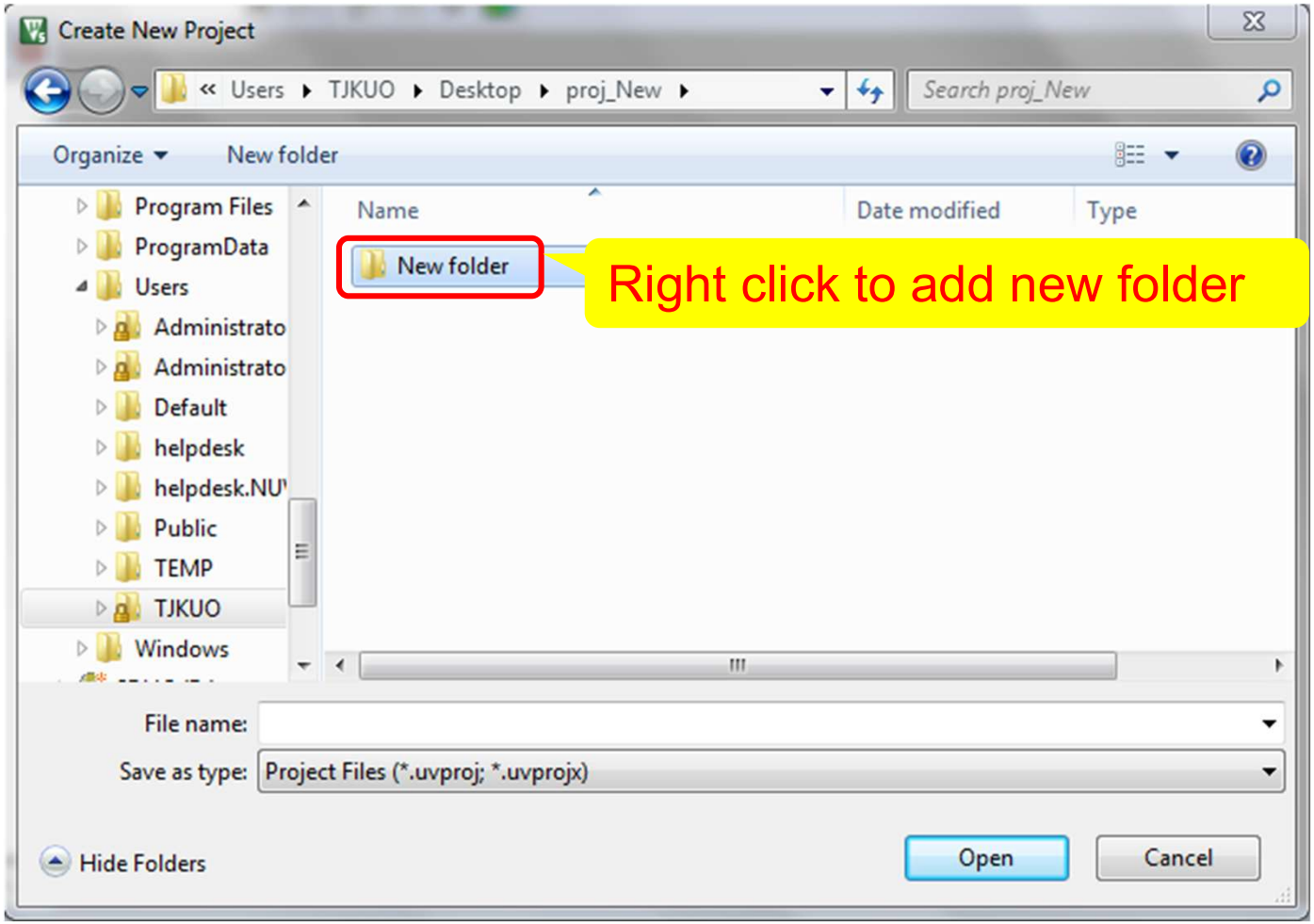
# New uVision Project



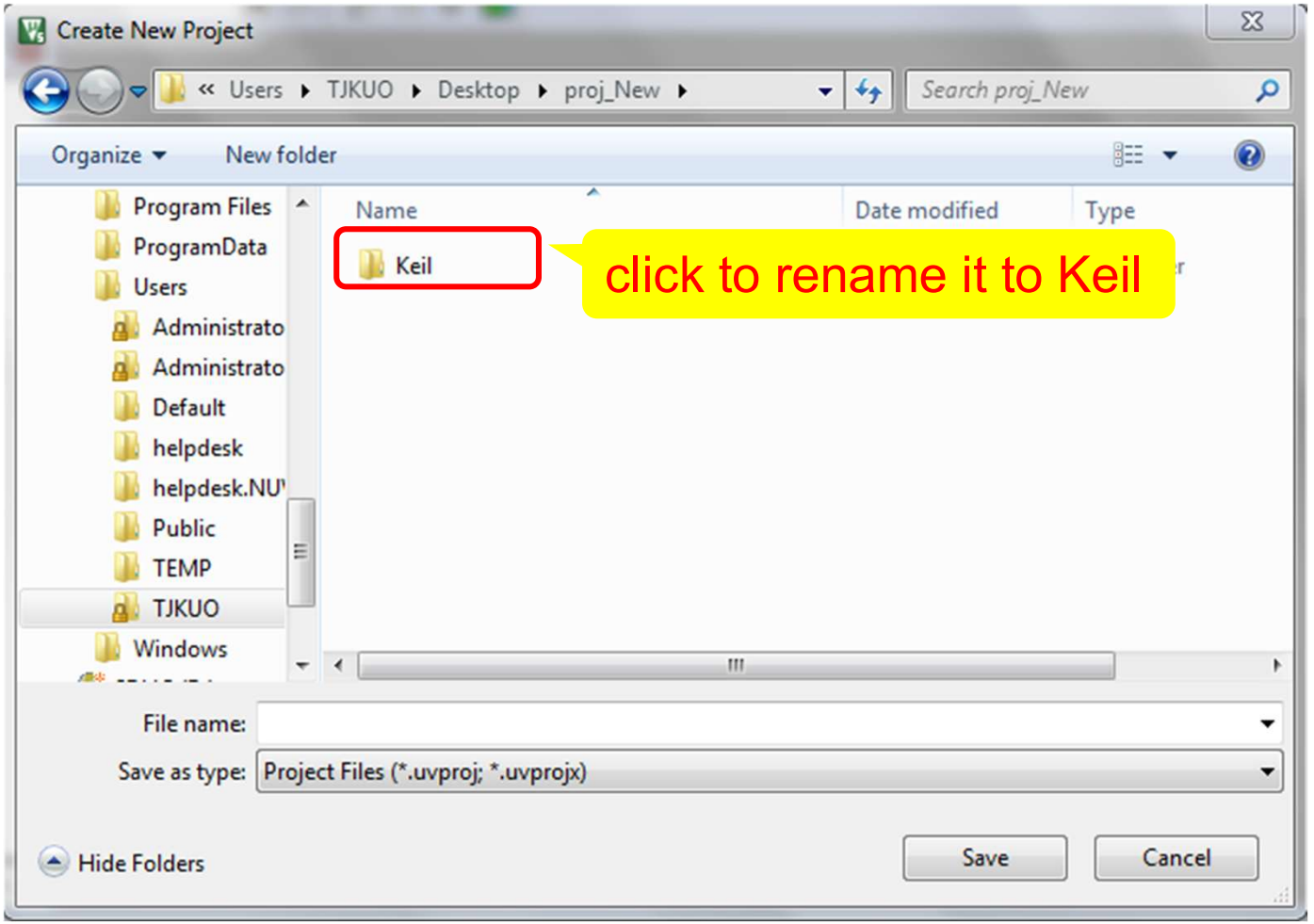
# New uVision Project



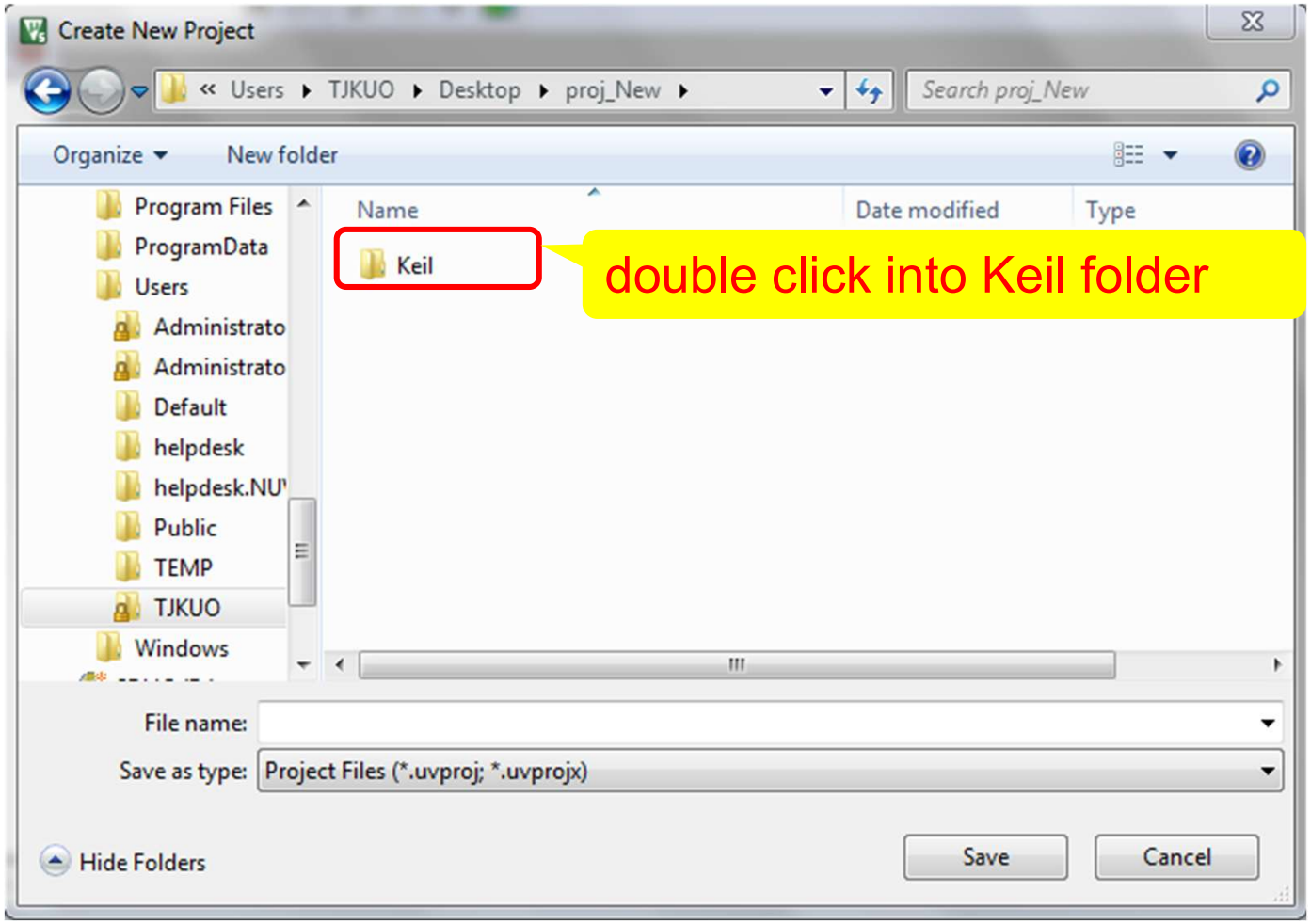
# create new folder



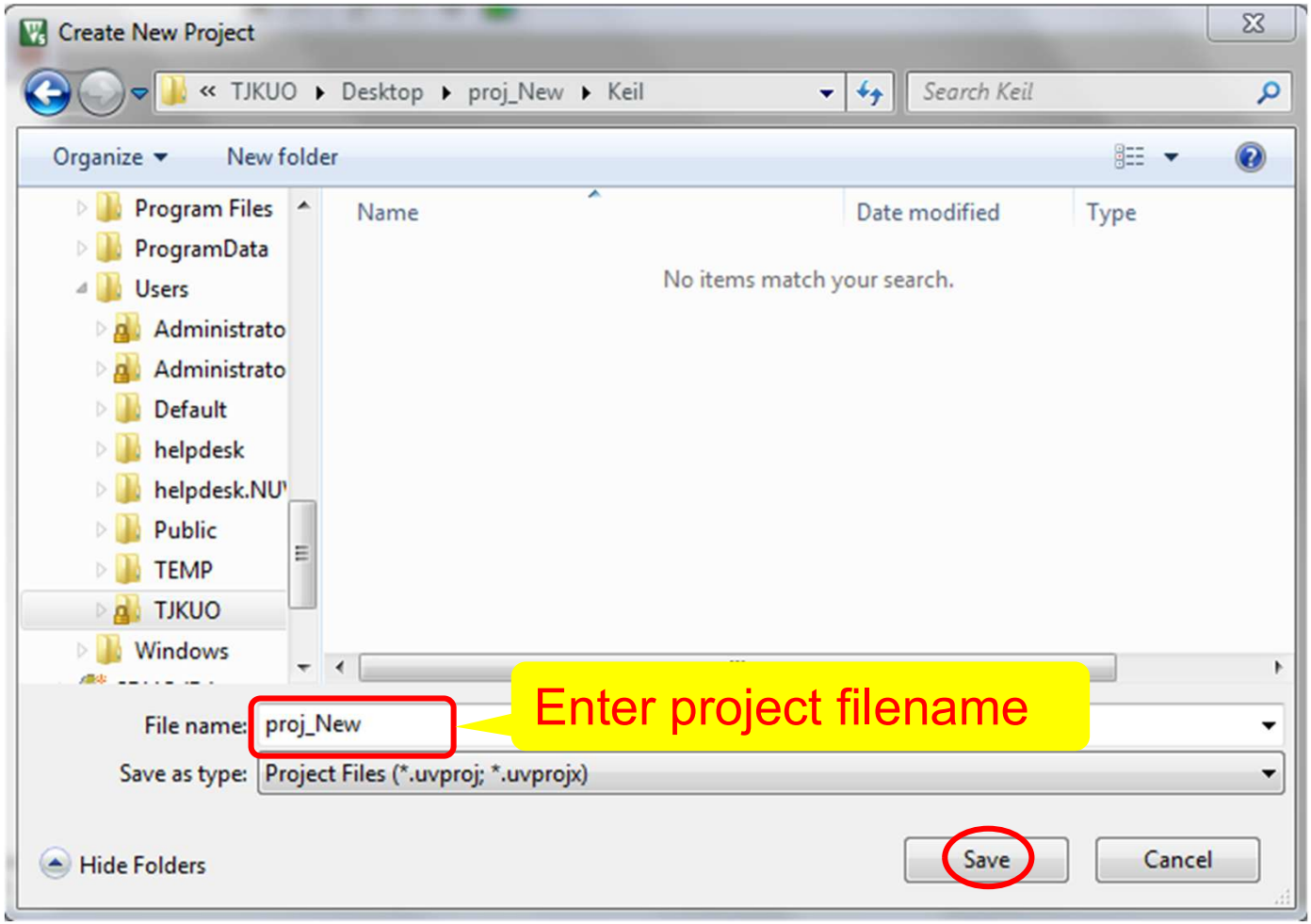
# rename new folder to Keil



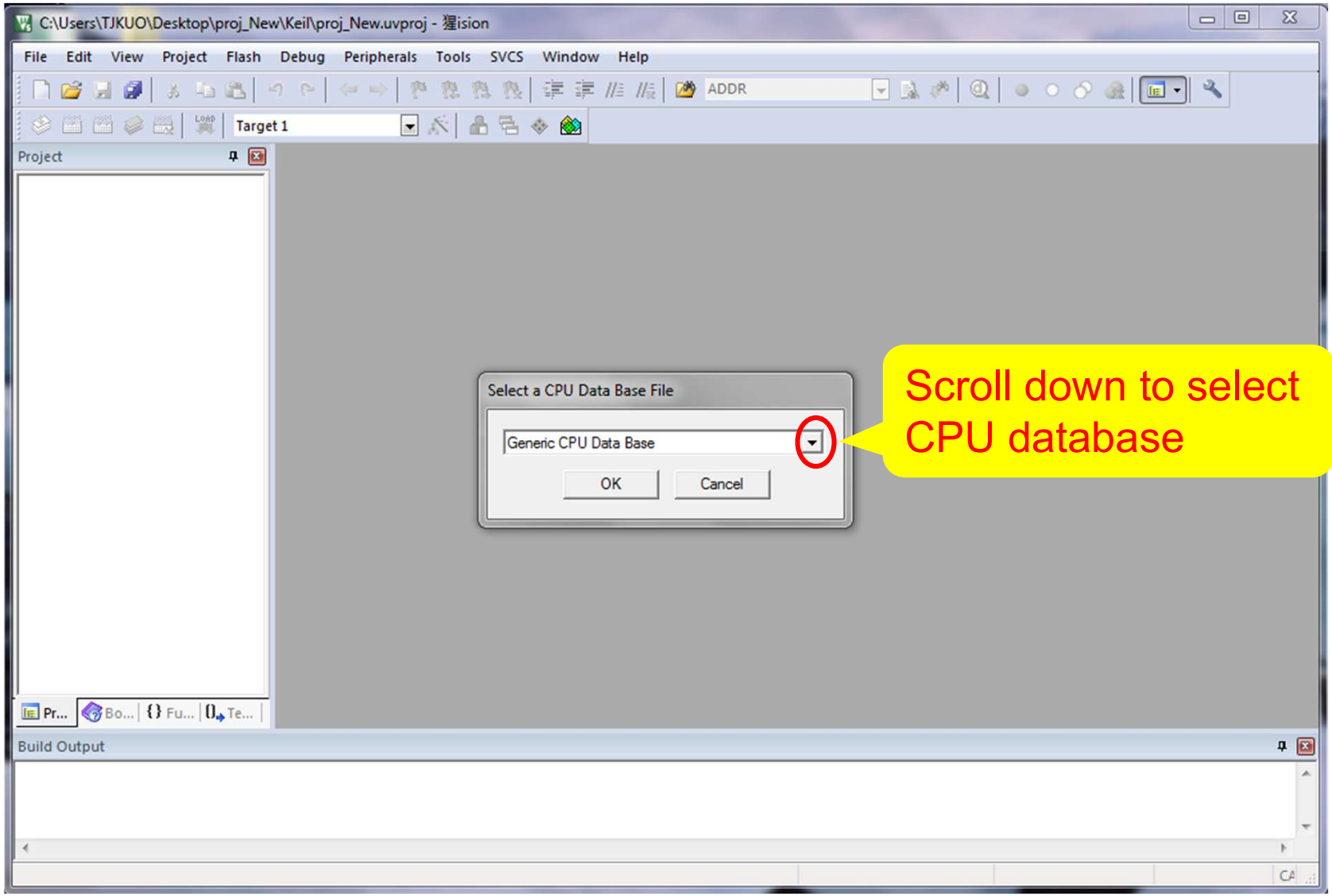
# click into Keil folder



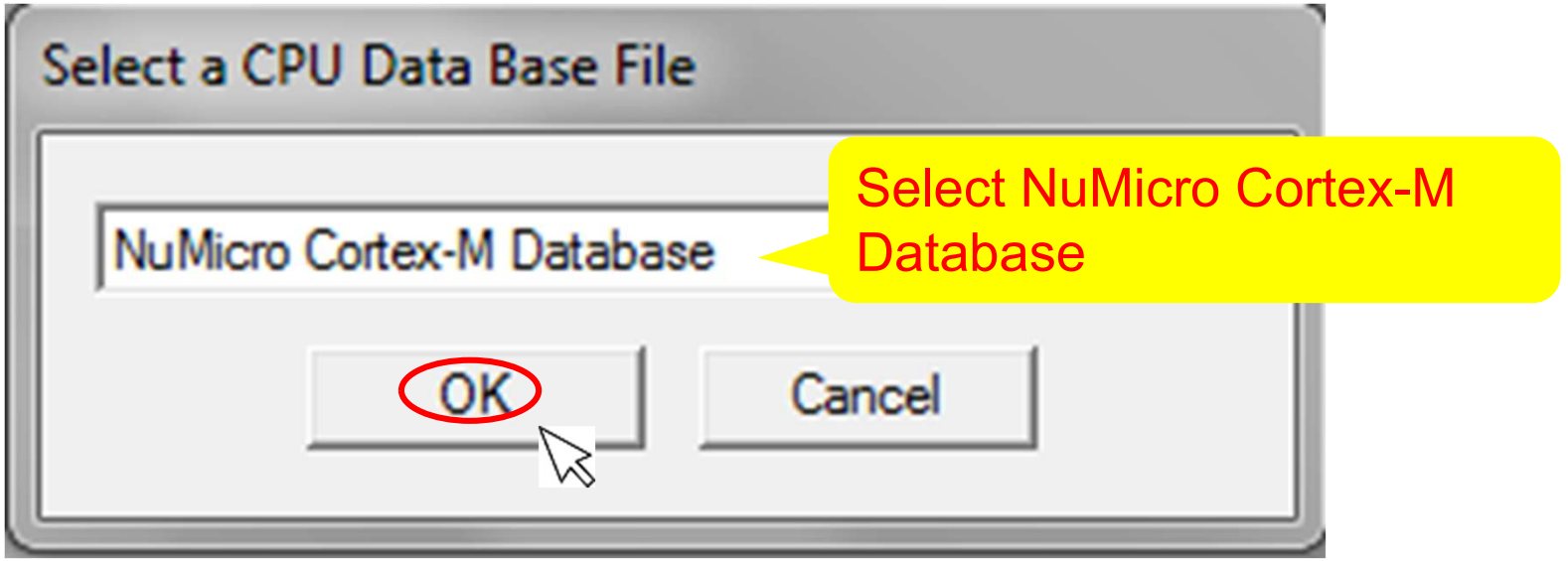
# Enter project filename & Save



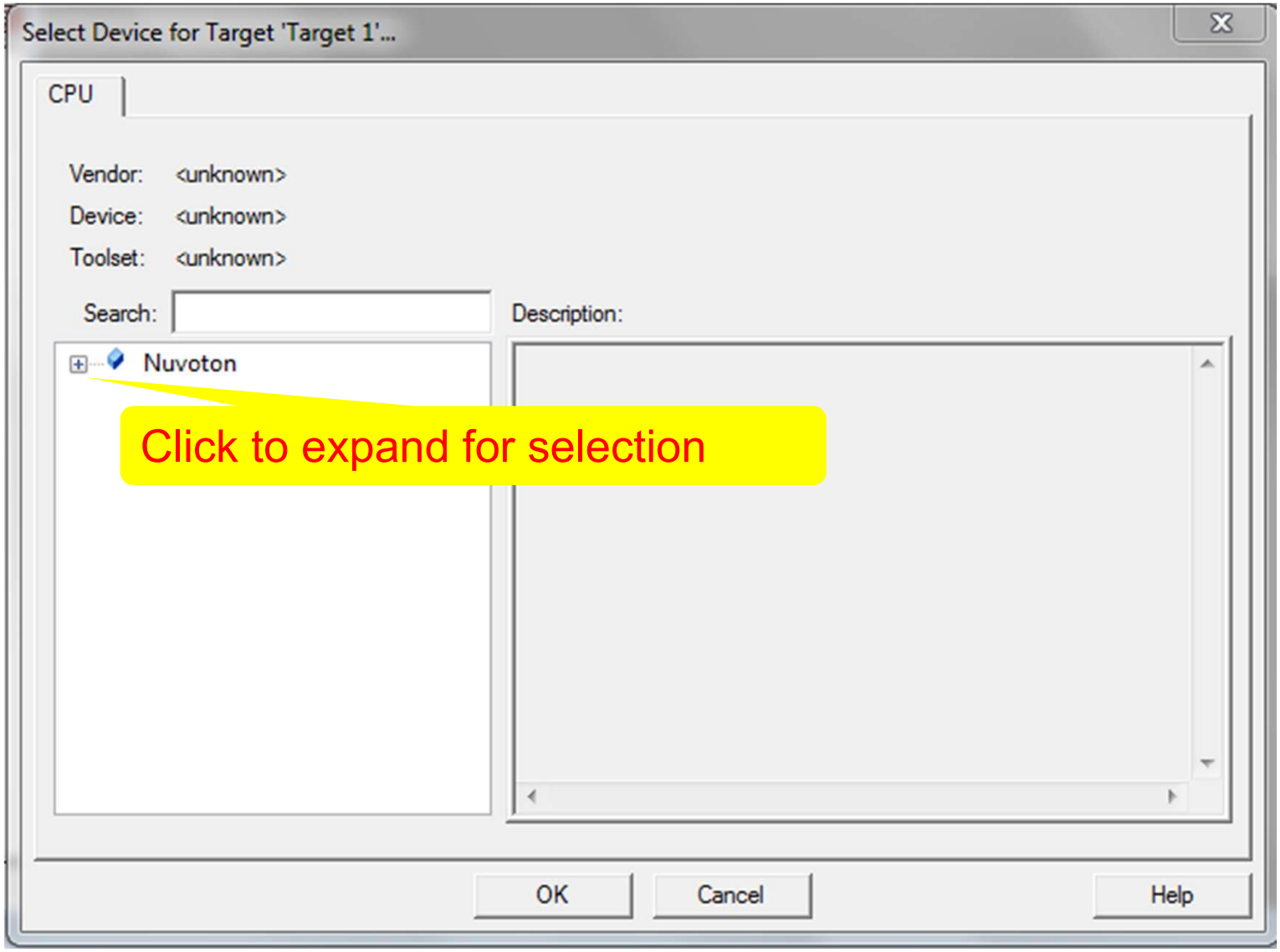
# Select CPU Data Base



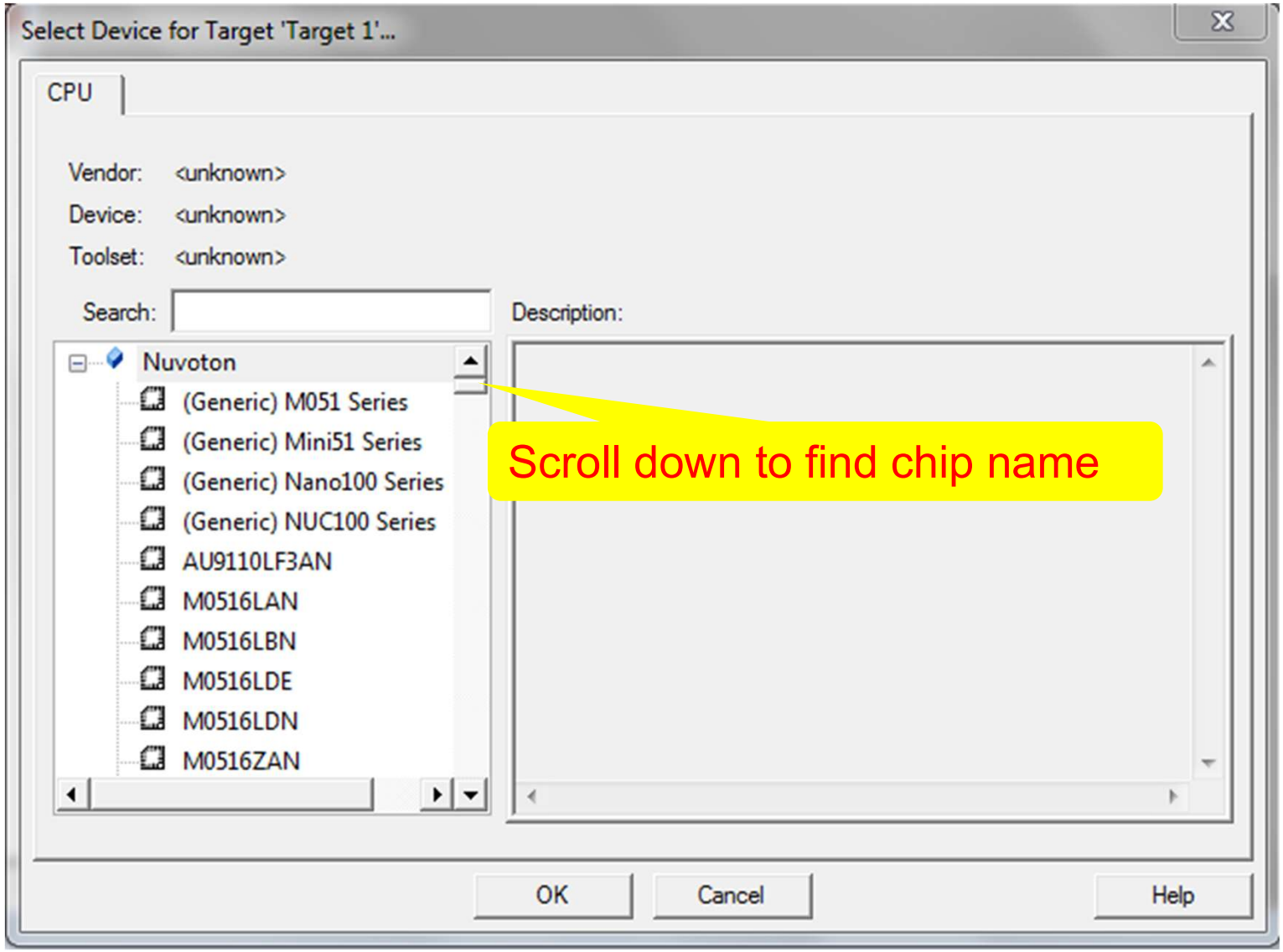
# Select NuMicro Cortex-M Database



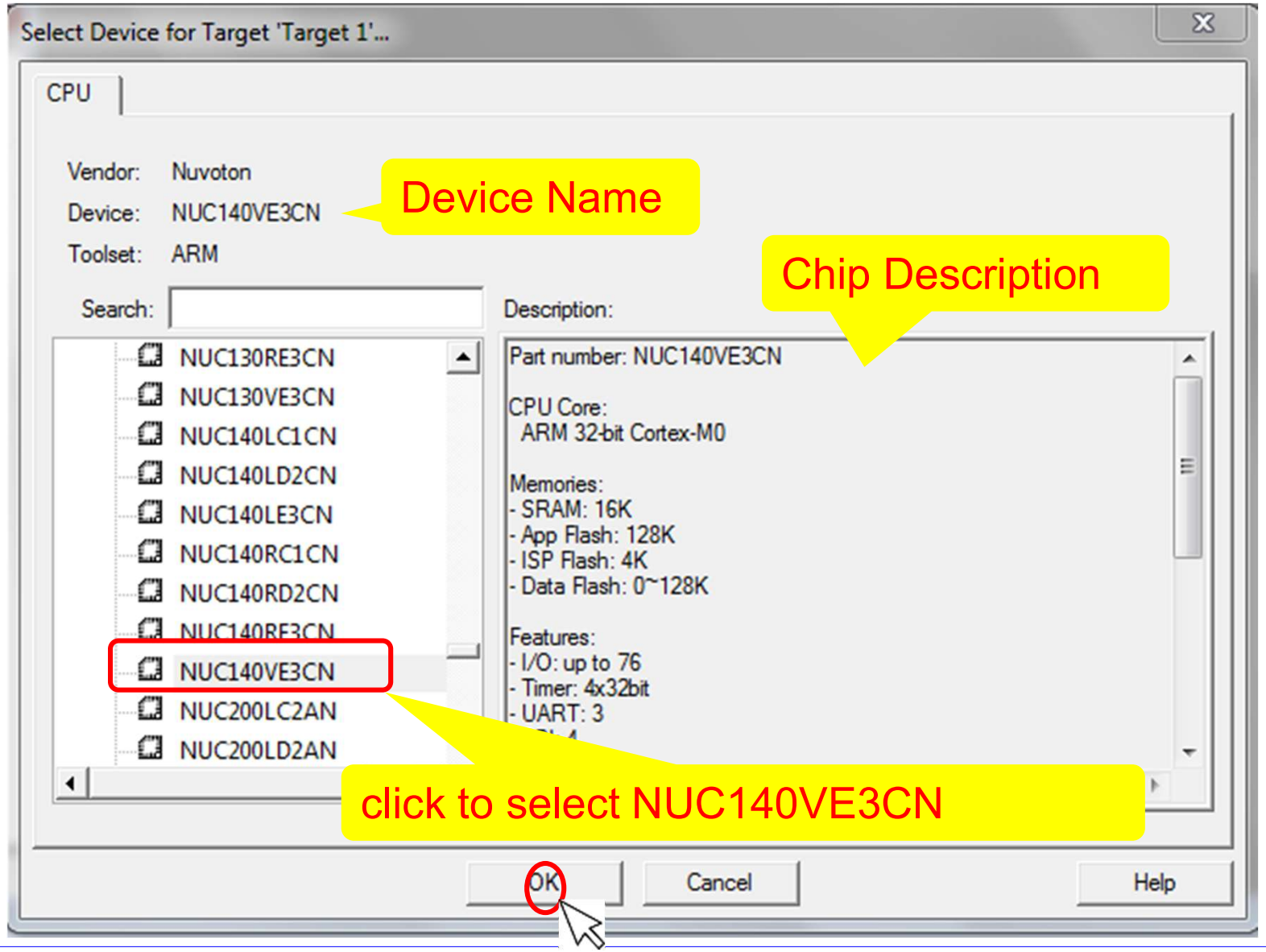
# Chip is not selected yet



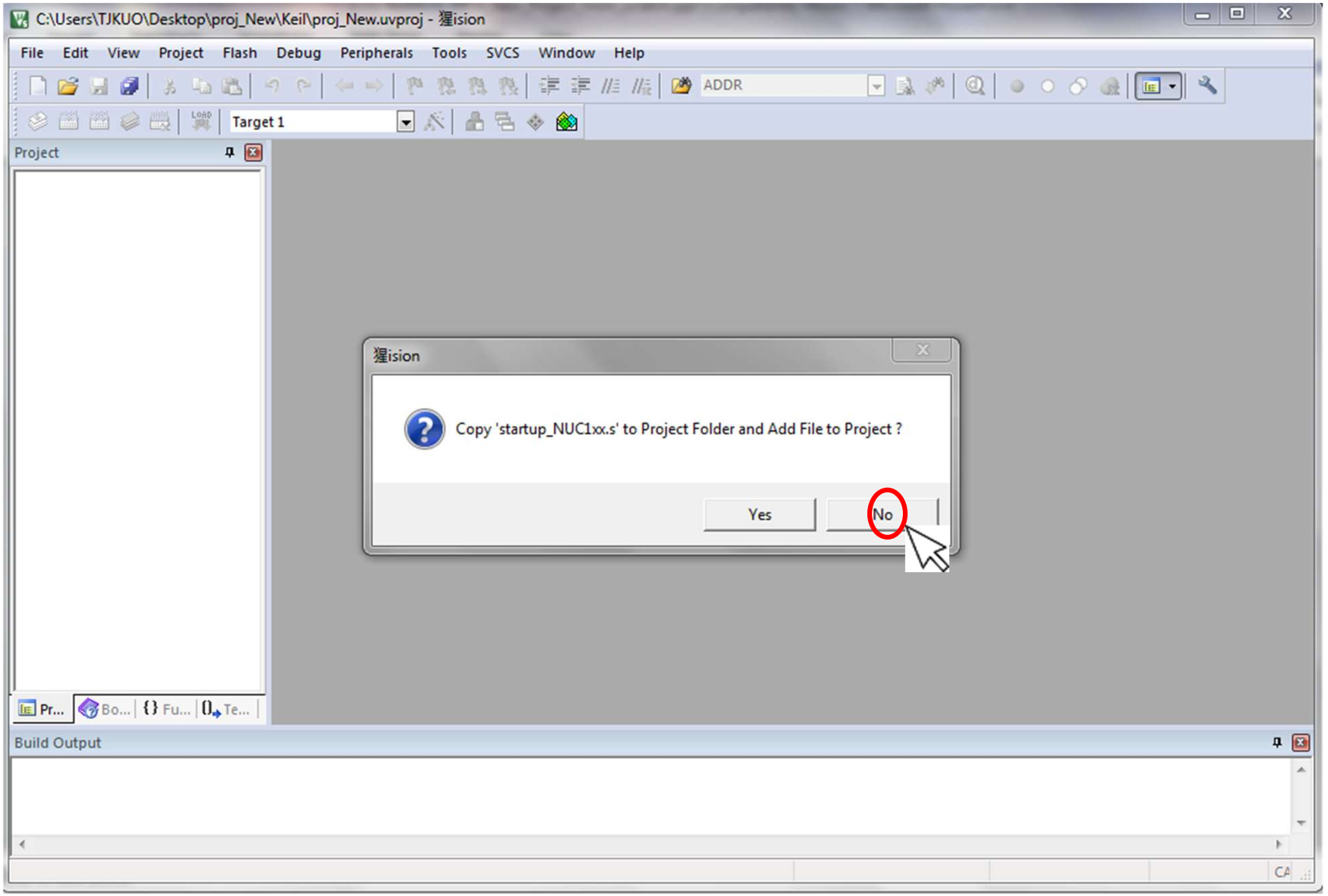
# Nuvoton chip selection



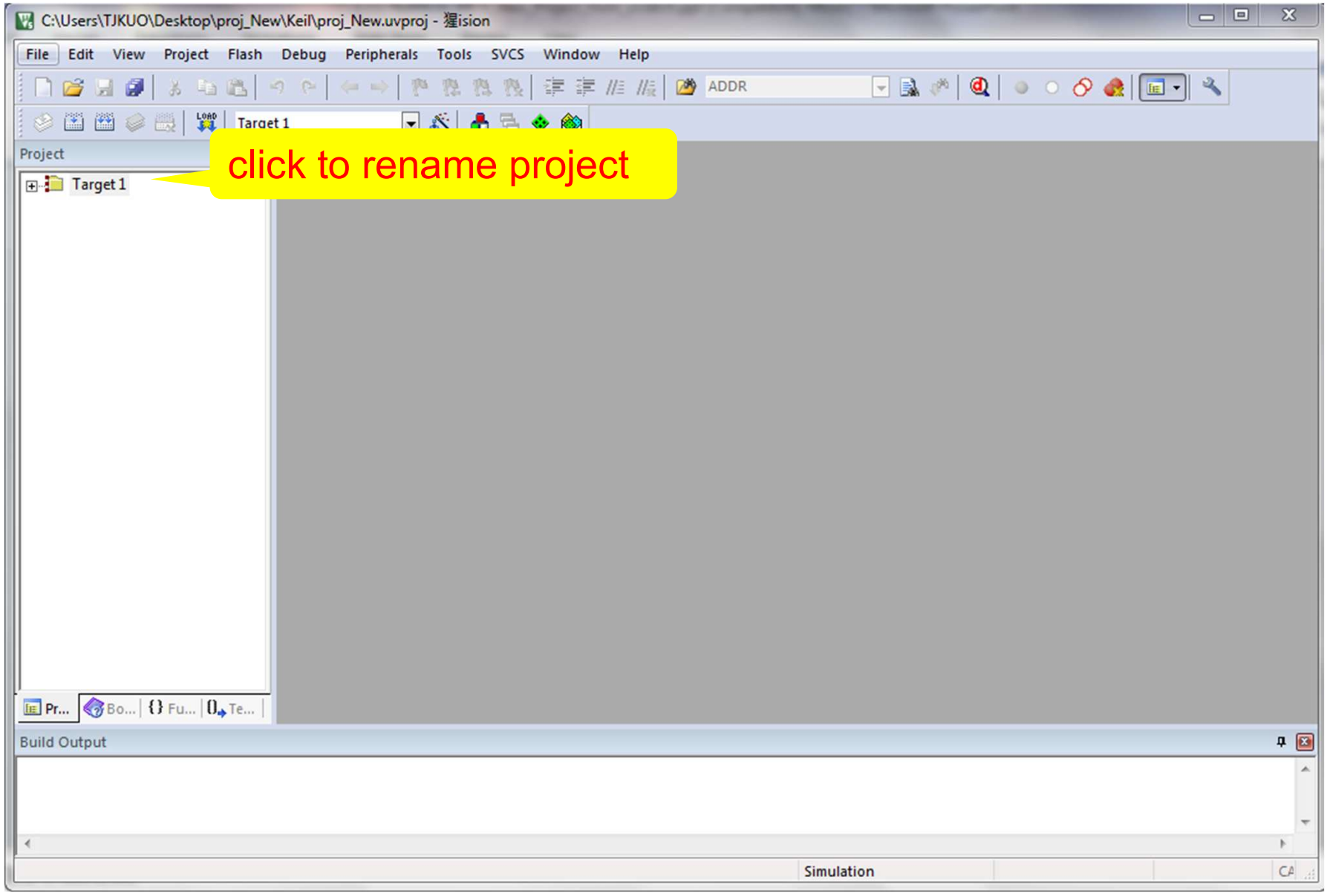
# Nuvoton chip selection



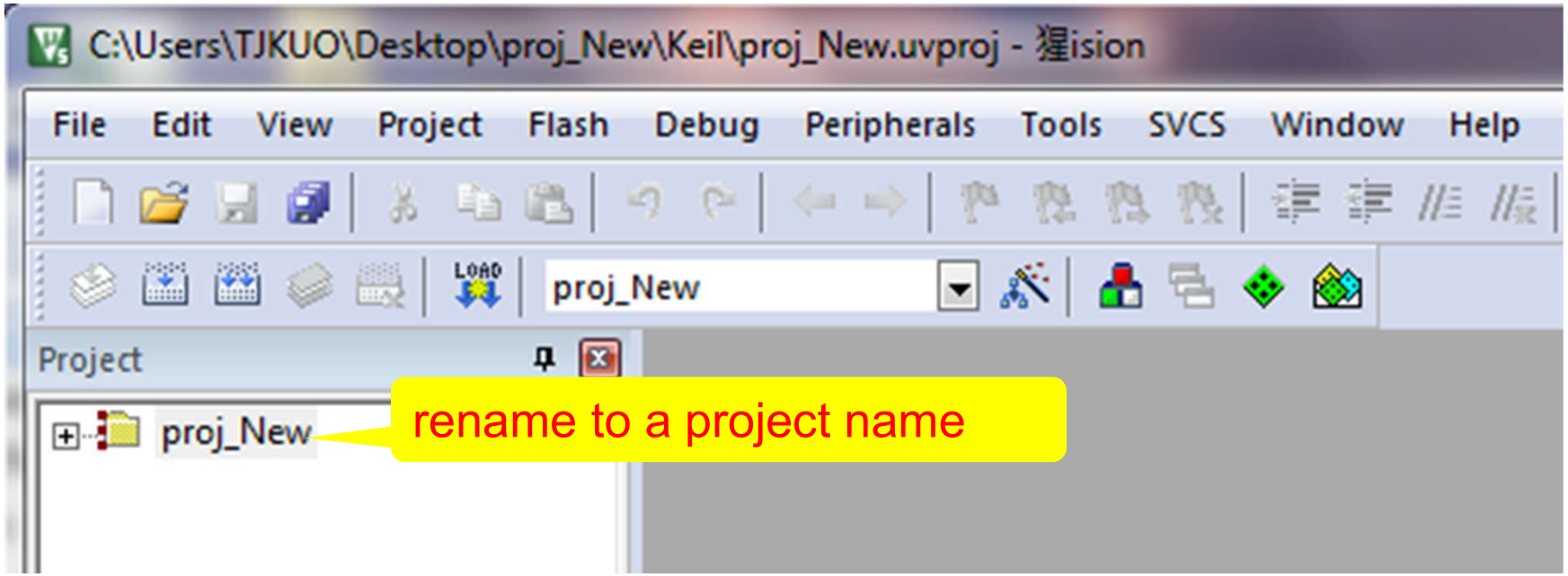
# No “copy startup_NUC1xx.s”



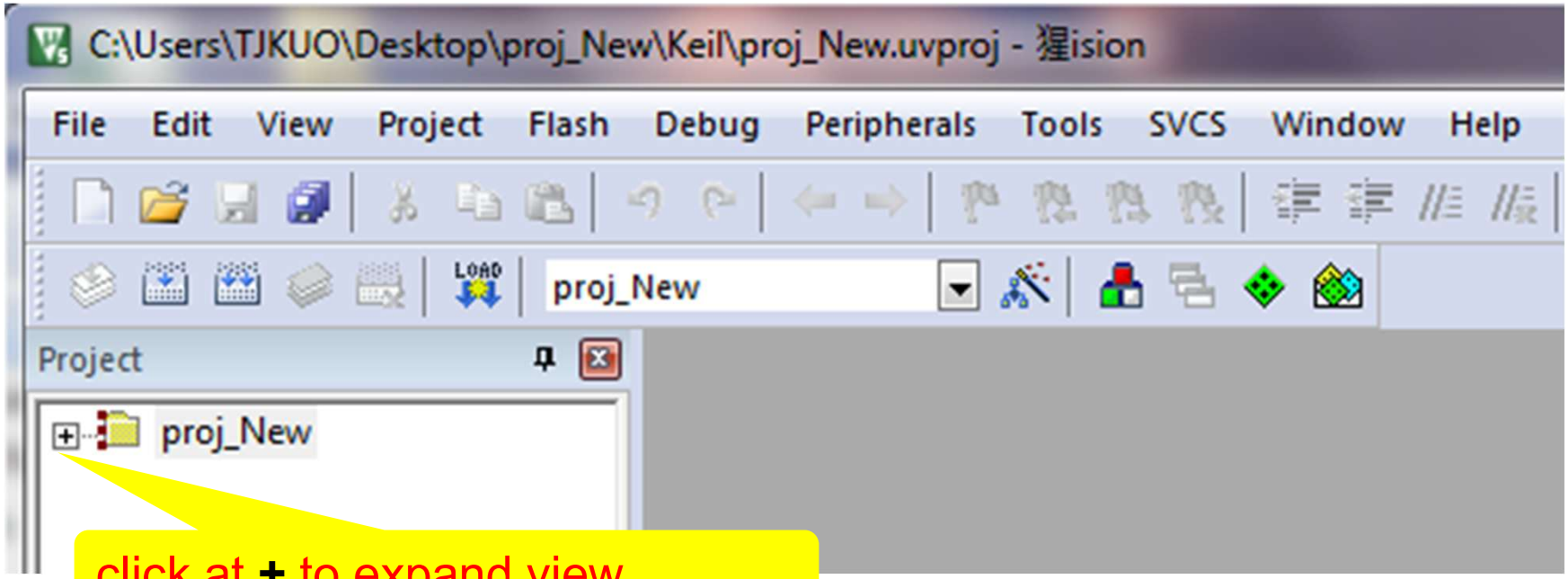
# Project created



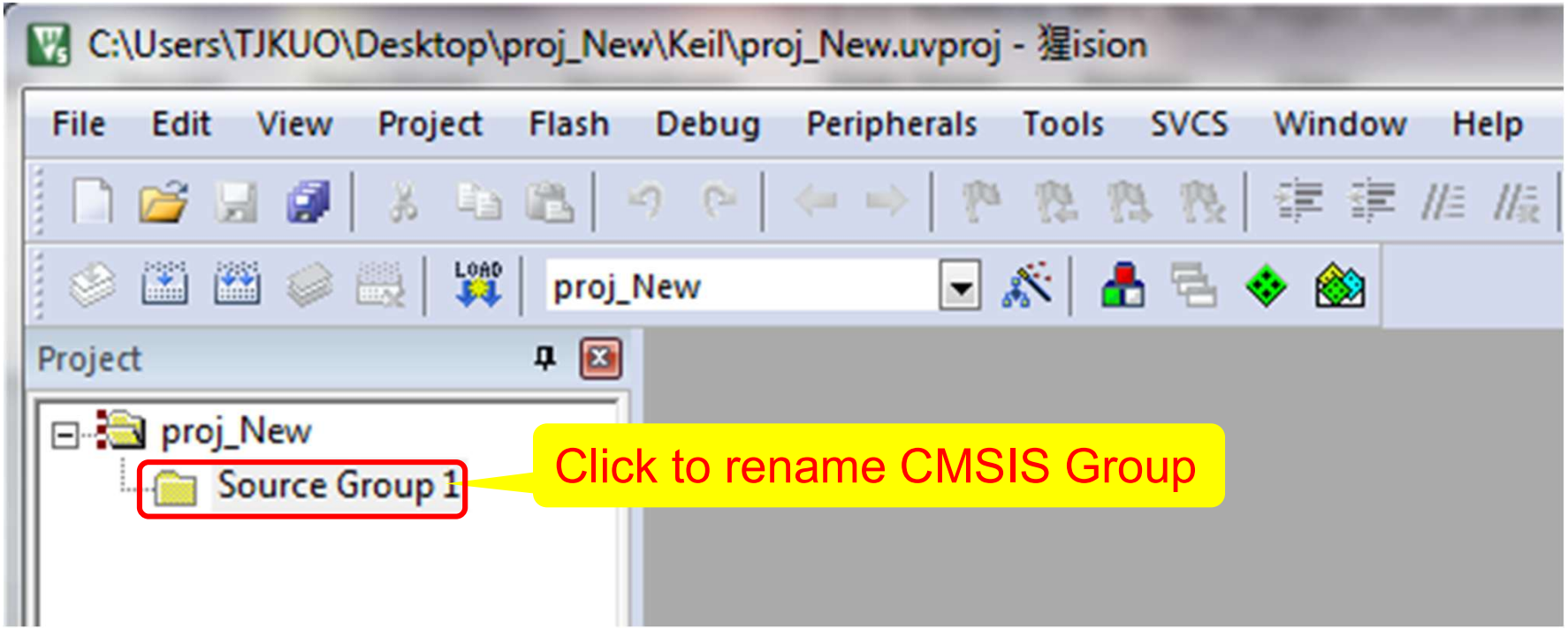
# Rename to a project name



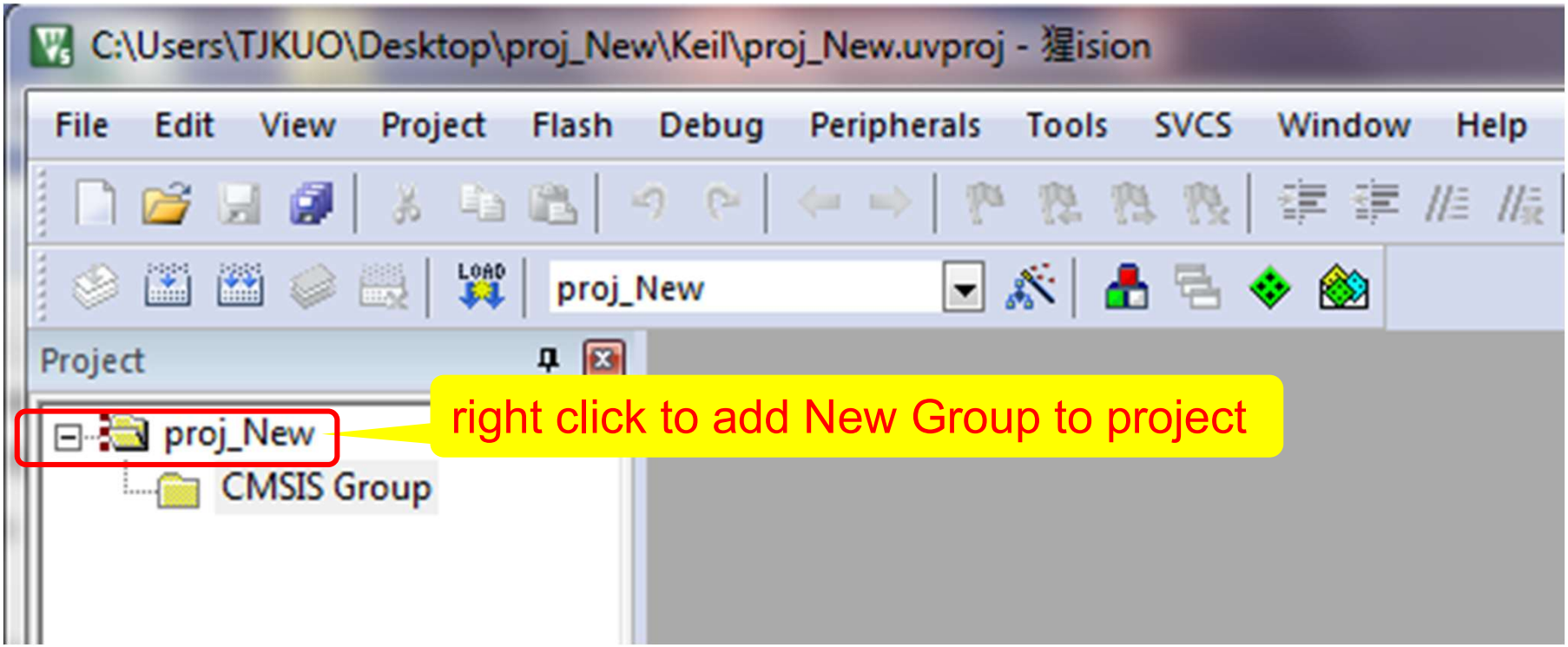
# click at + to expand view



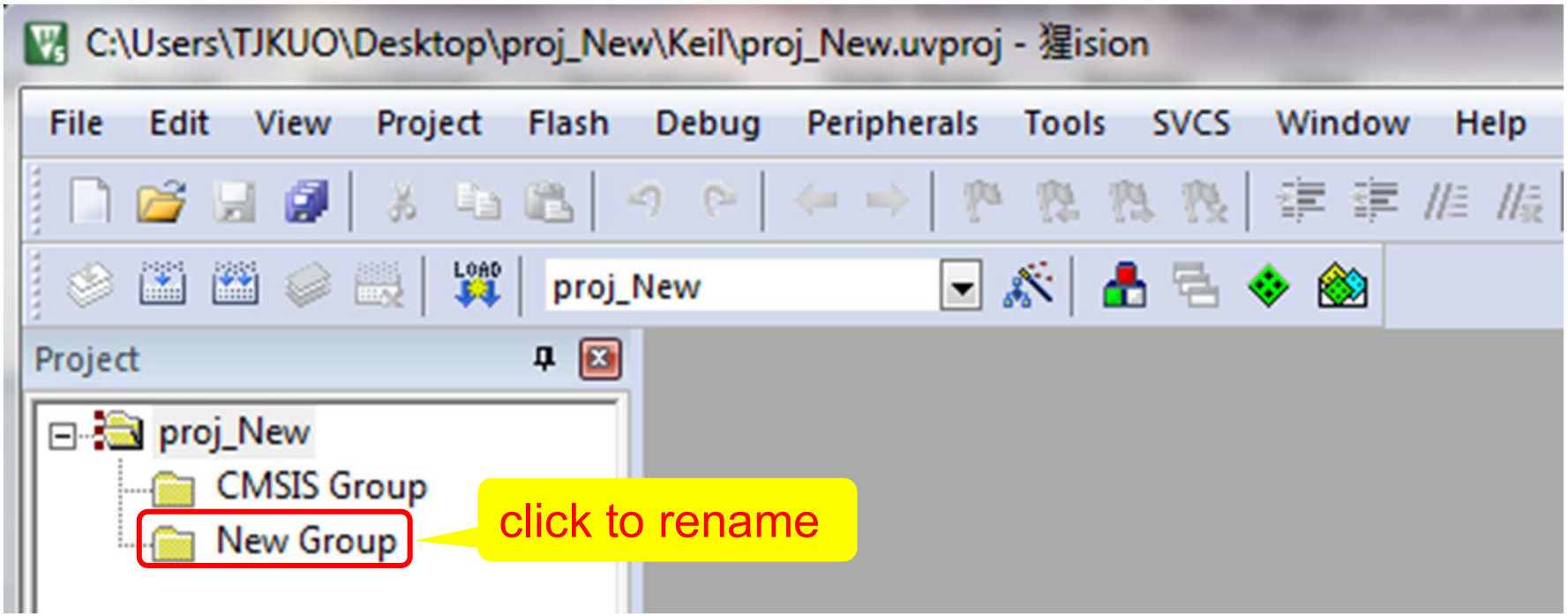
# Rename Group



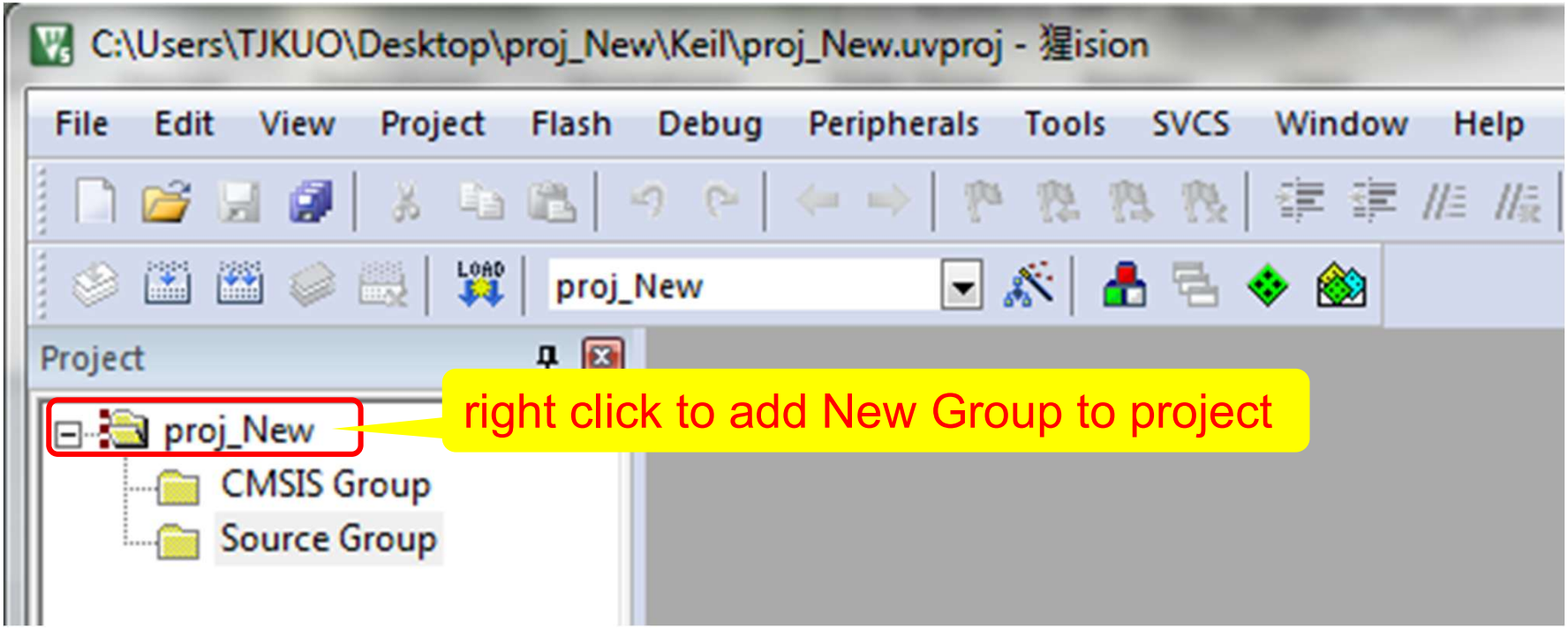
# Add New Group



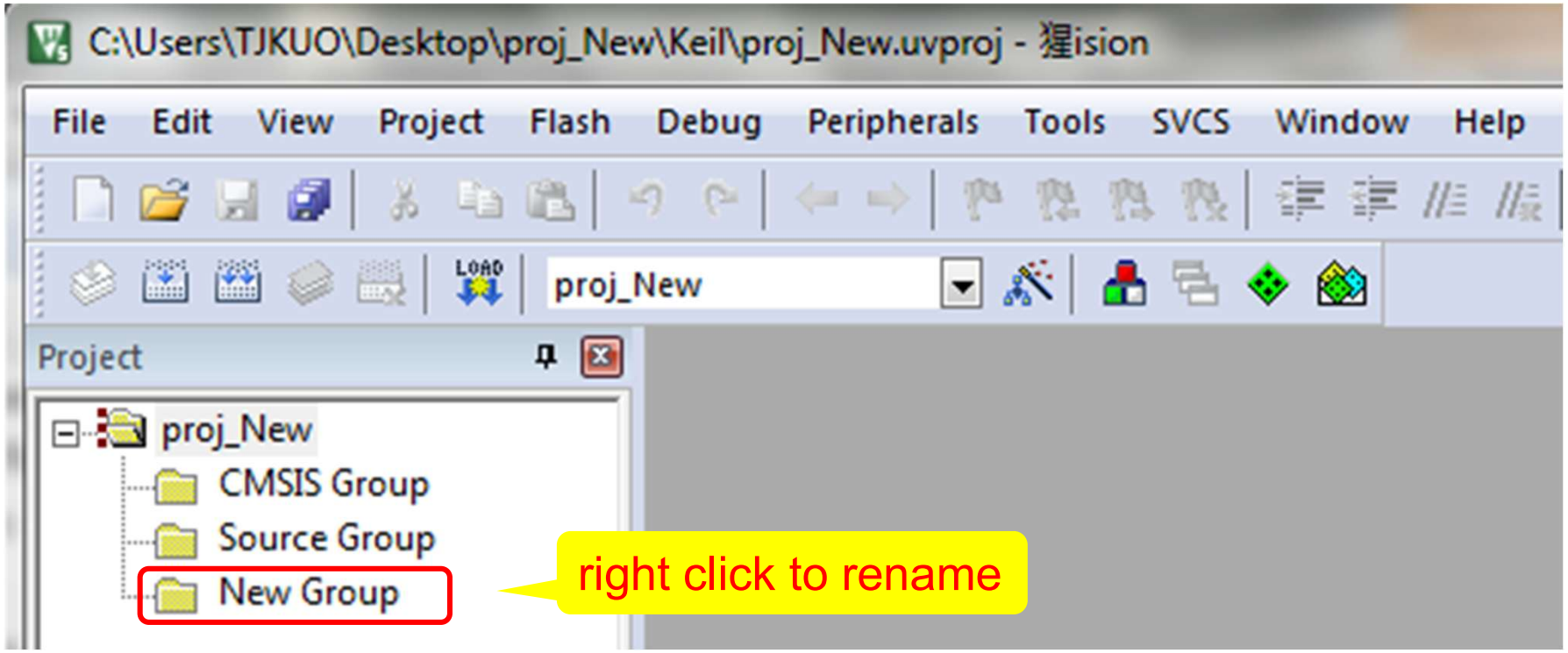
# Rename **New Group** to **Source Group**



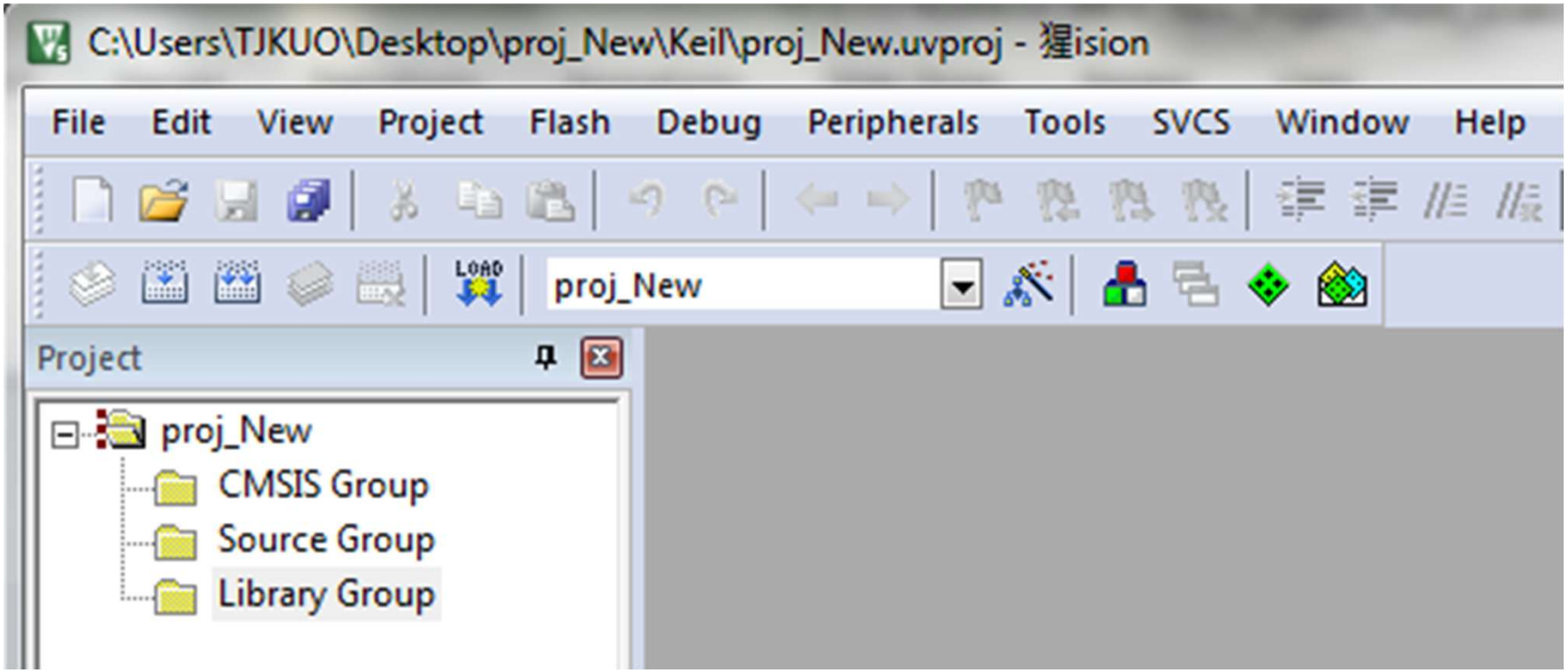
# Add New Group



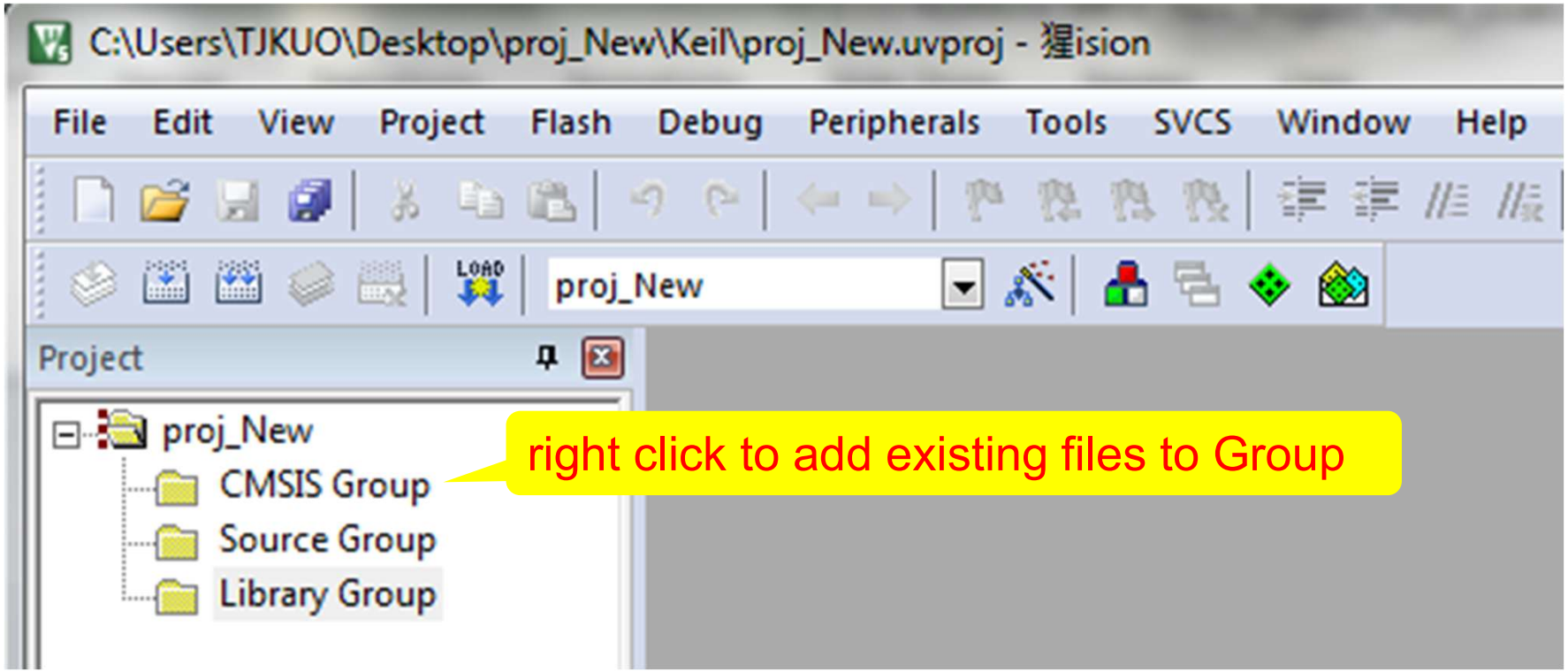
# Rename **New Group** to **Library Group**



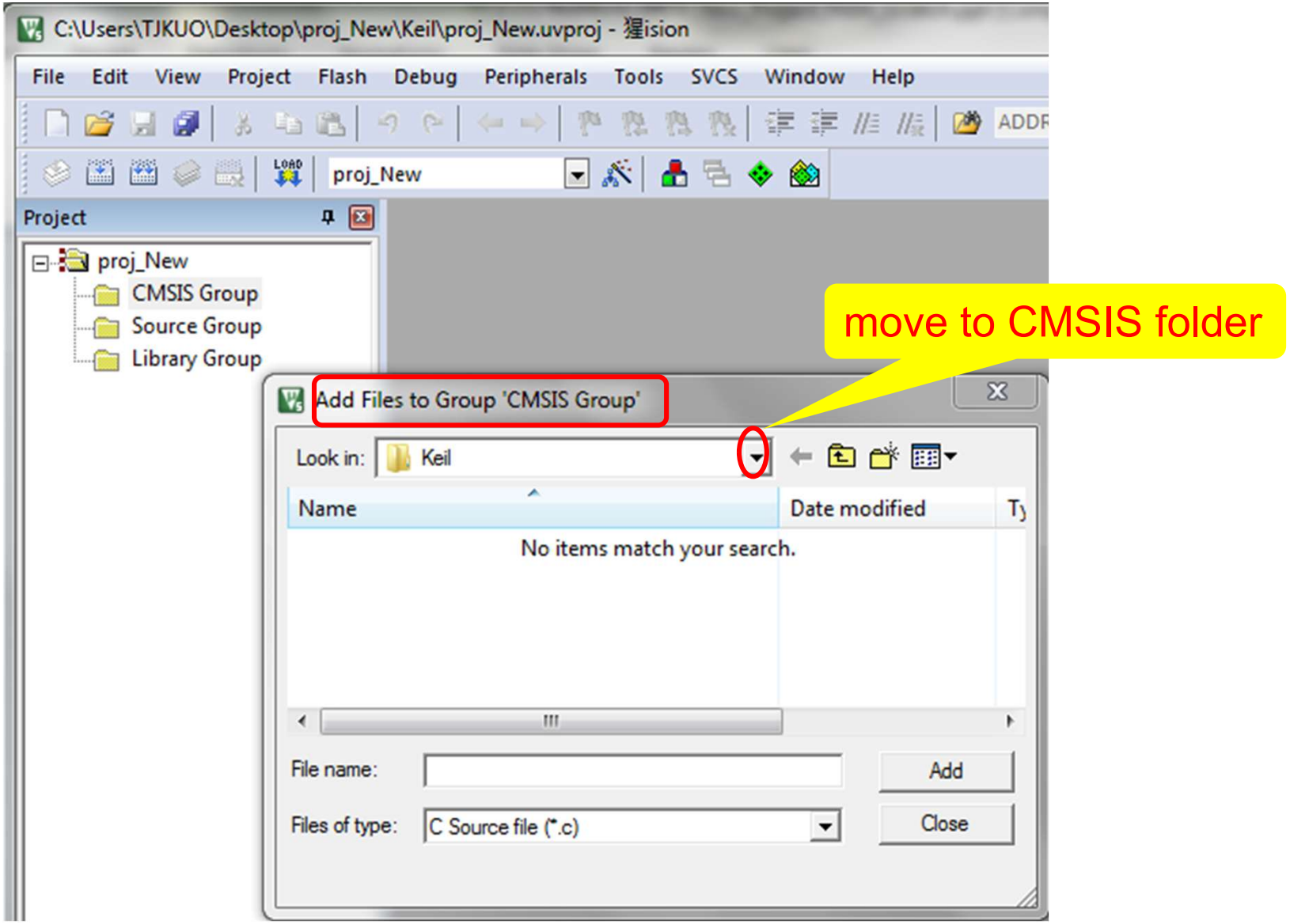
# 3 Groups created



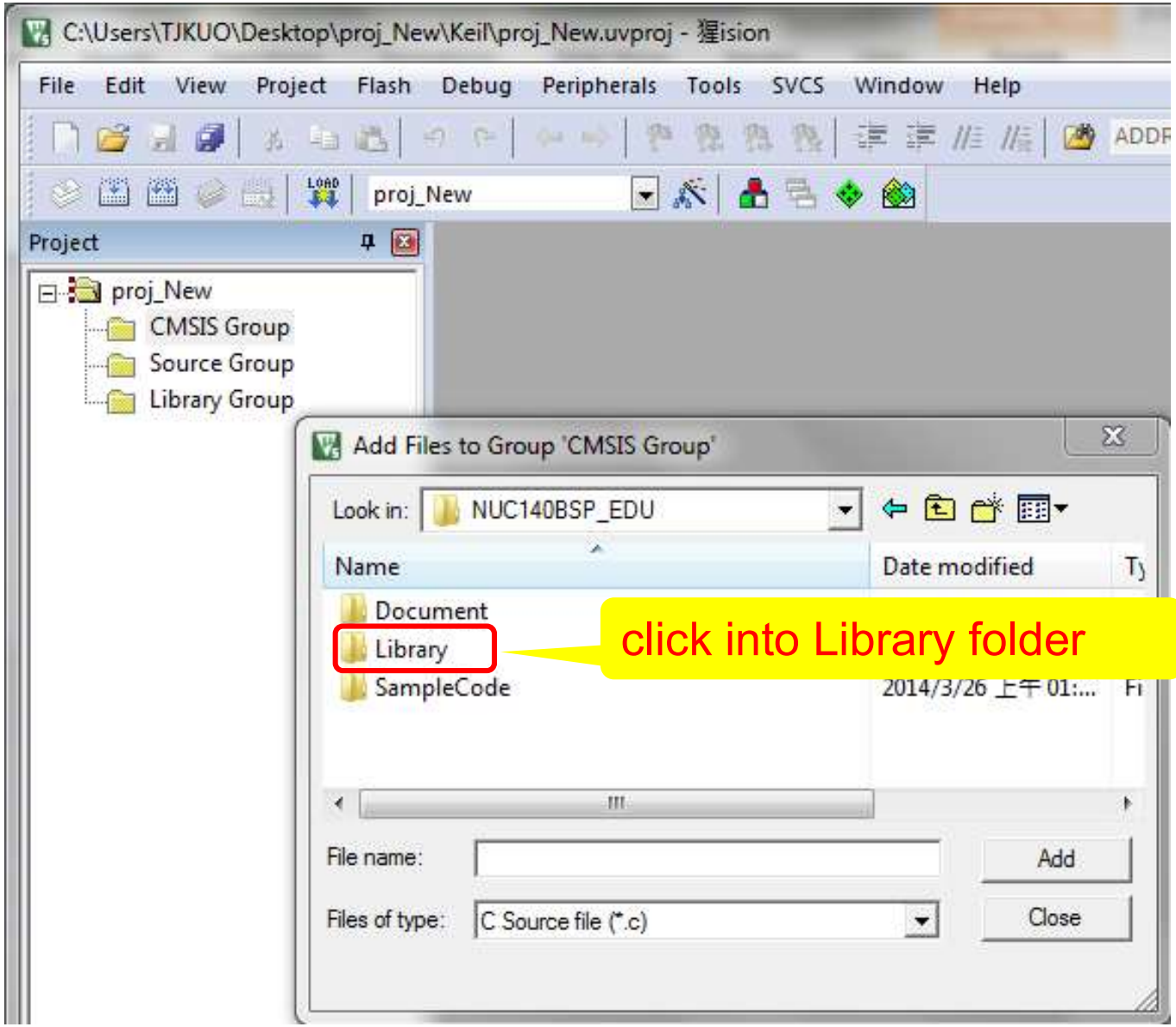
# Add Files to **CMSIS Group**



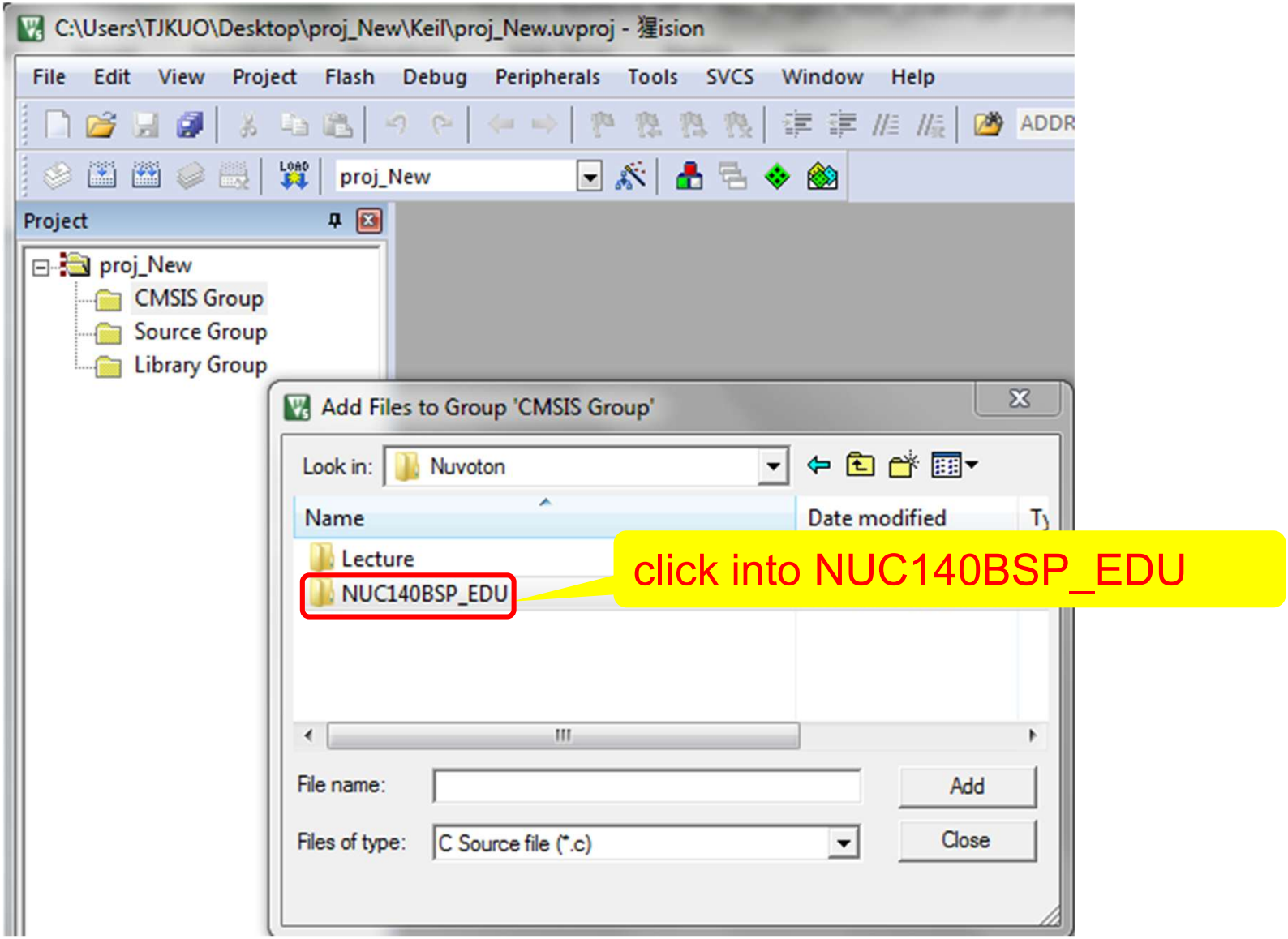
# Add Files to Group



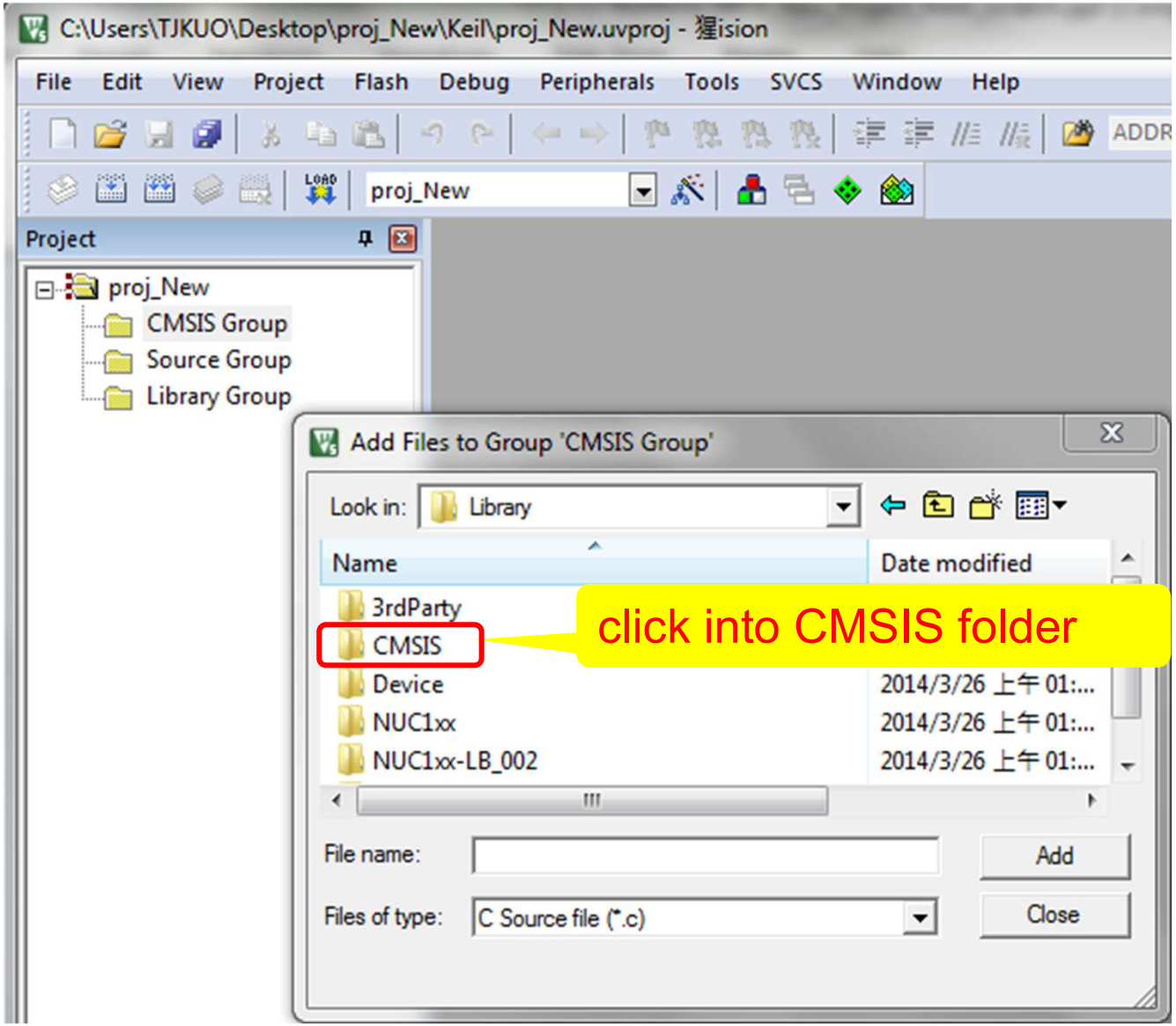
# change to **Library** folder



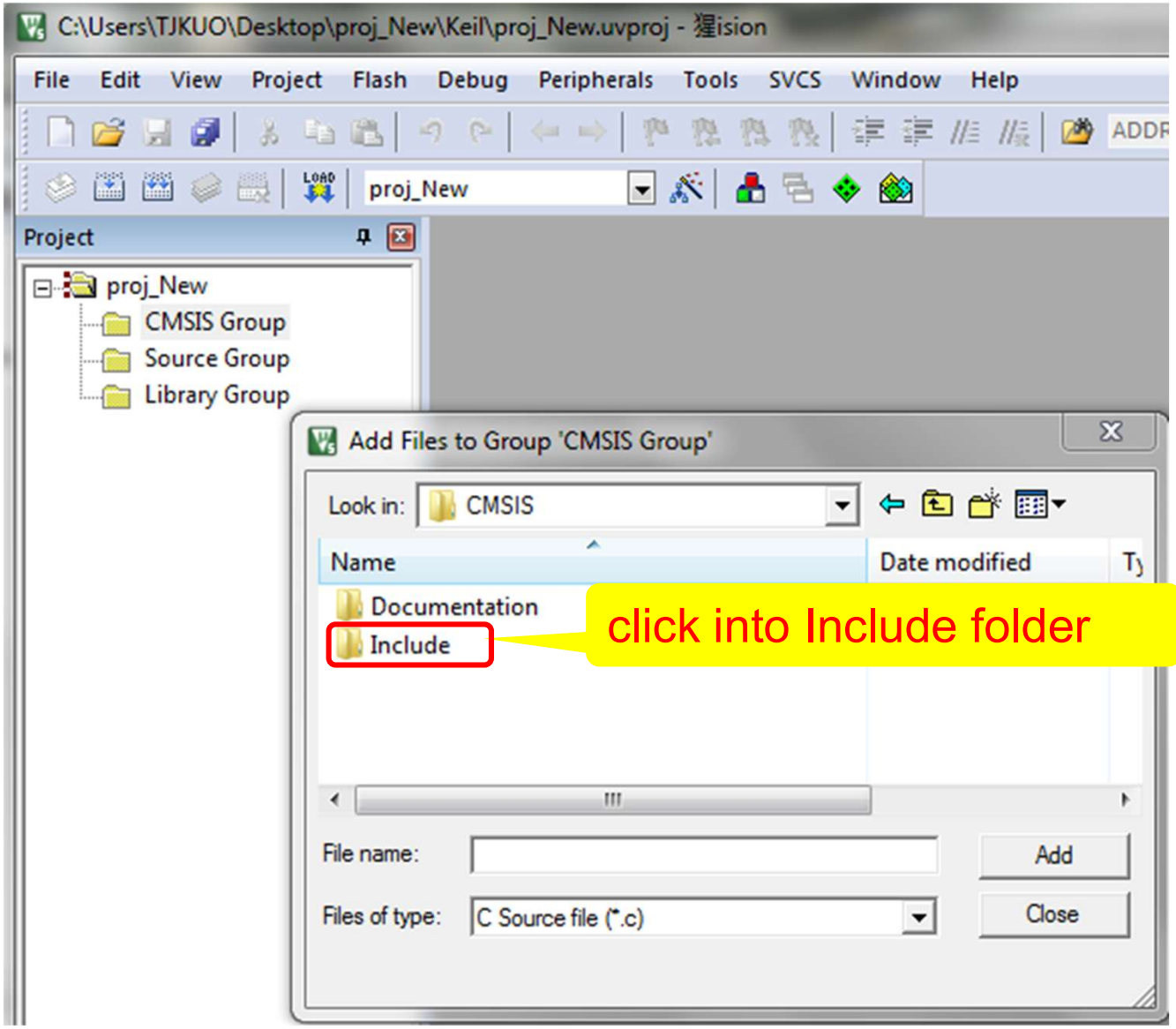
# change to **NUC140BSP_EDU**



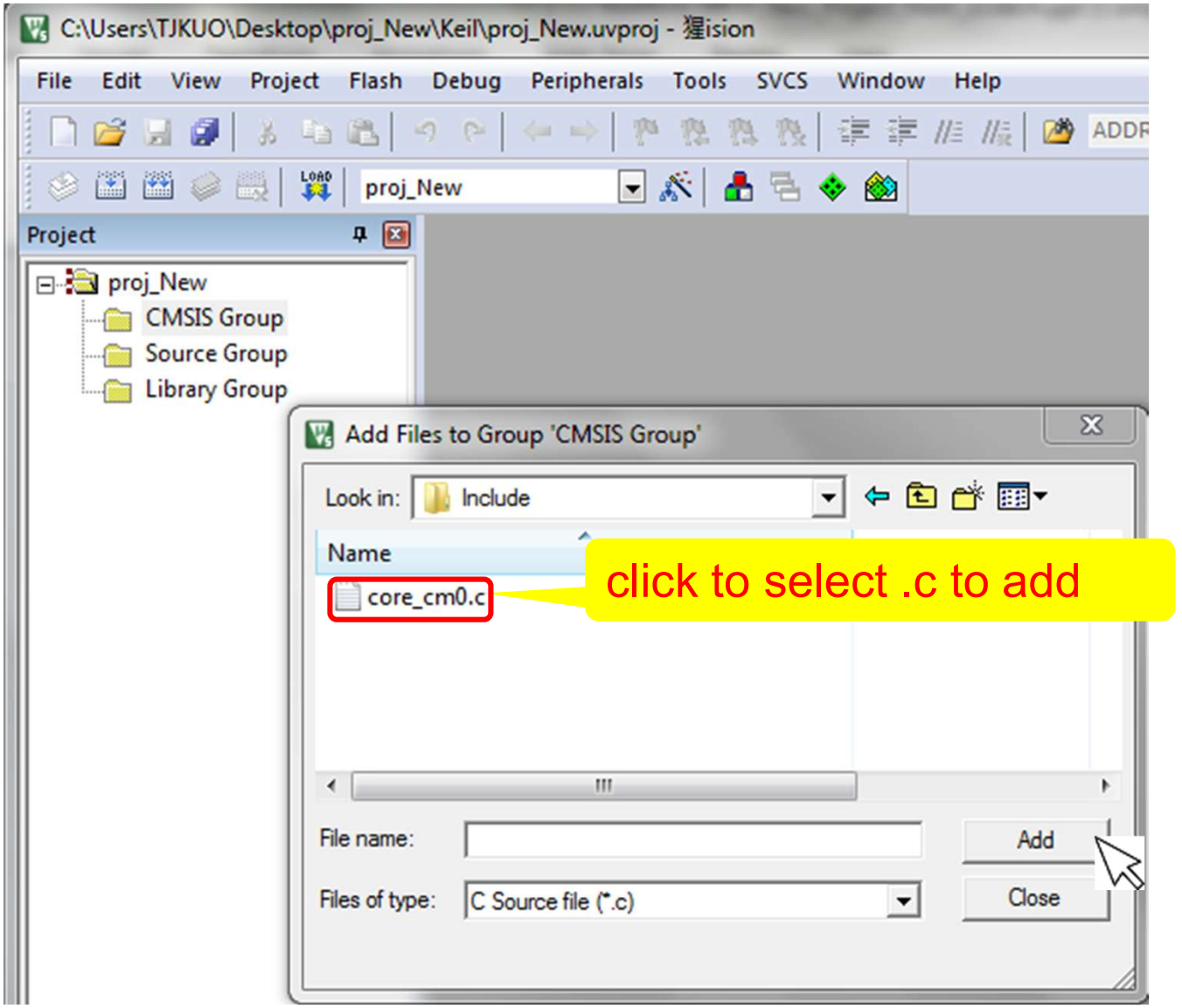
# change to **CMSIS** folder



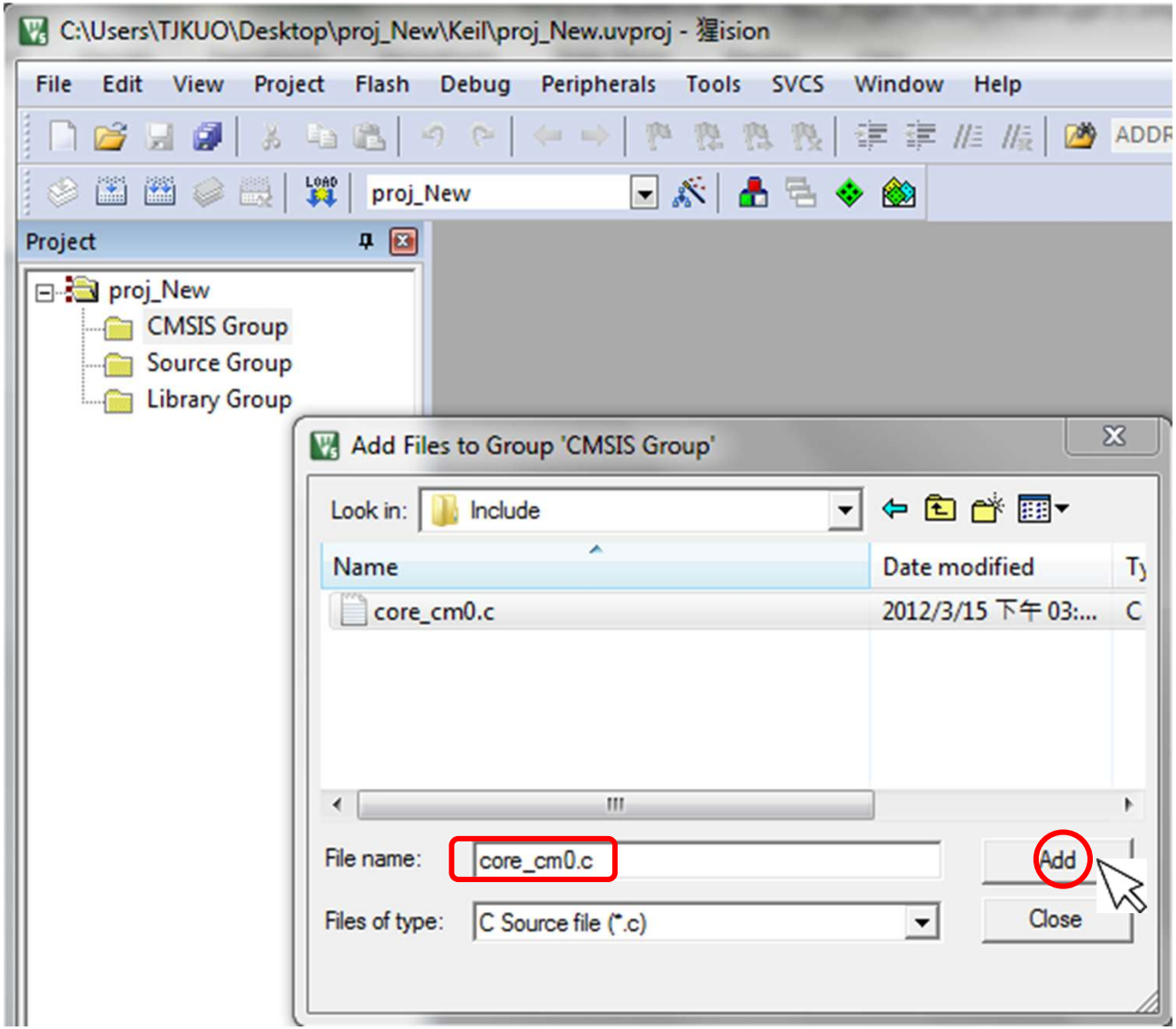
# change to **Include** folder



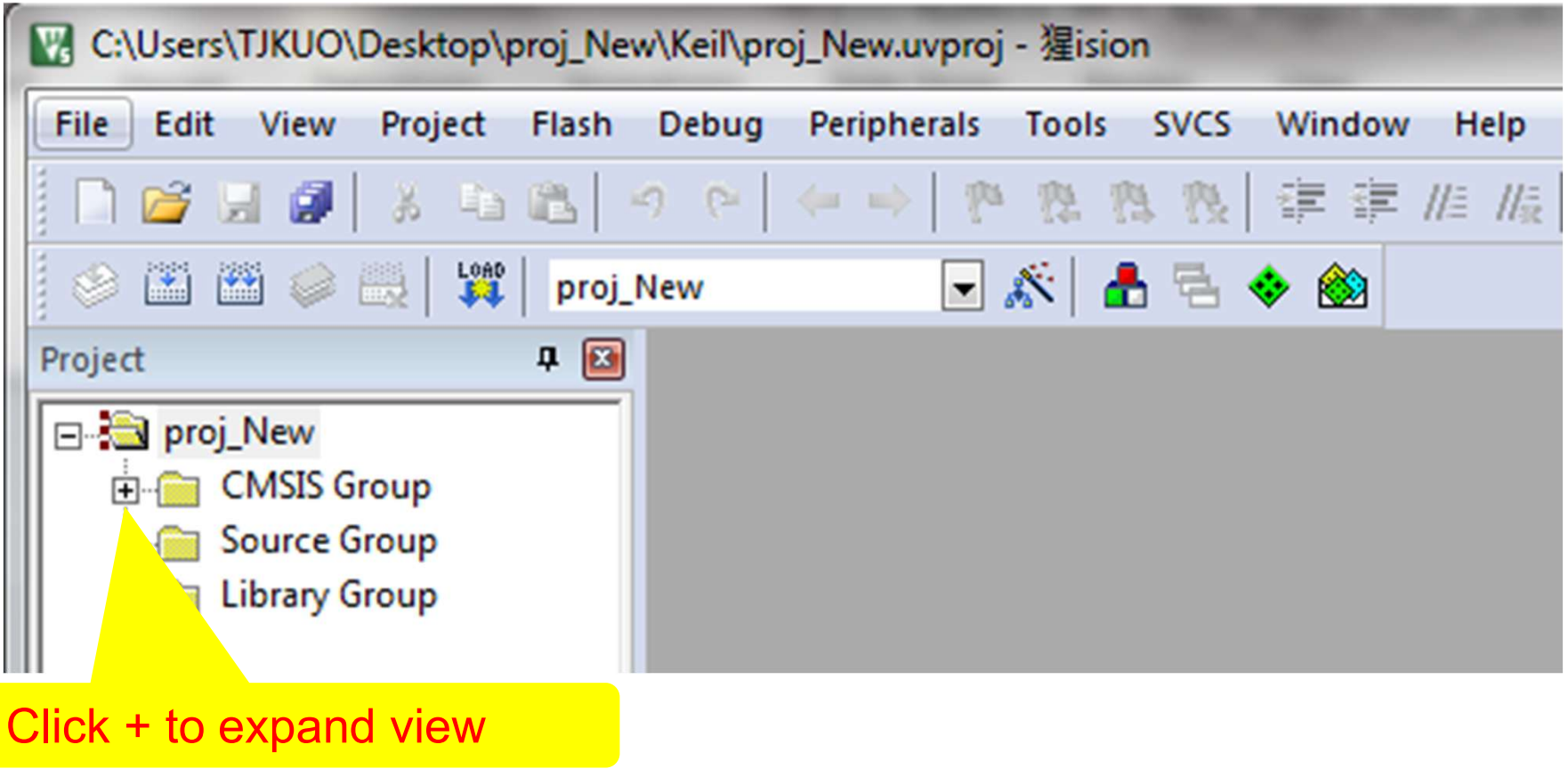
# select **core_cm0.c** to add



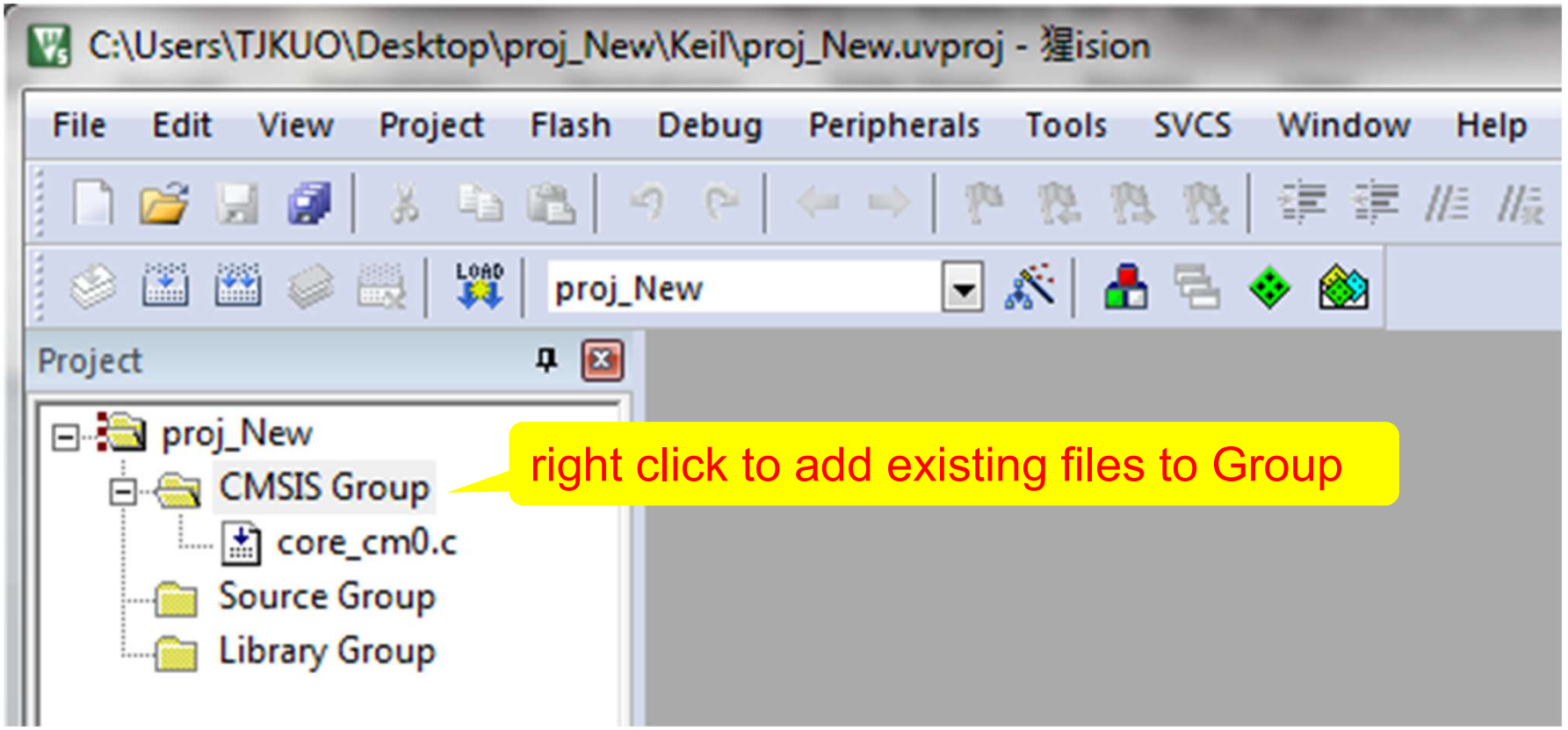
# add **core_cm0.c** to CMSIS Group



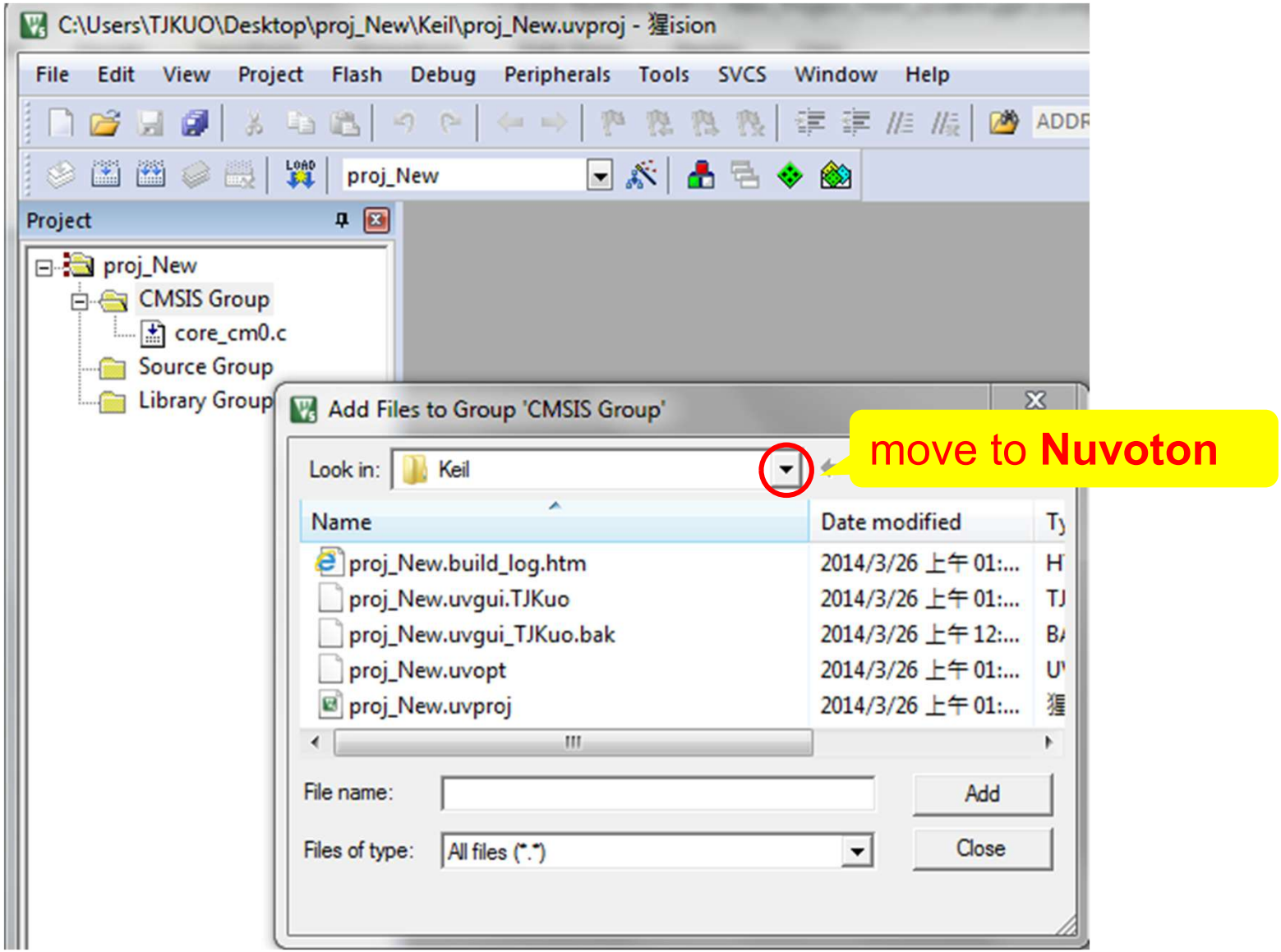
# click to expand **CMSIS Group**



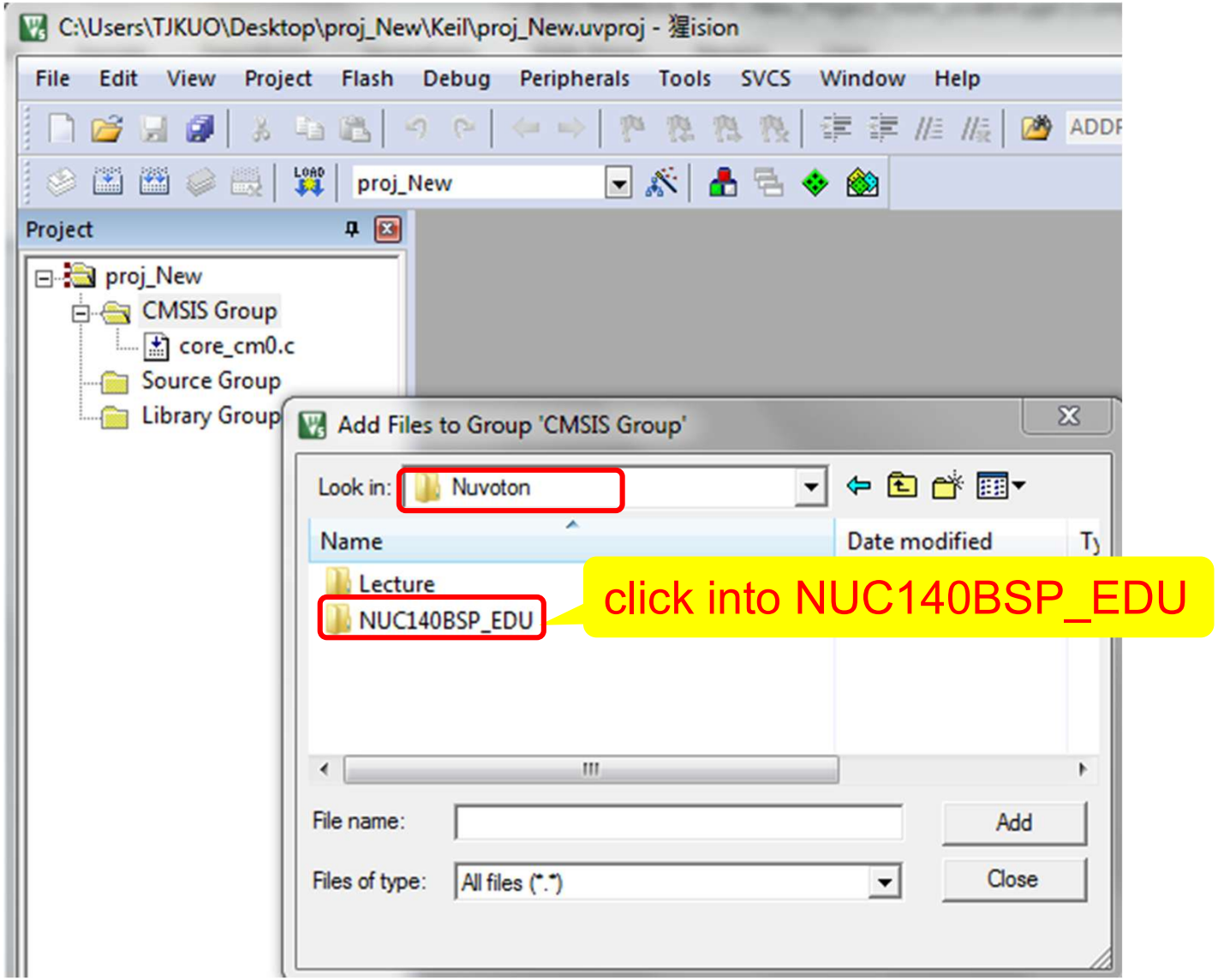
# Add Existing Files to Group



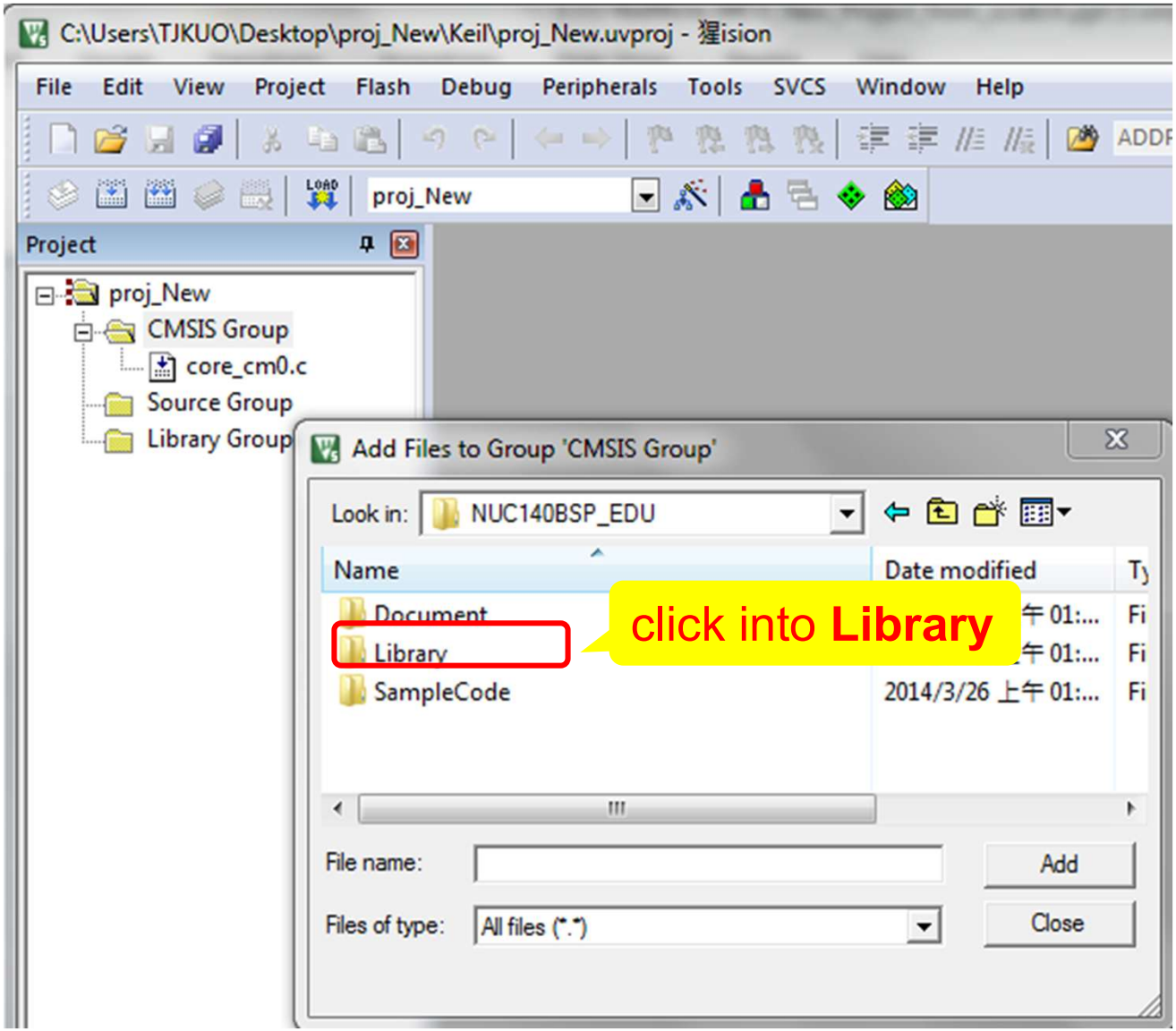
# Change to Nuvoton directory



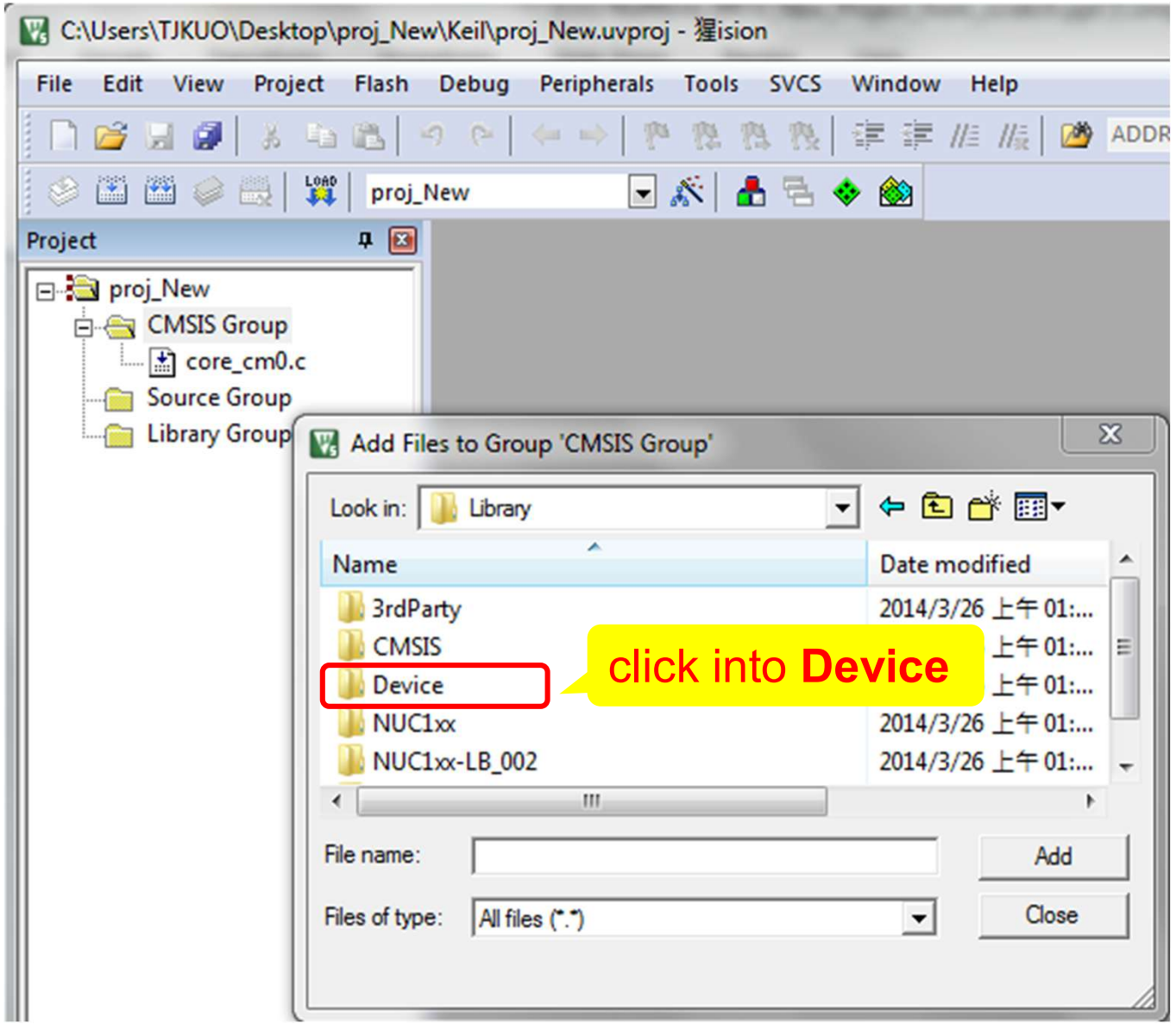
# Change to **NUC140BSP_EDU** directory



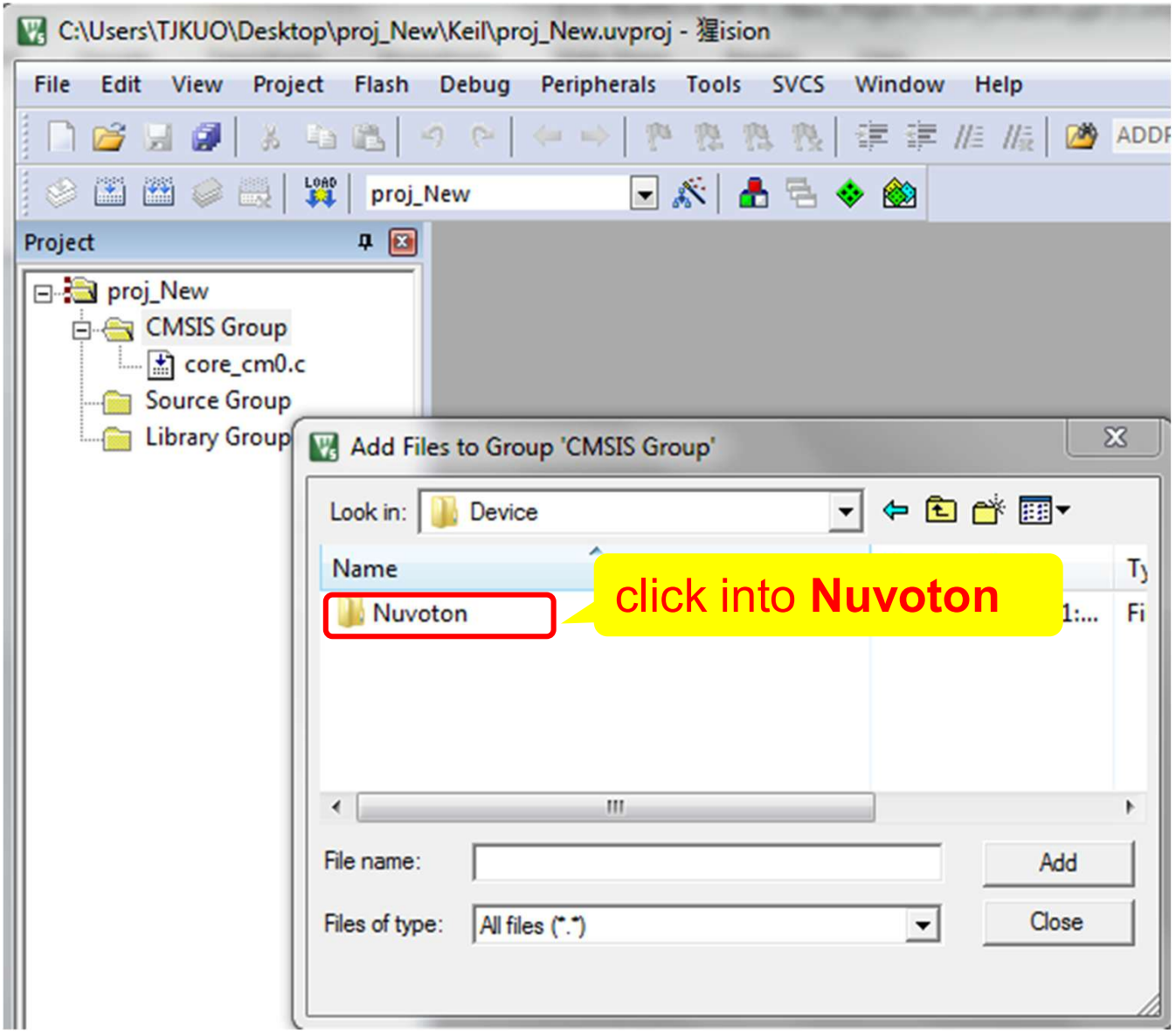
# change to **Library** directory



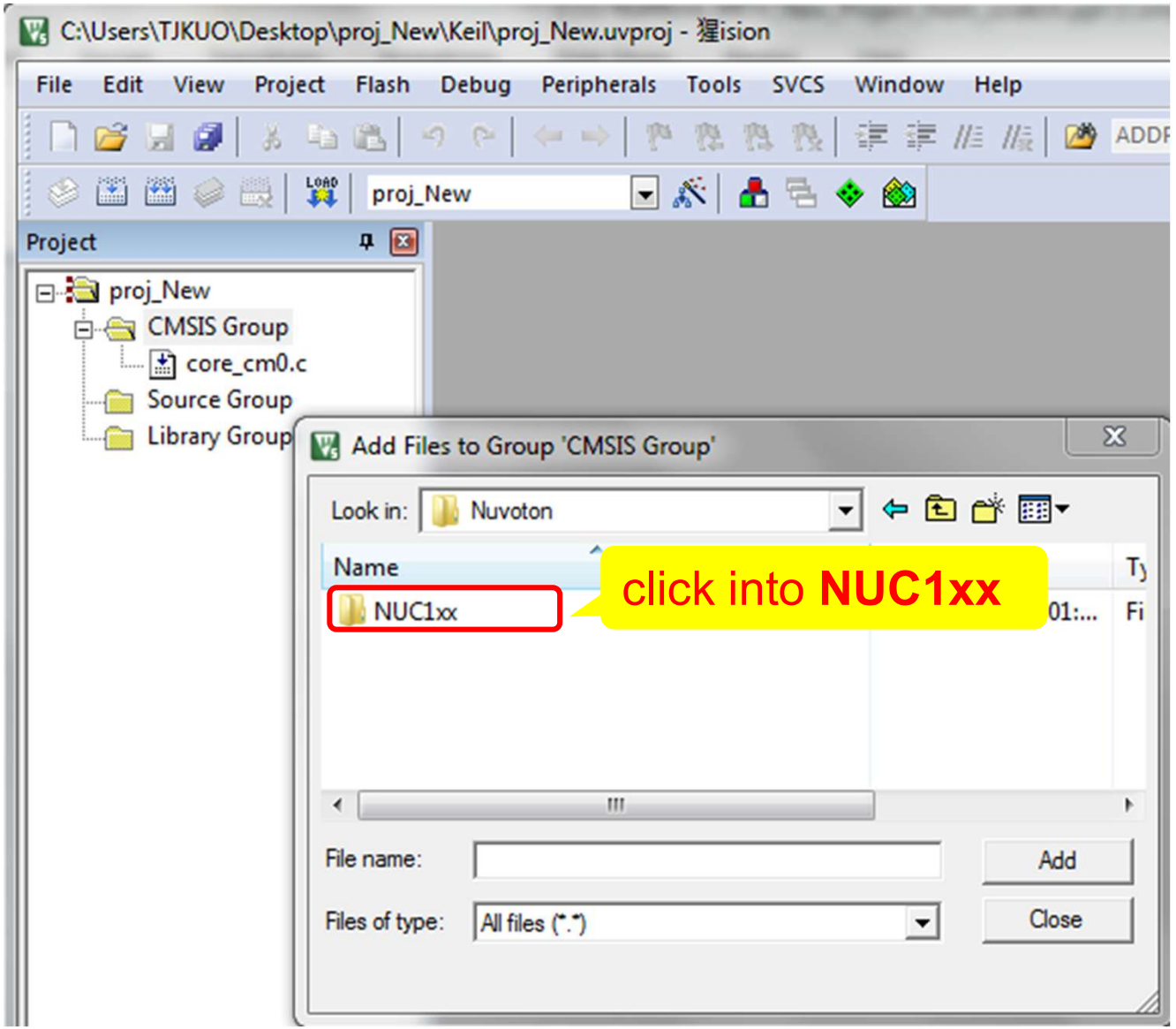
# change to **Device** directory



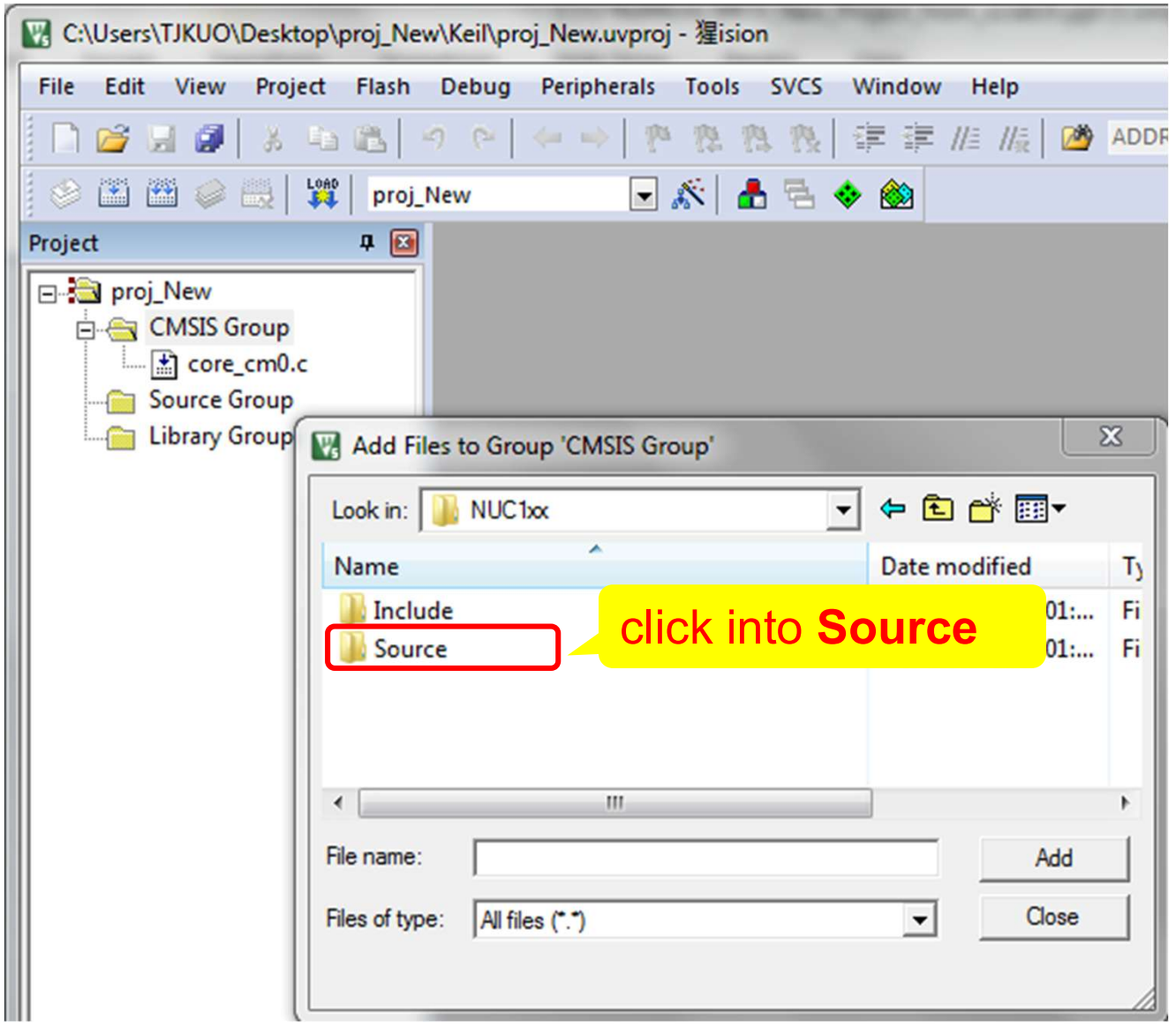
# change to **Nuvoton** directory



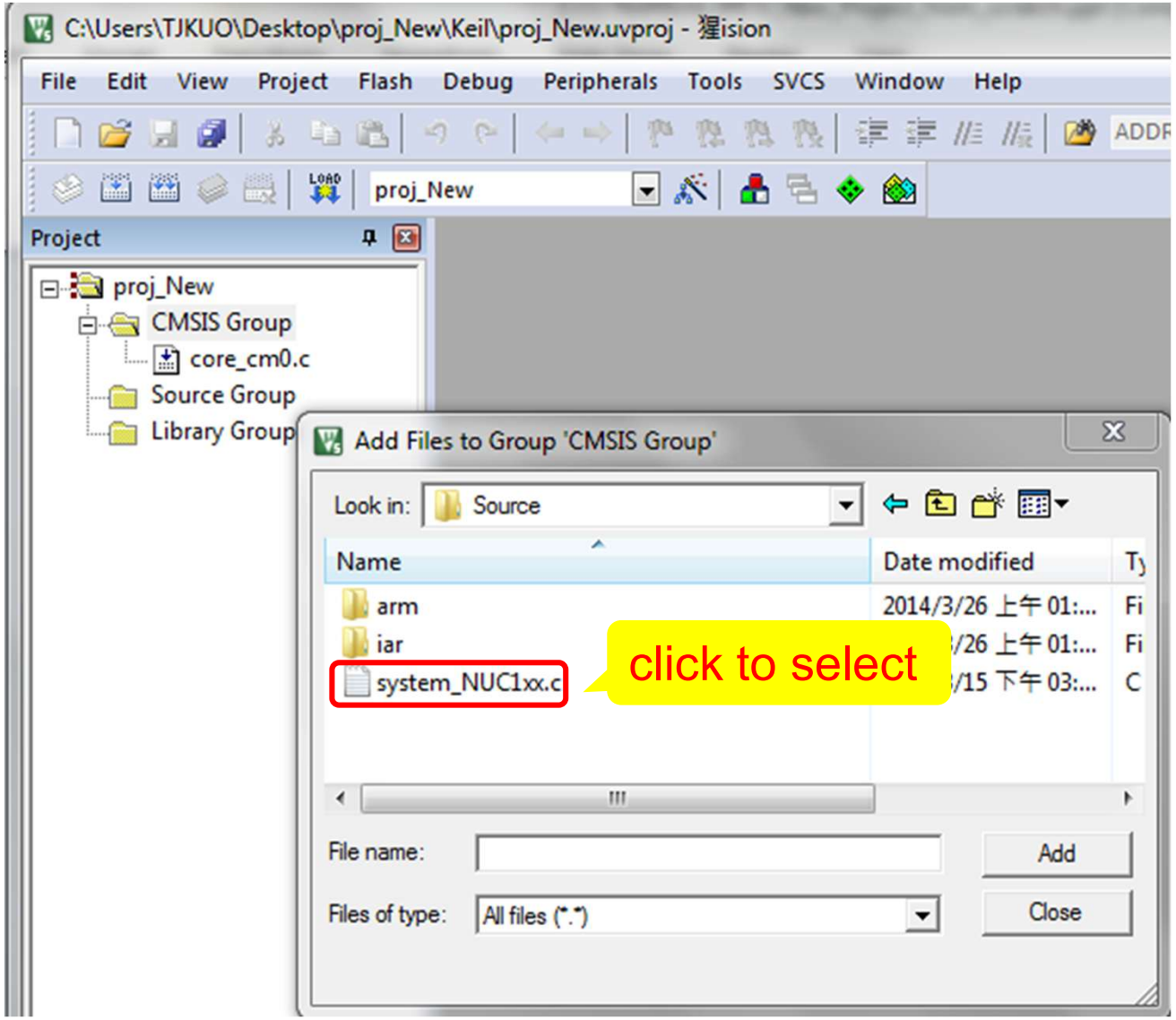
# change to **NUC1xx** directory



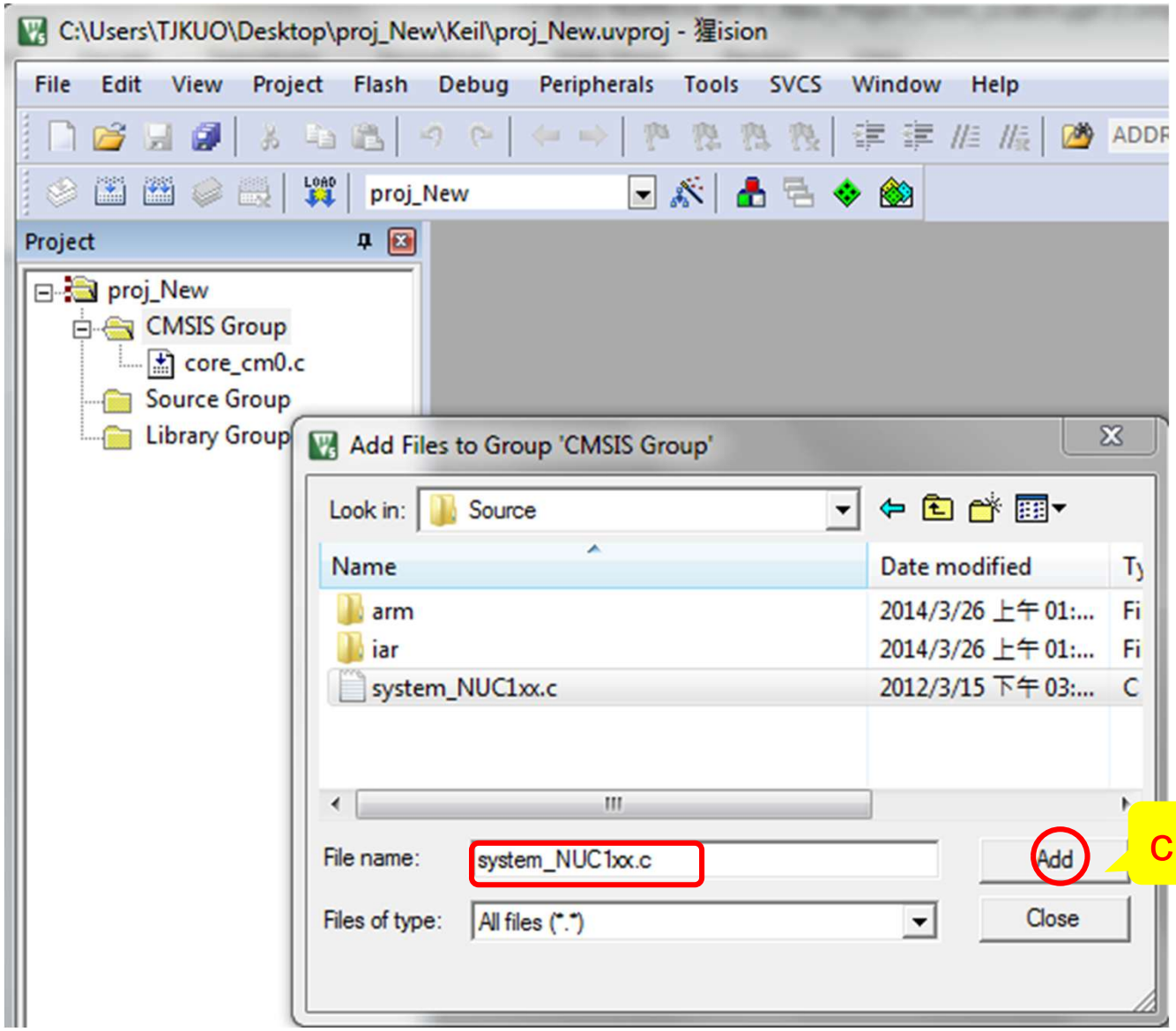
# change to **Source** directory



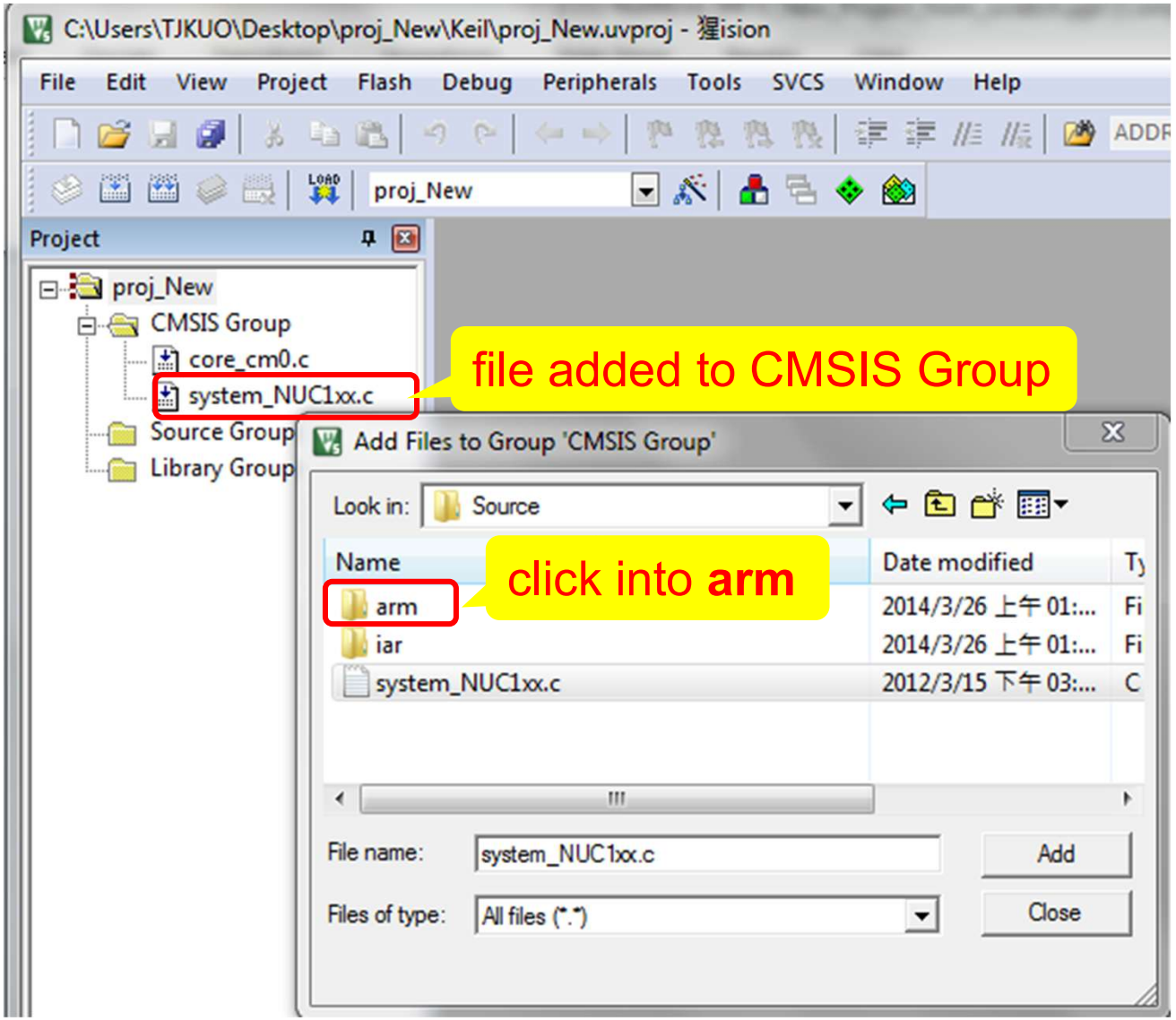
# click to select system_NUC1xx.c



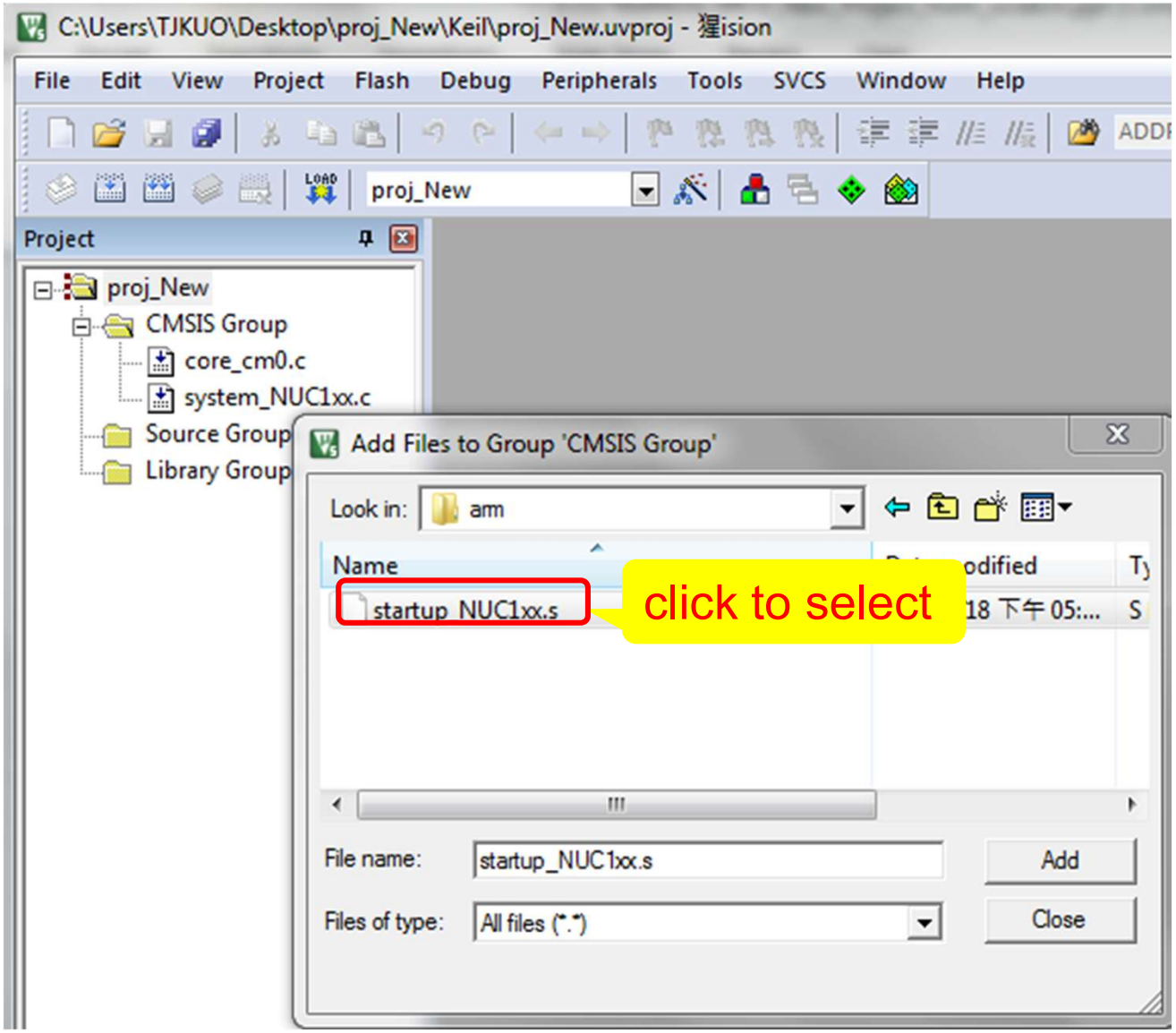
# click to add **system_NUC1xx.c**



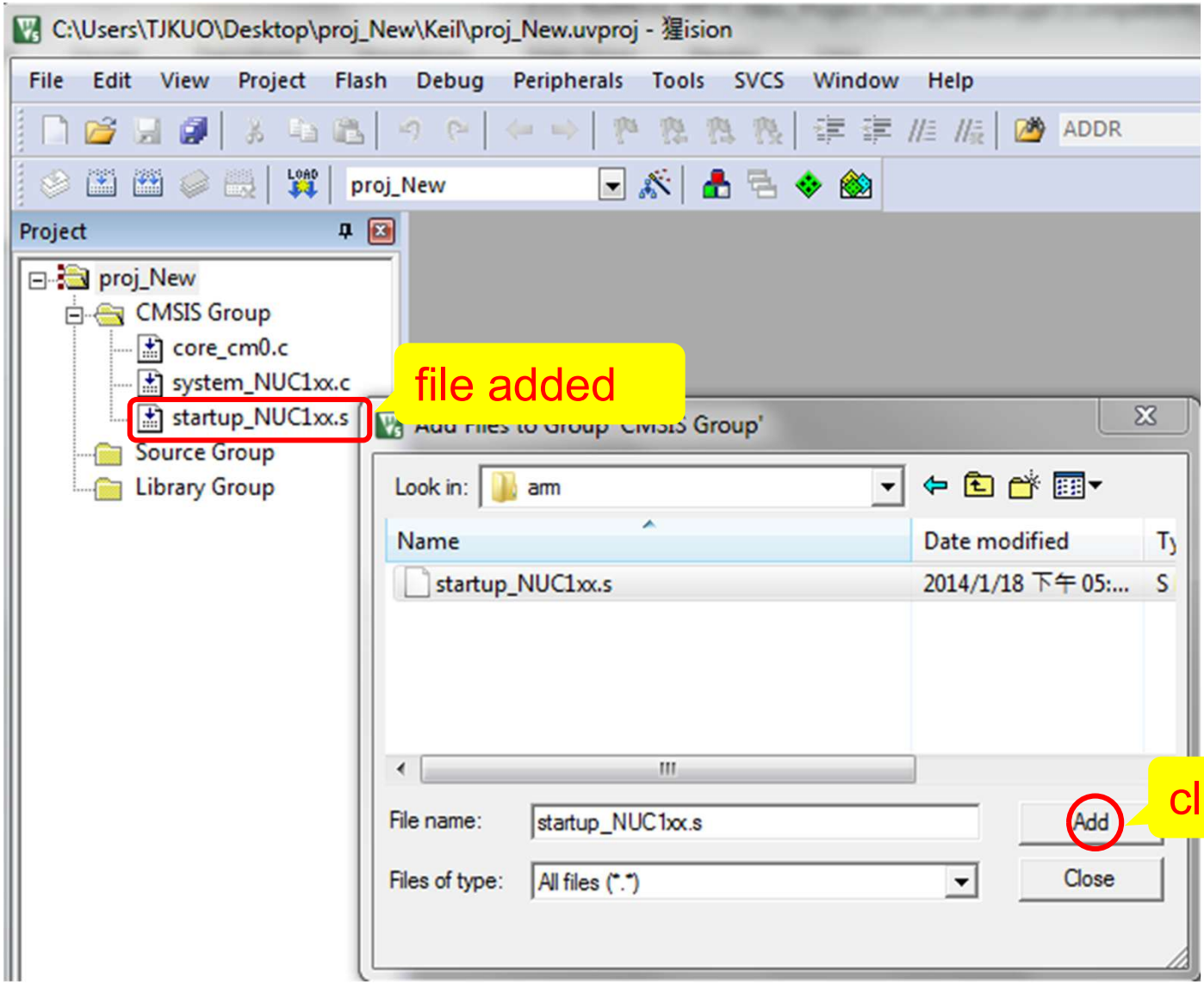
# change to **arm** directory



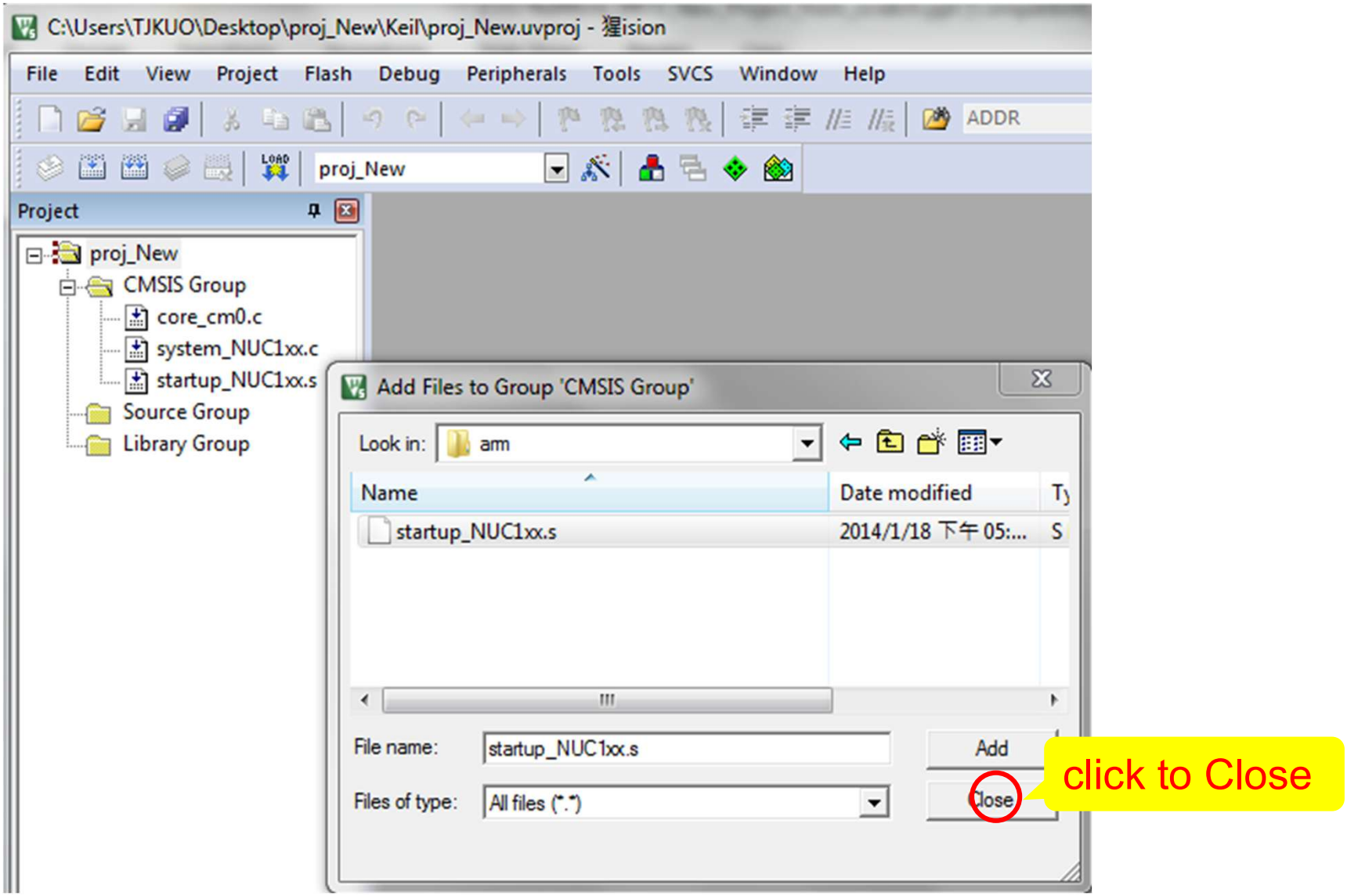
# select **startup_NUC1xx.s** to add



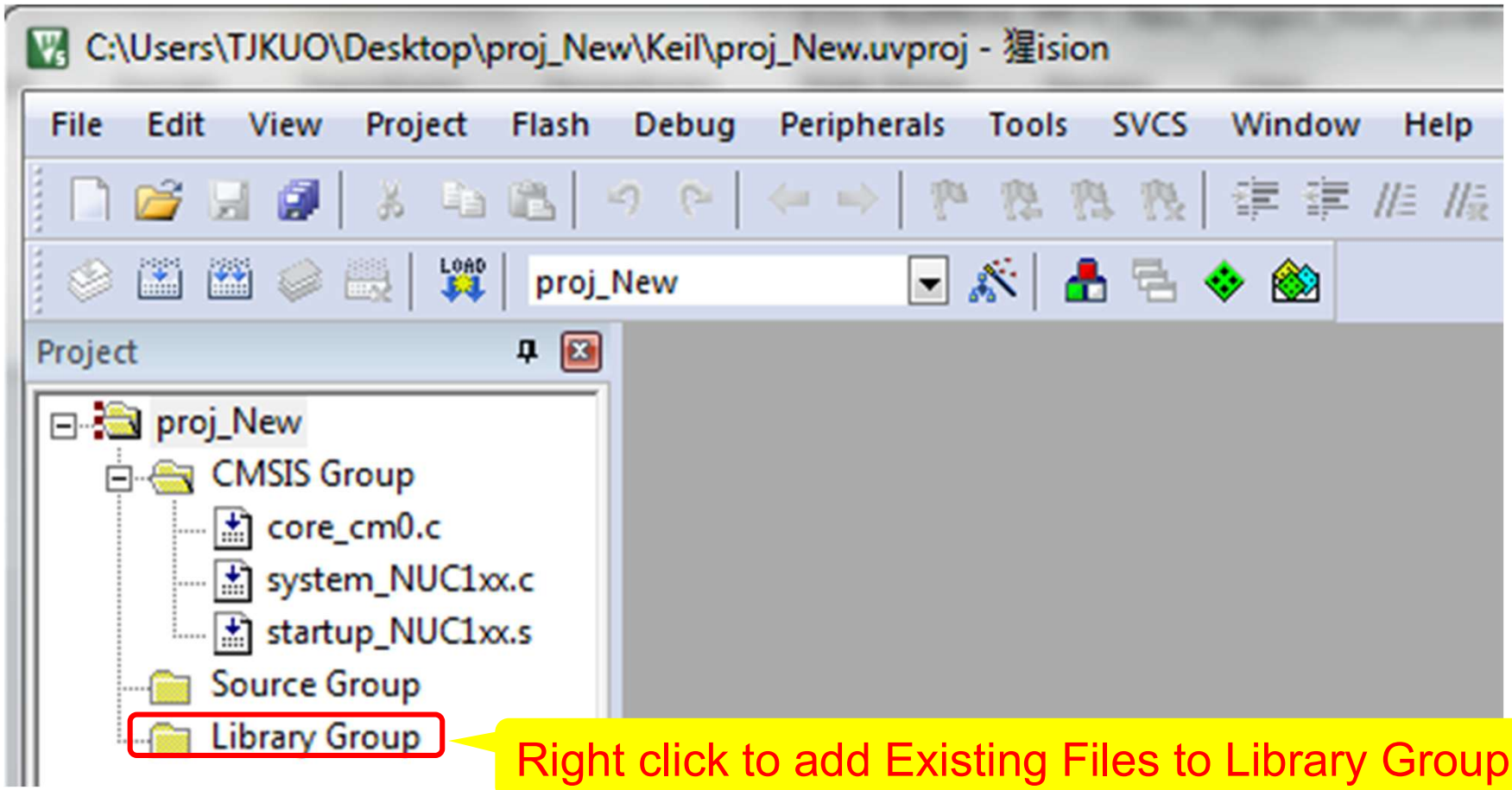
# add **startup_NUC1xx.s** to **CMSIS Group**



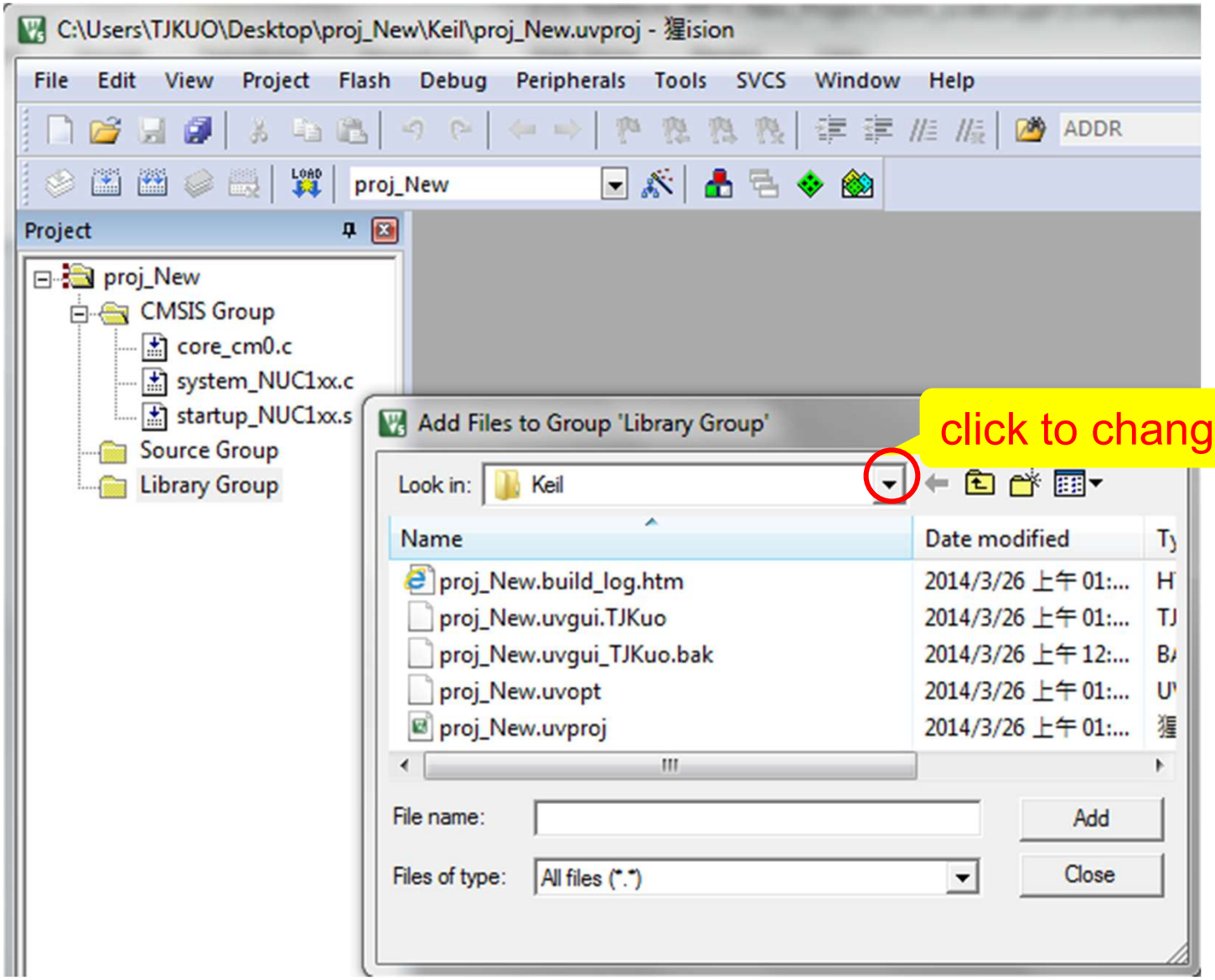
# Close Add Files to **CMSIS Group**



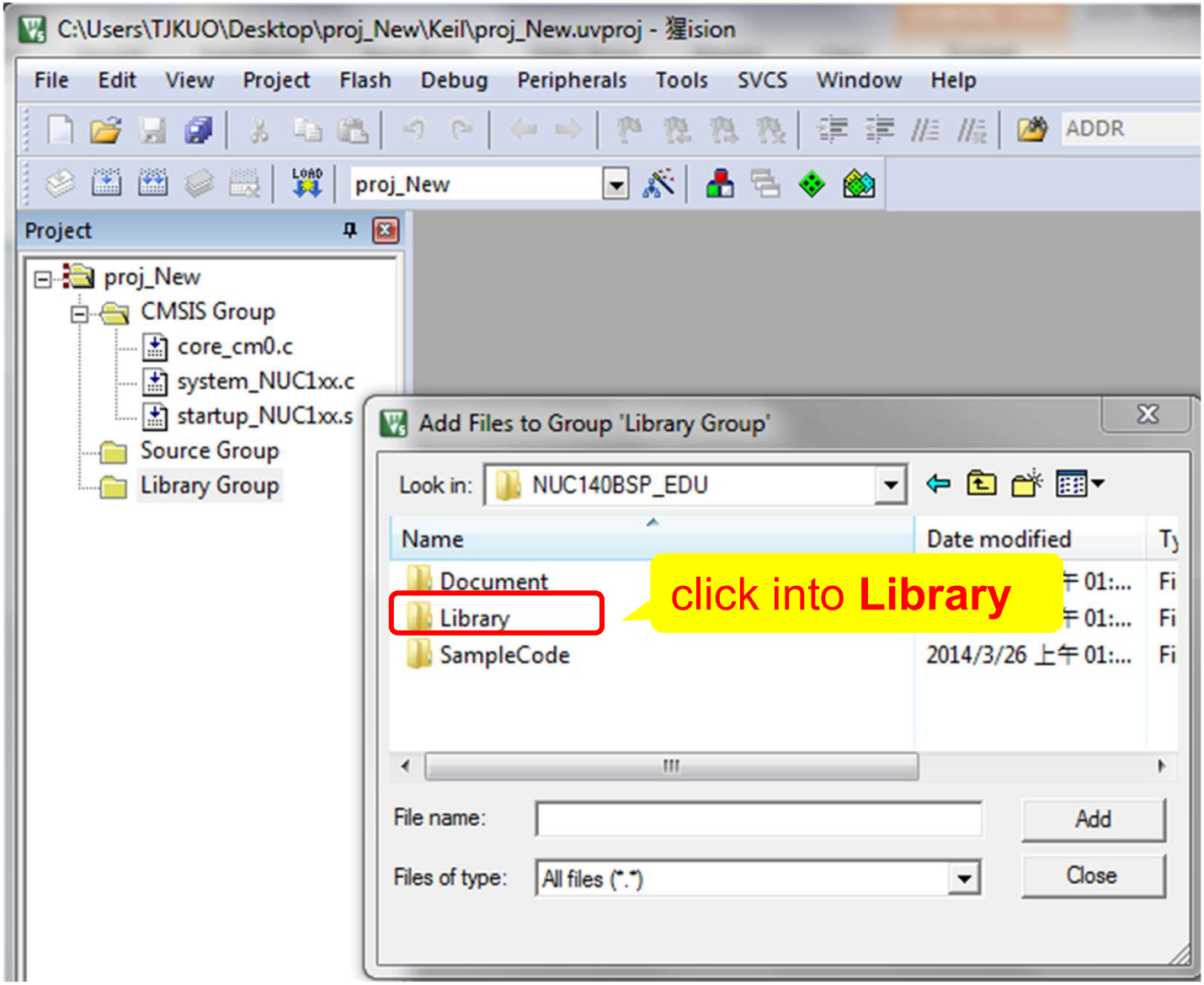
# Add **Interface Driver** to **Library Group**



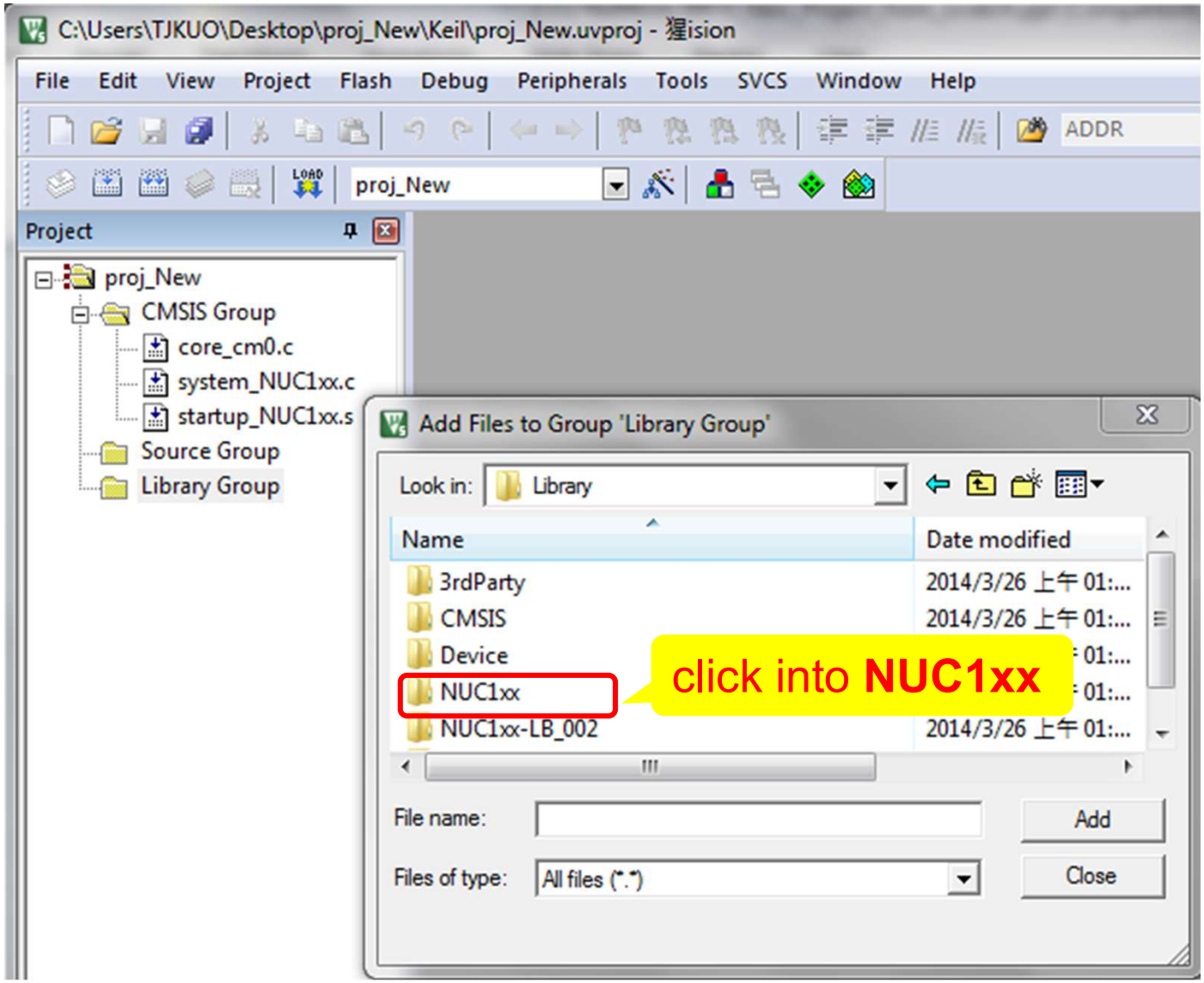
# Change to NUC140BSP_EDU directory



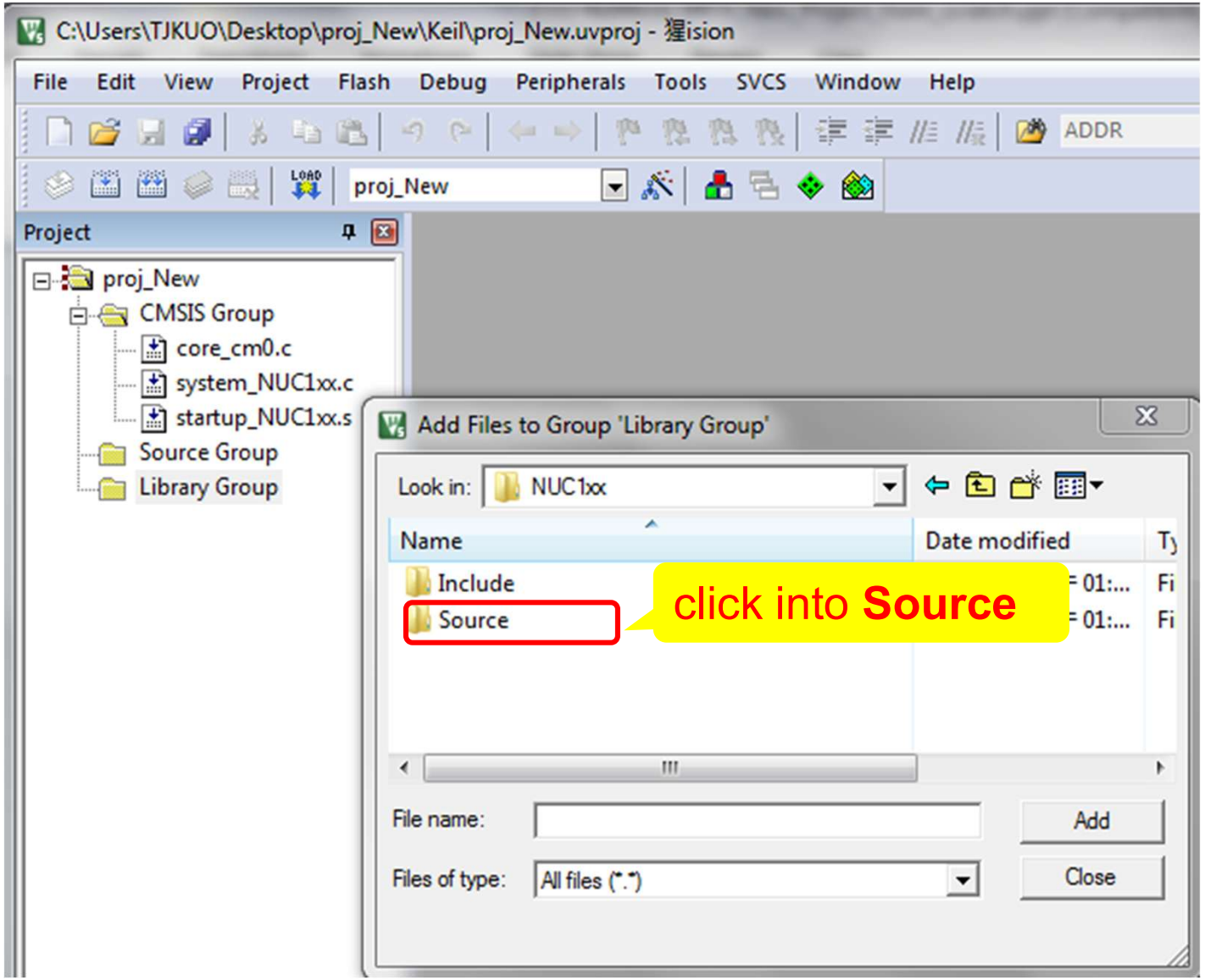
# change to **Library** directory



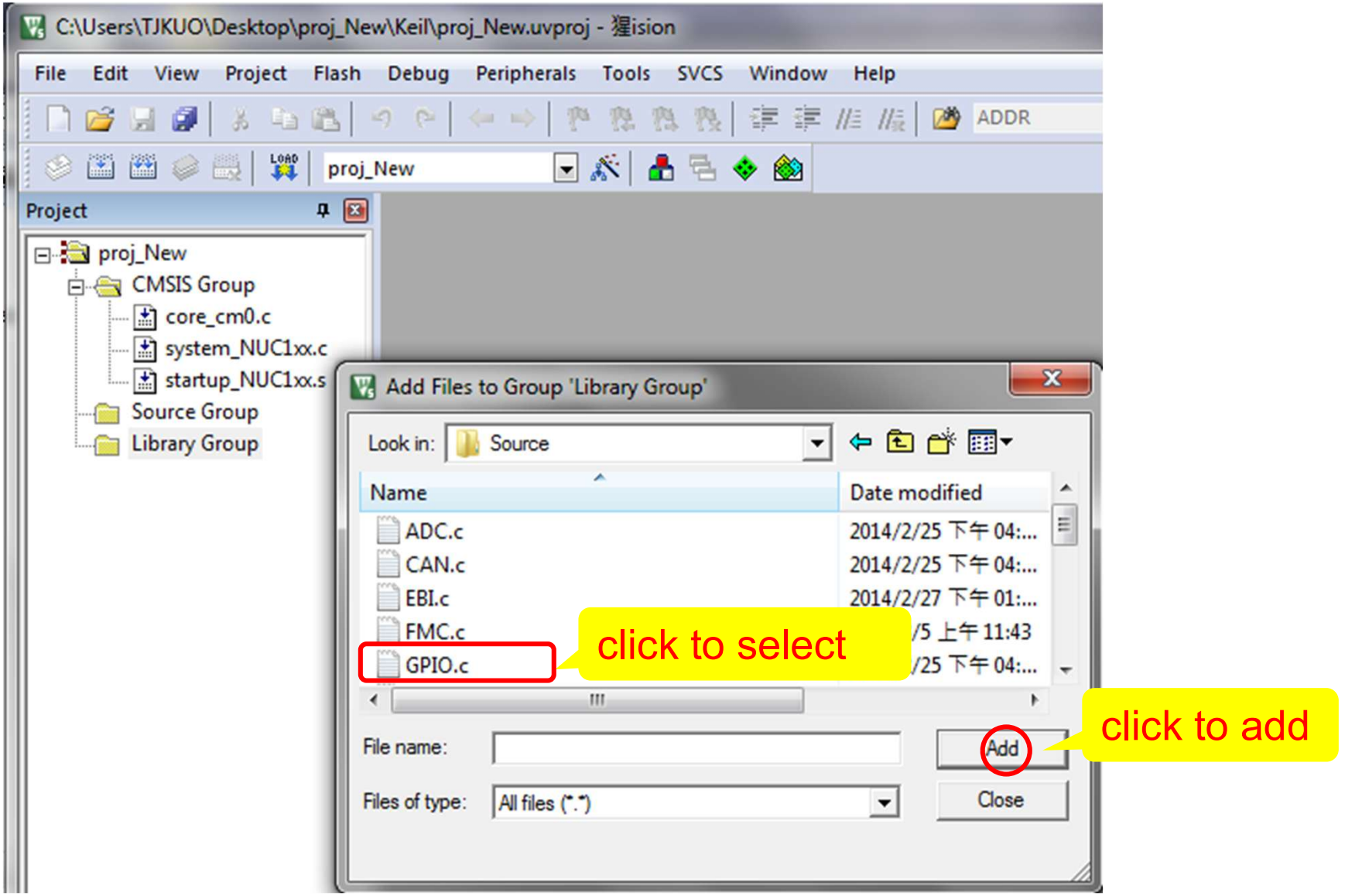
# change to **NUC1xx** directory



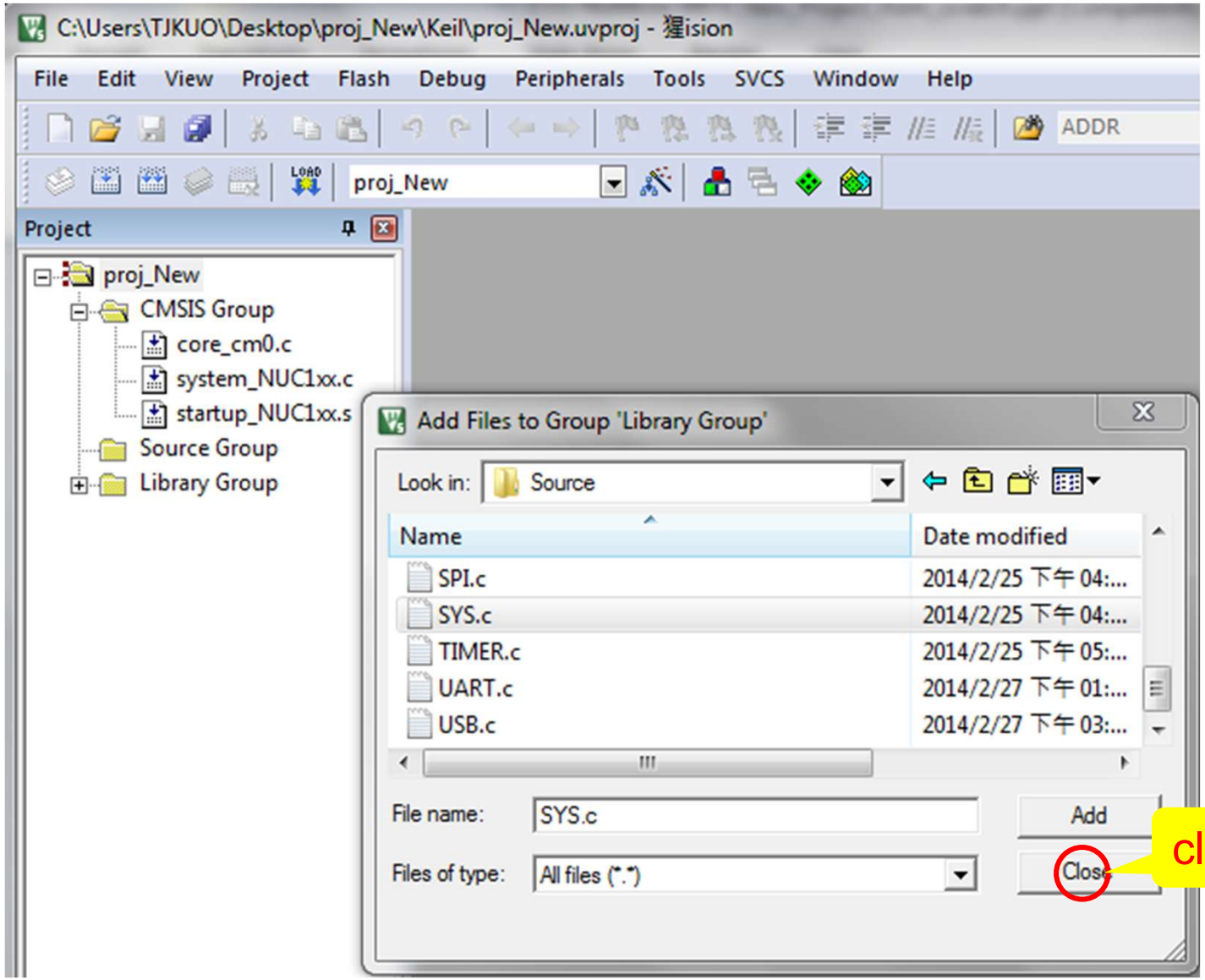
# change to **Source** directory



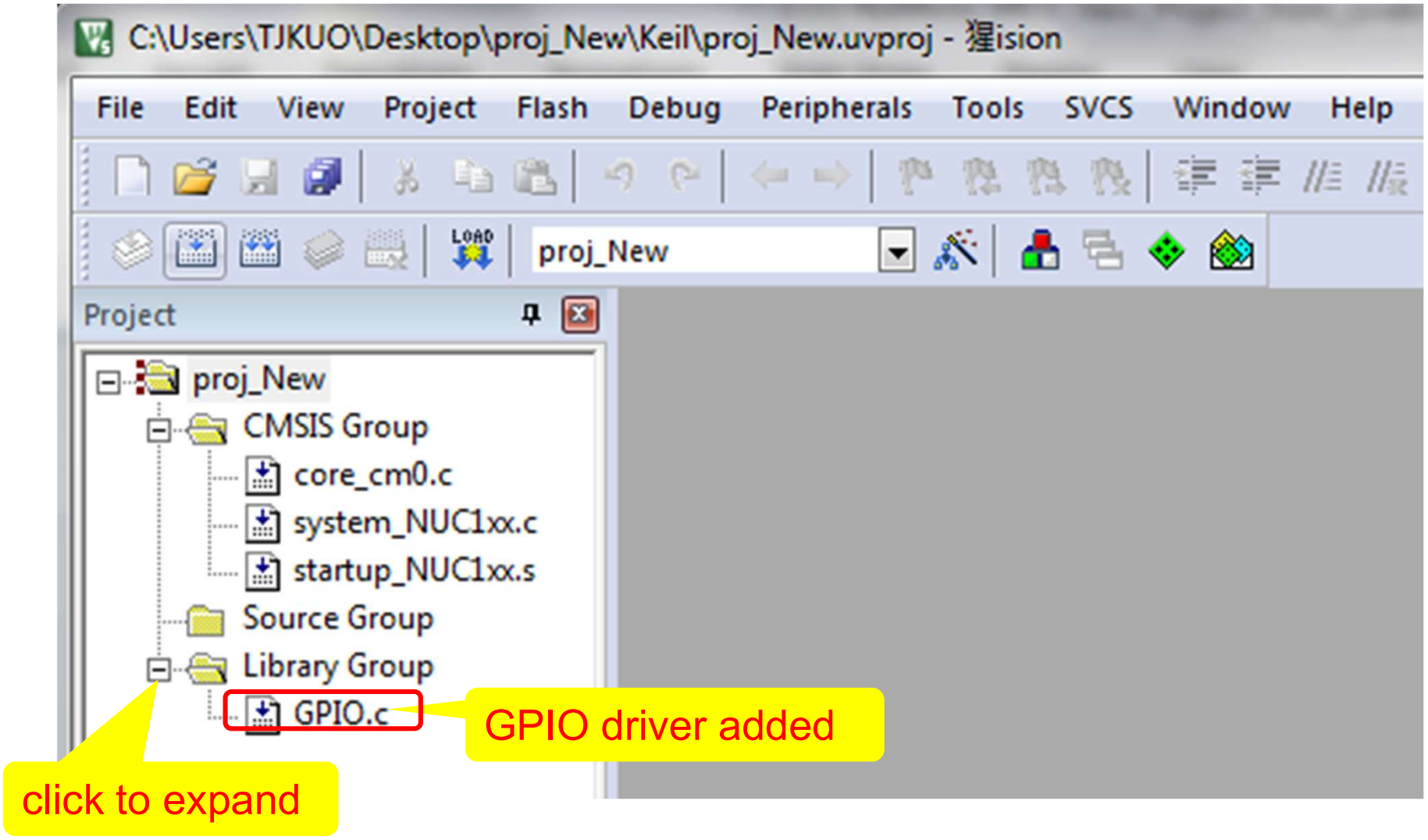
# Add Interface Driver to **Library Group**



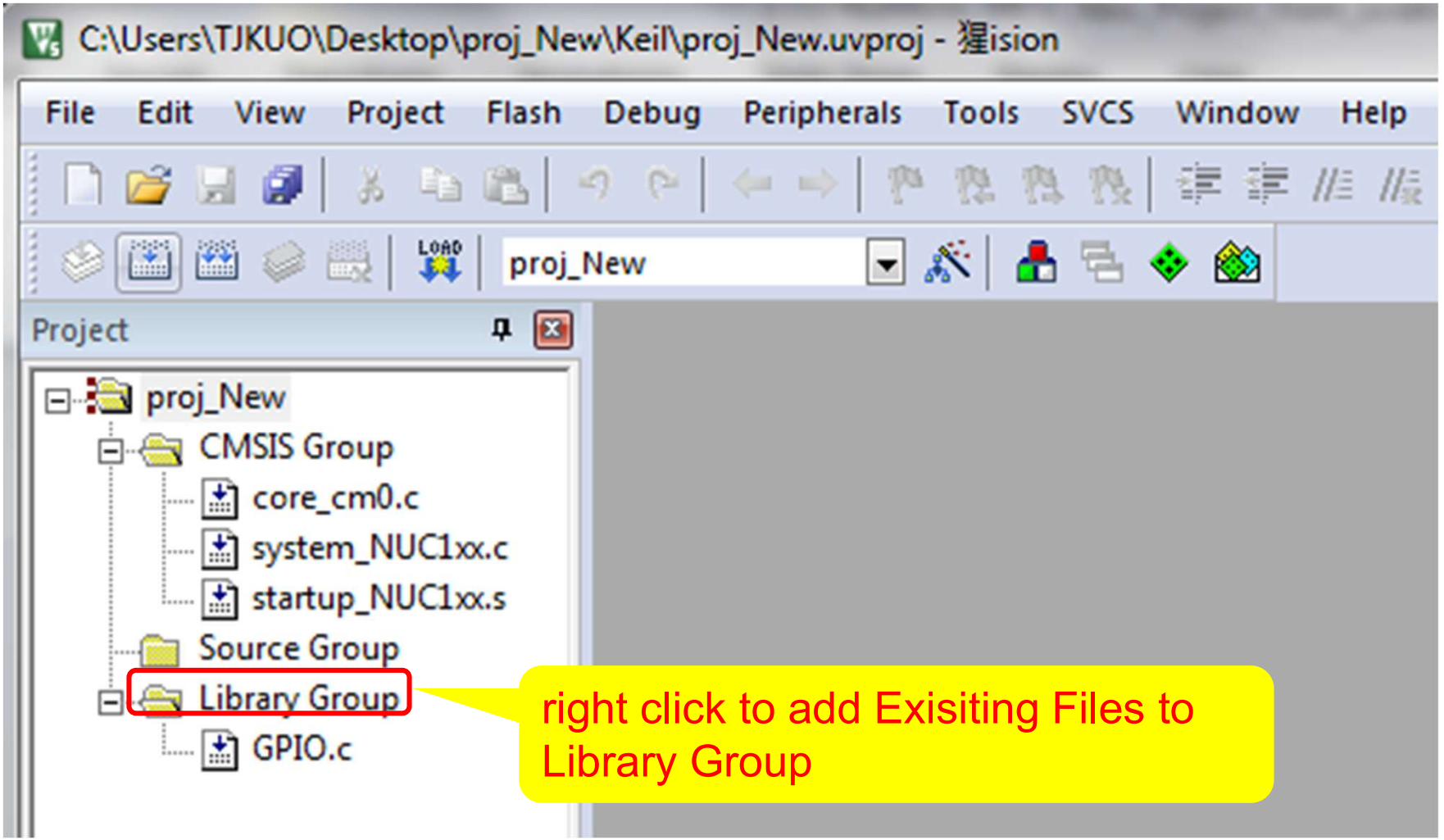
# Close Add Files



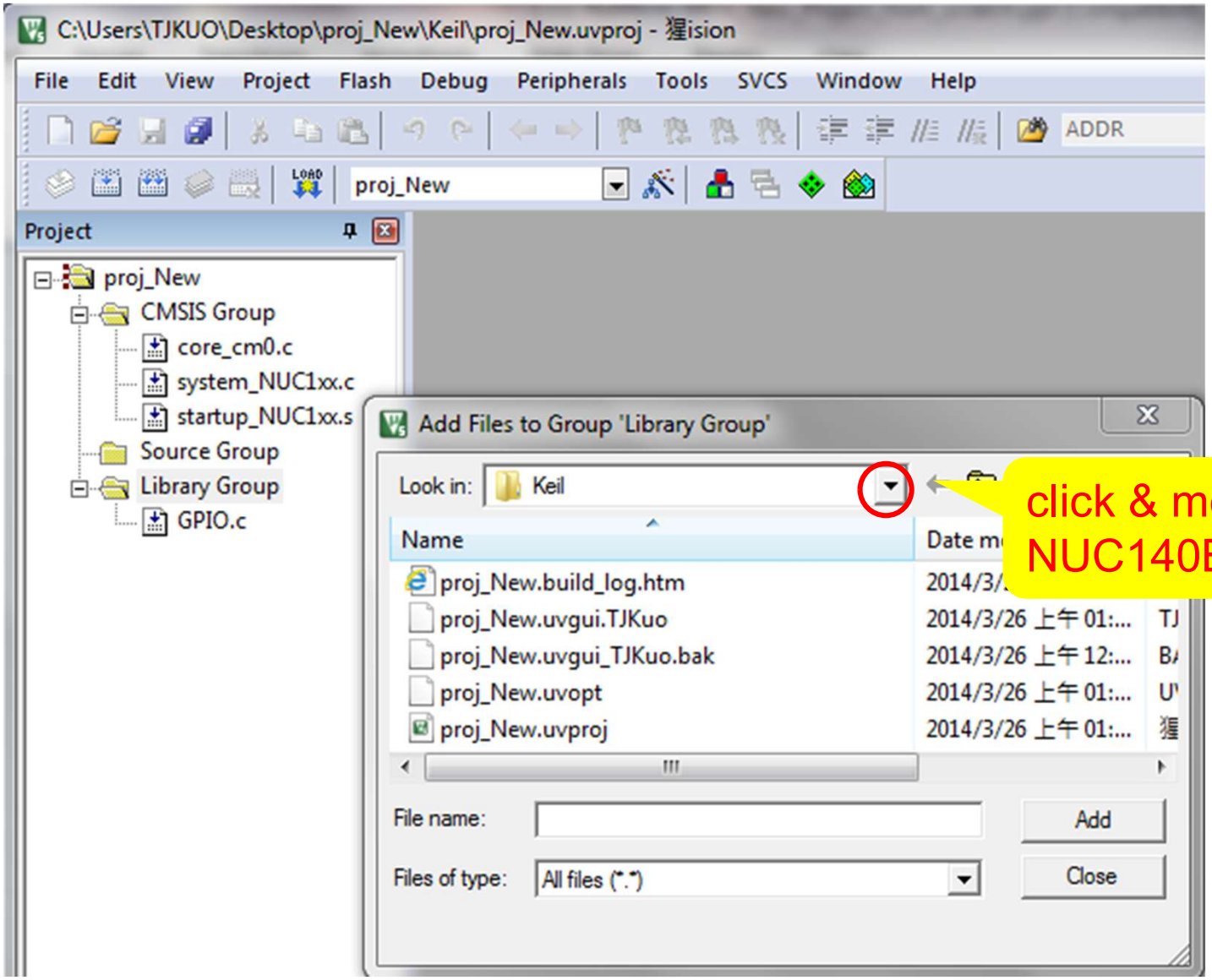
# Interface Driver added



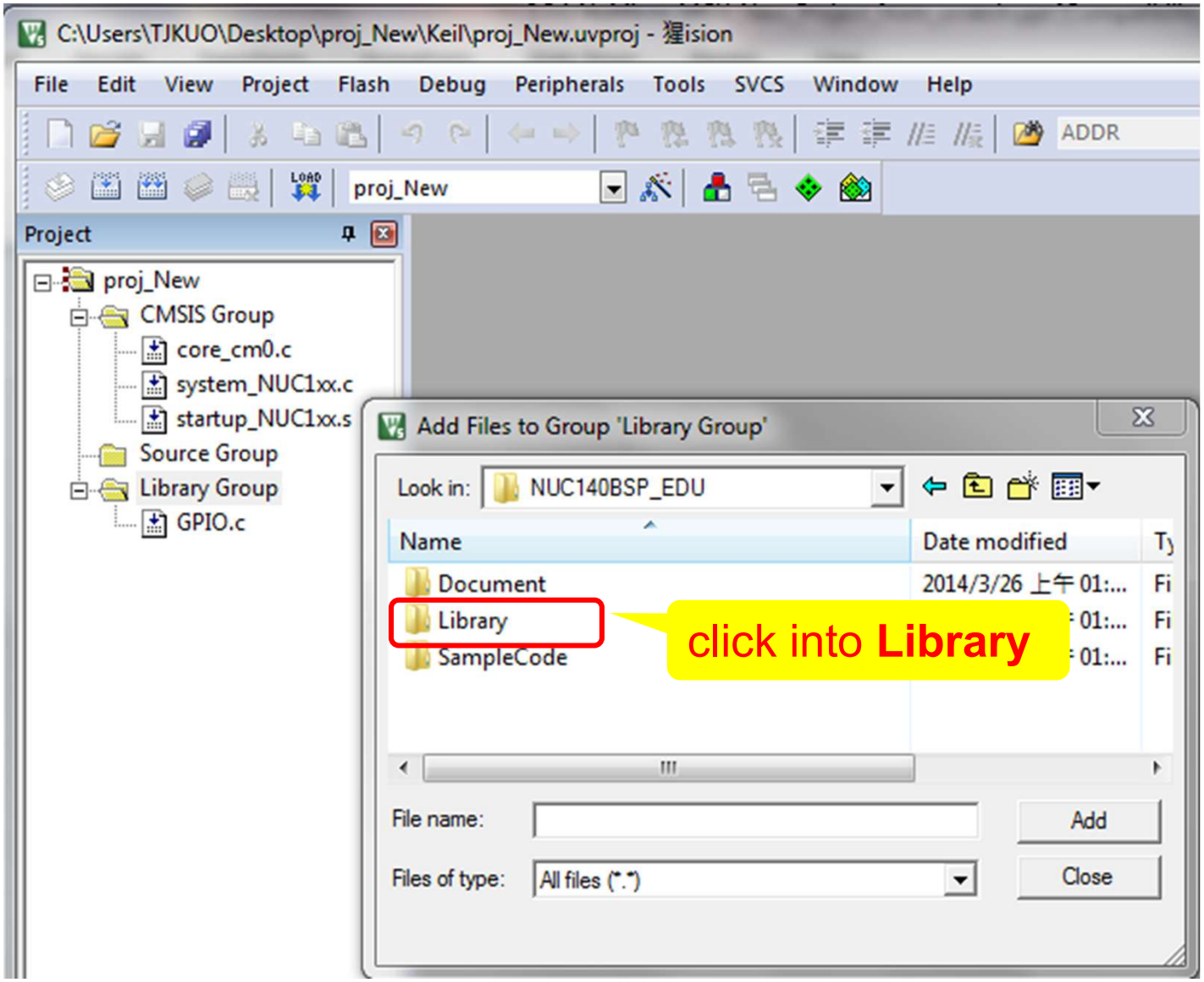
# Adding more **Interface Drivers**



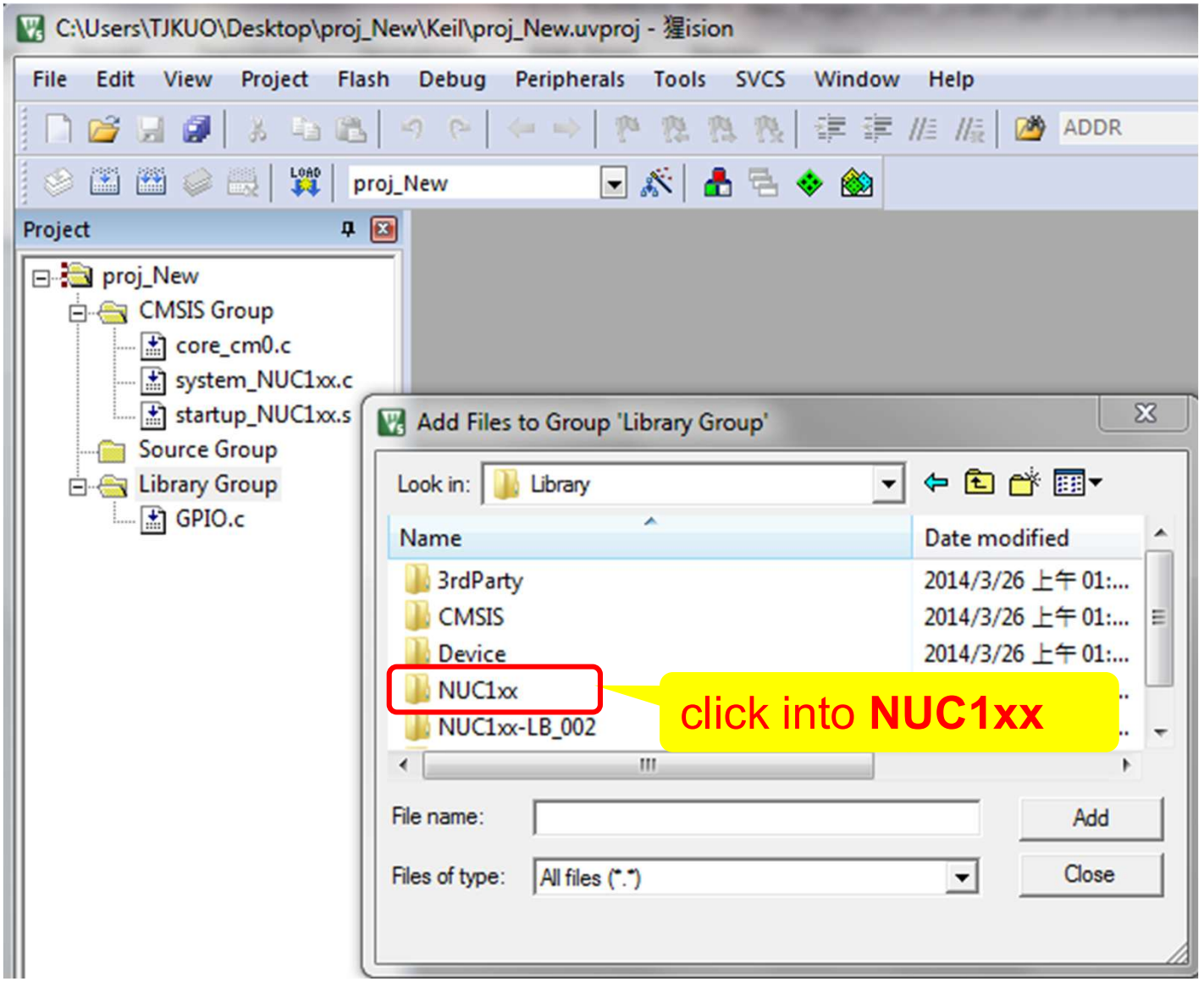
# Change to NUC140BSP_EDU directory



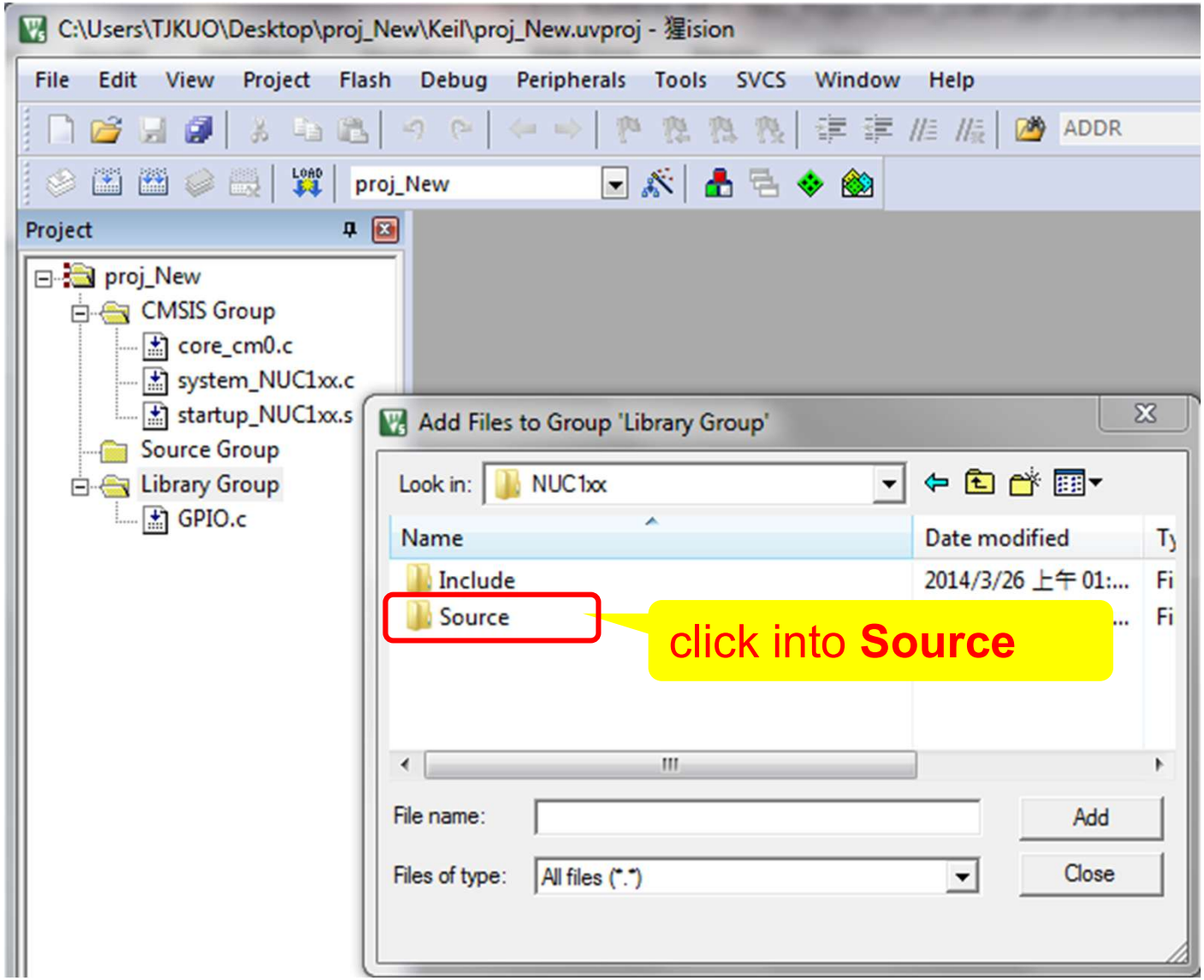
# Change to **Library** directory



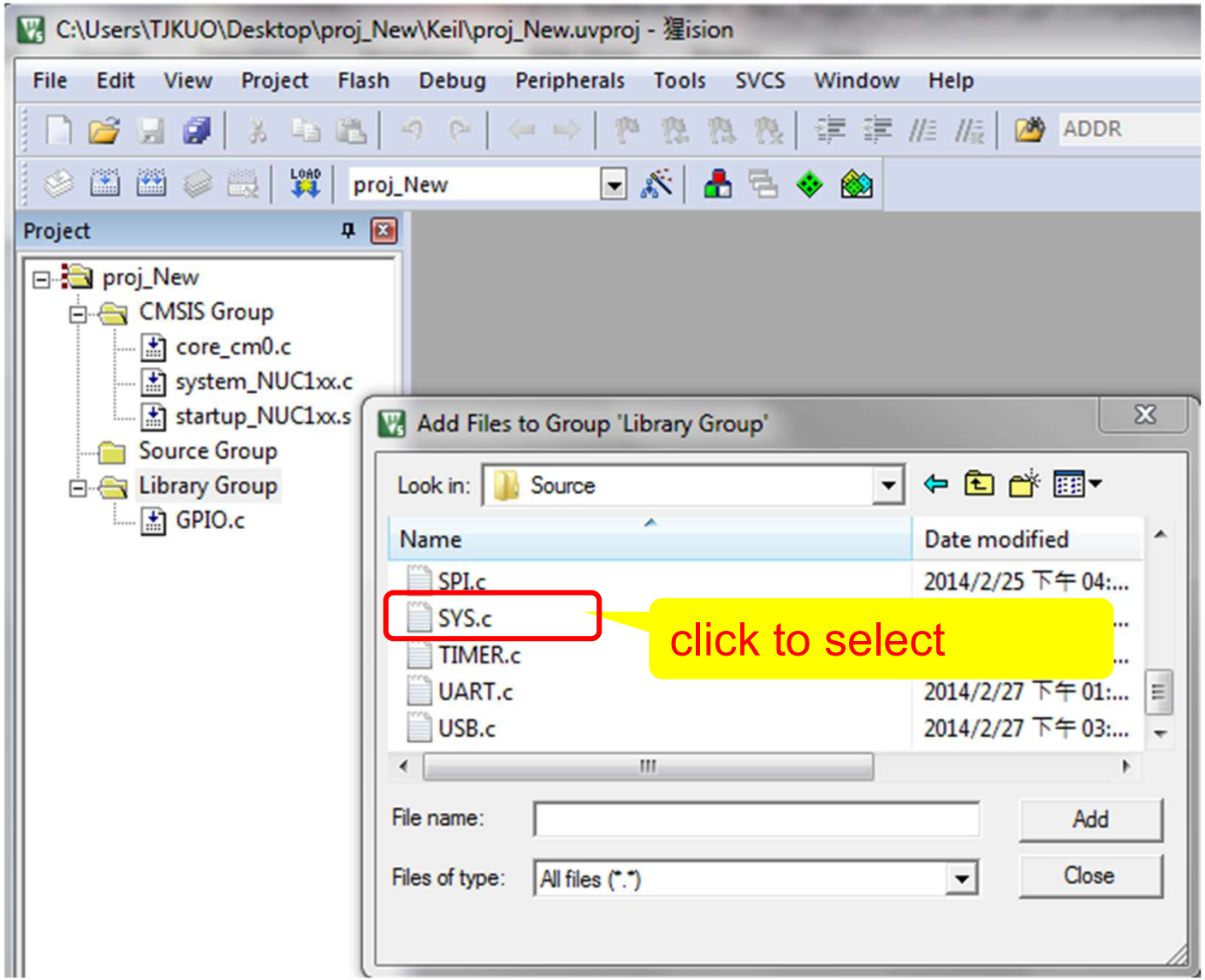
# Change to **NUC1xx** directory



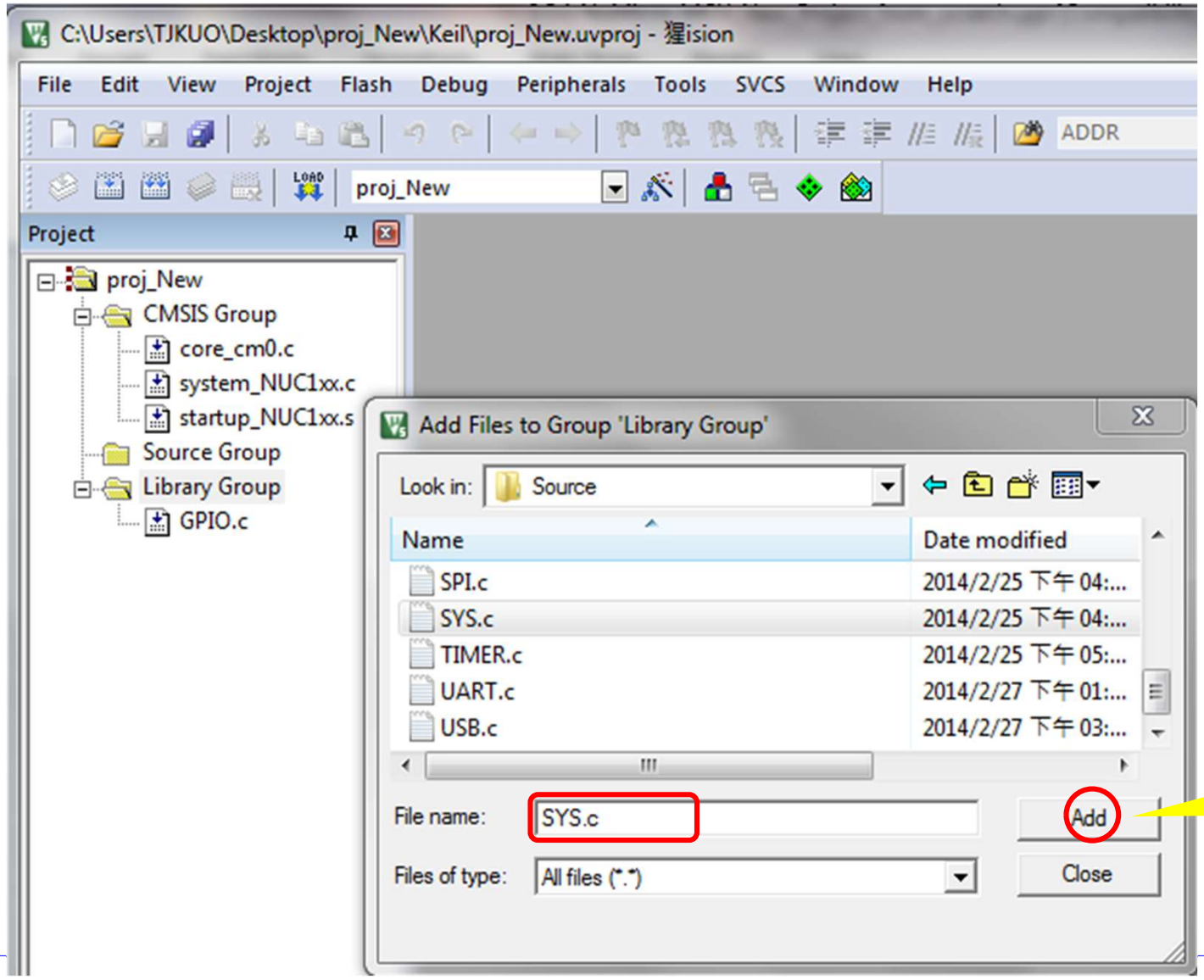
# Change to **Source** directory



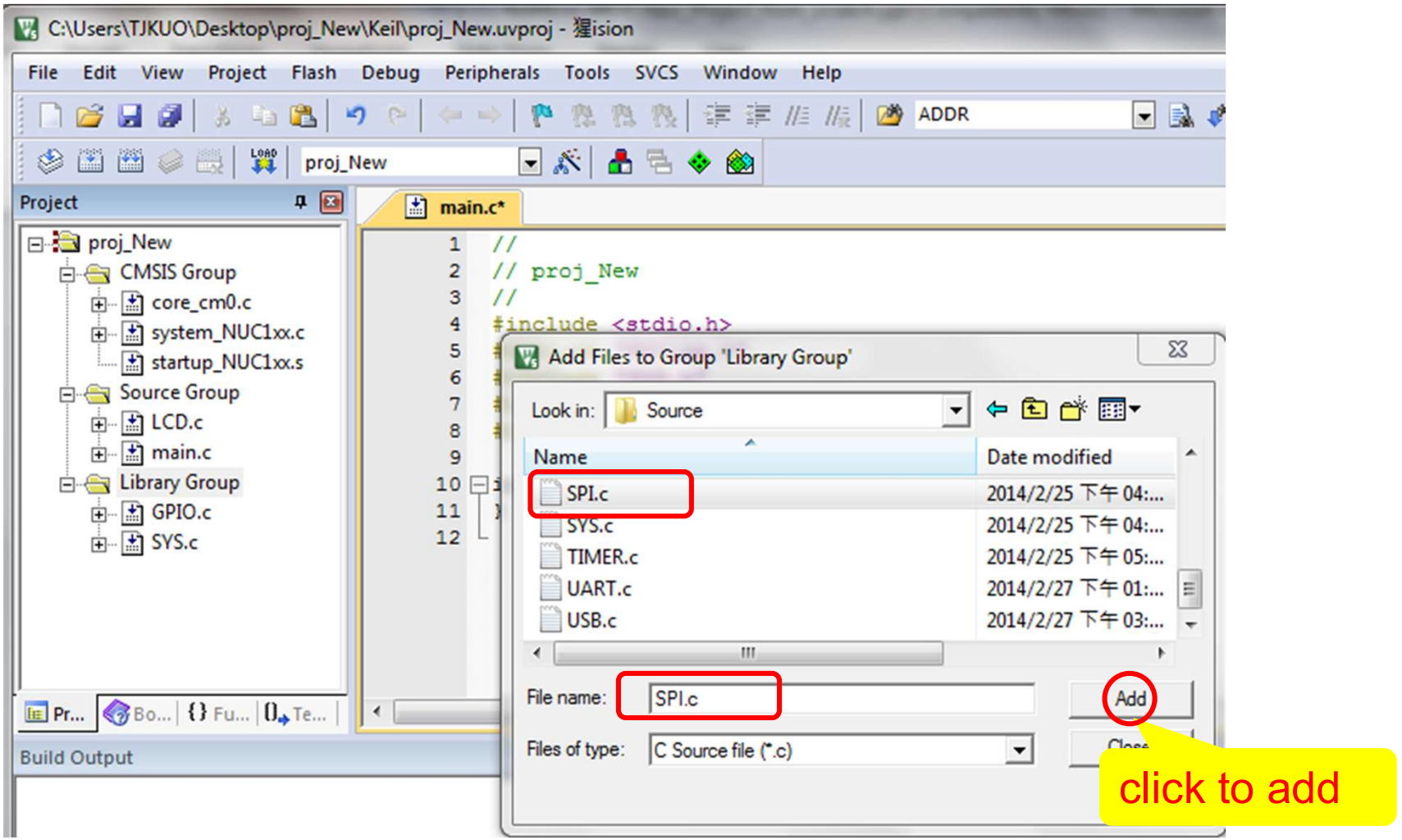
# Select interface driver to add [1]



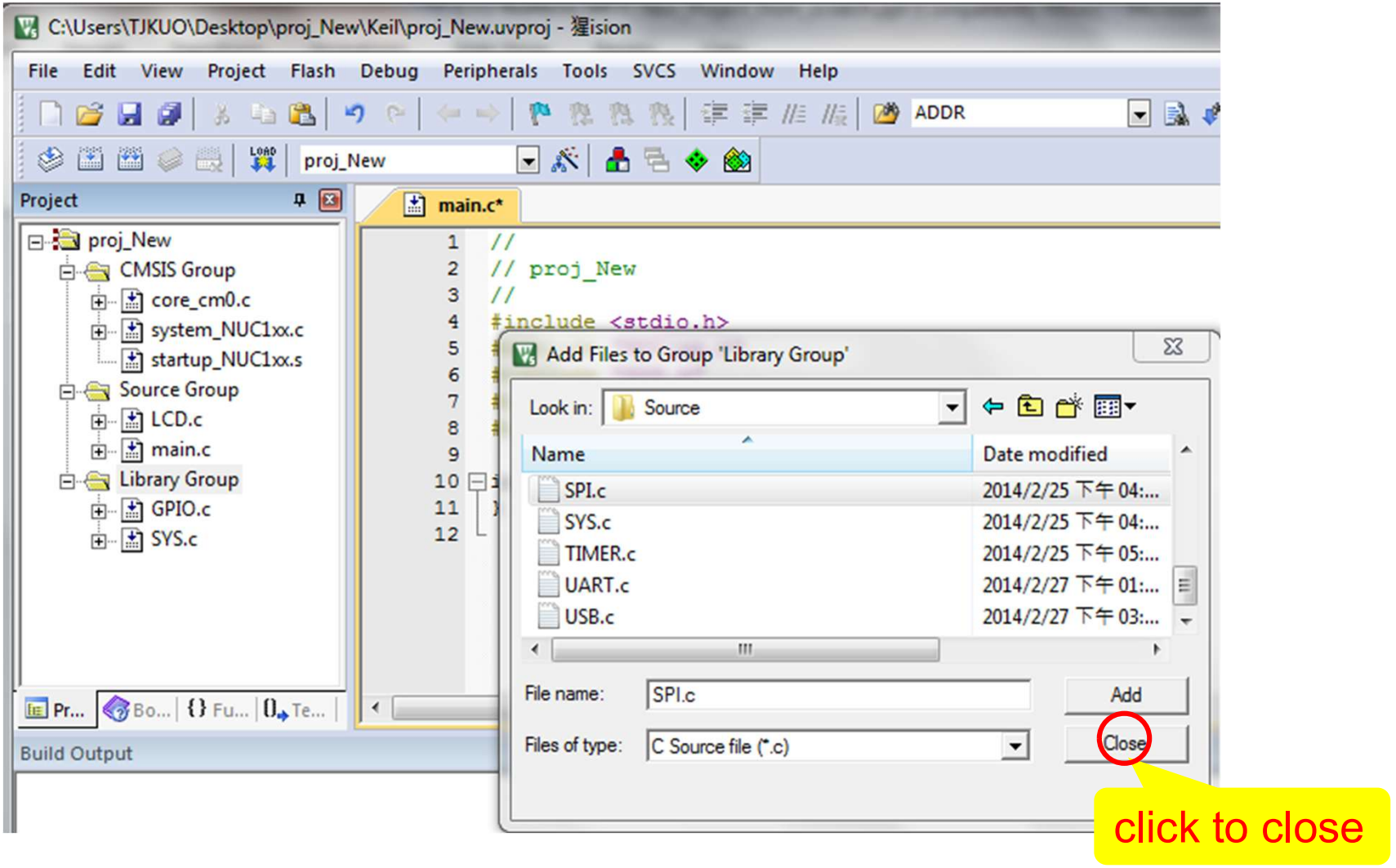
# Select interface driver to add [2]



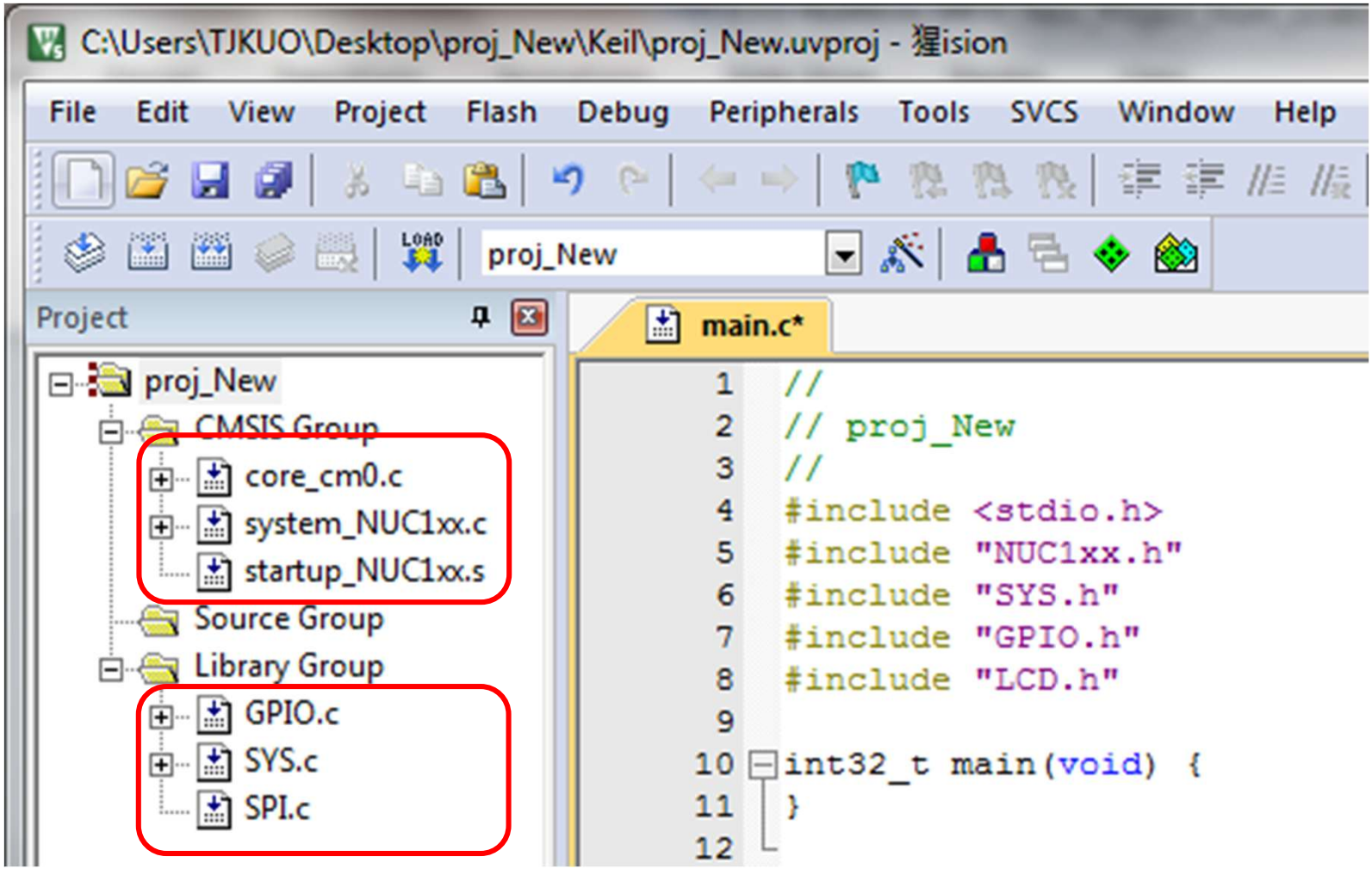
# Select interface driver to add [3]



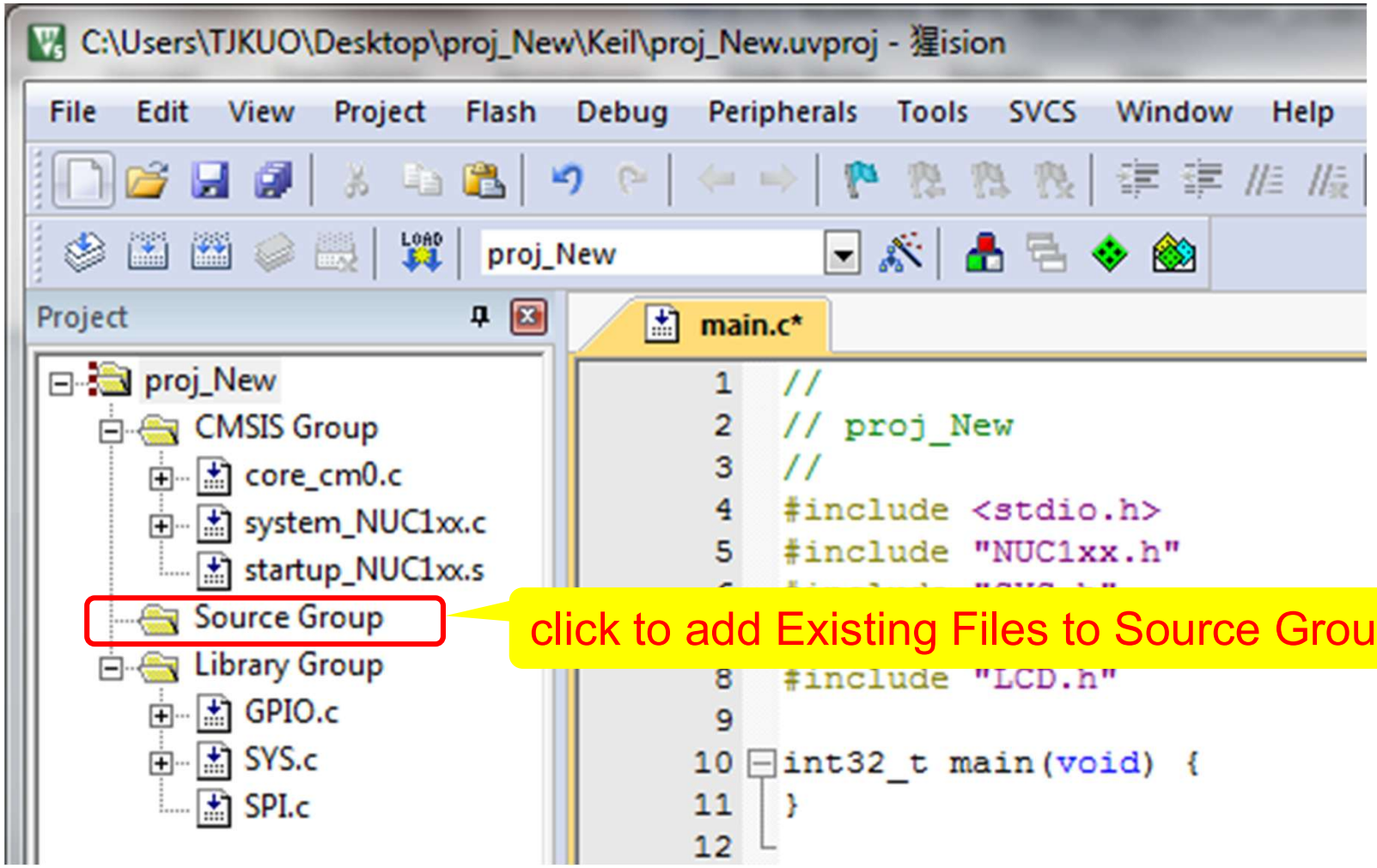
# Close Add Files



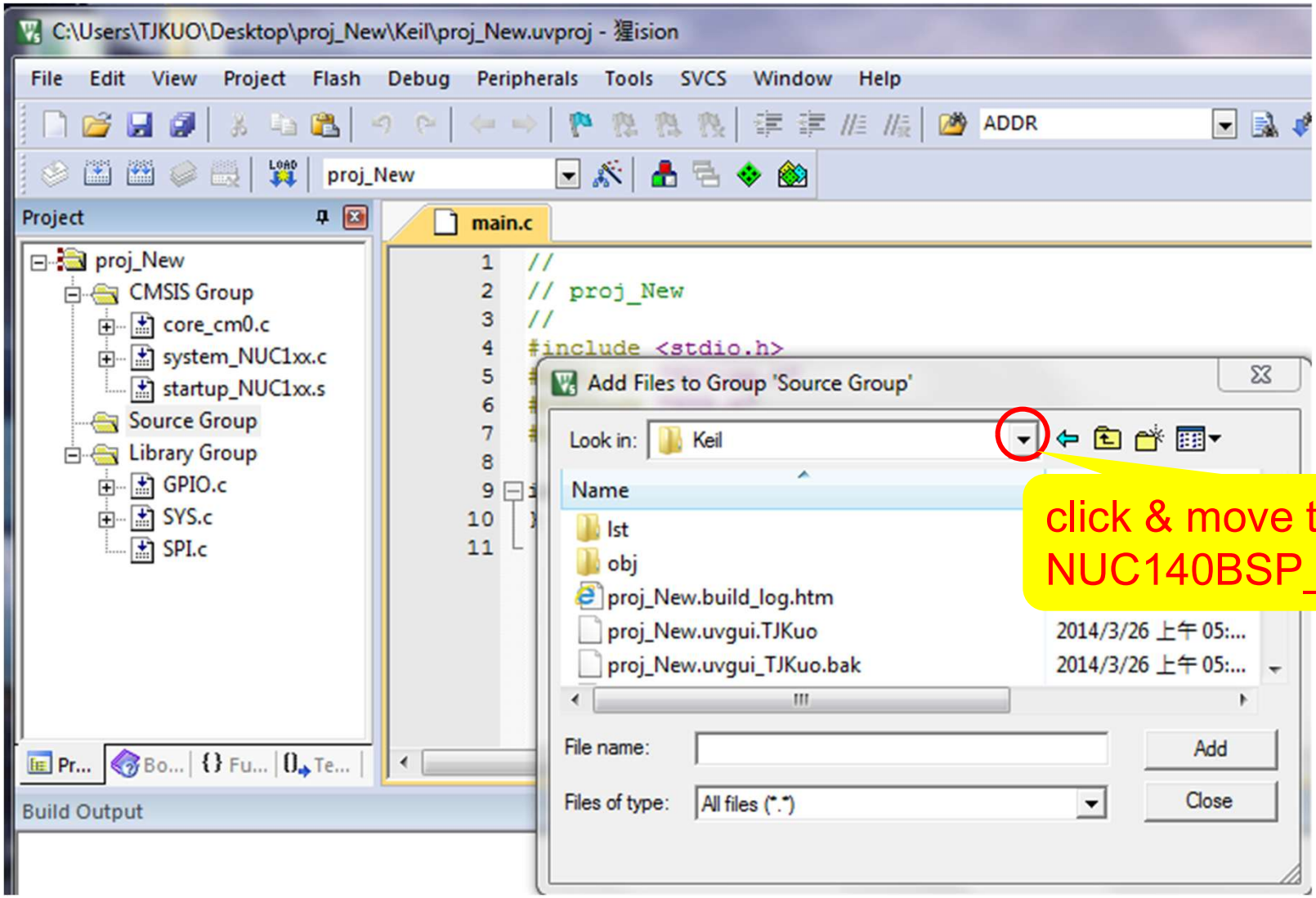
# **CMSIS & Library Group** files added



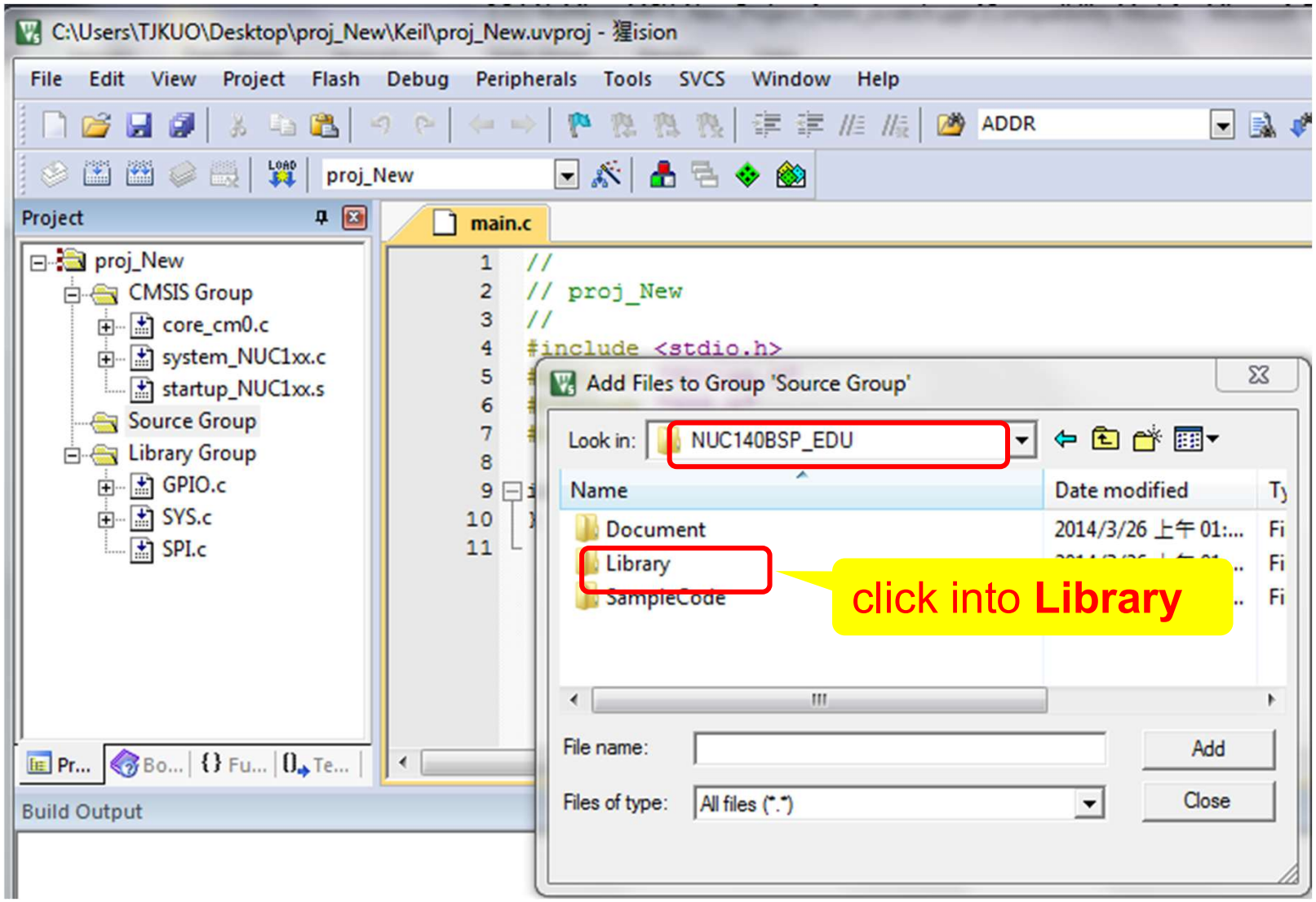
# To add **Peripheral Driver**



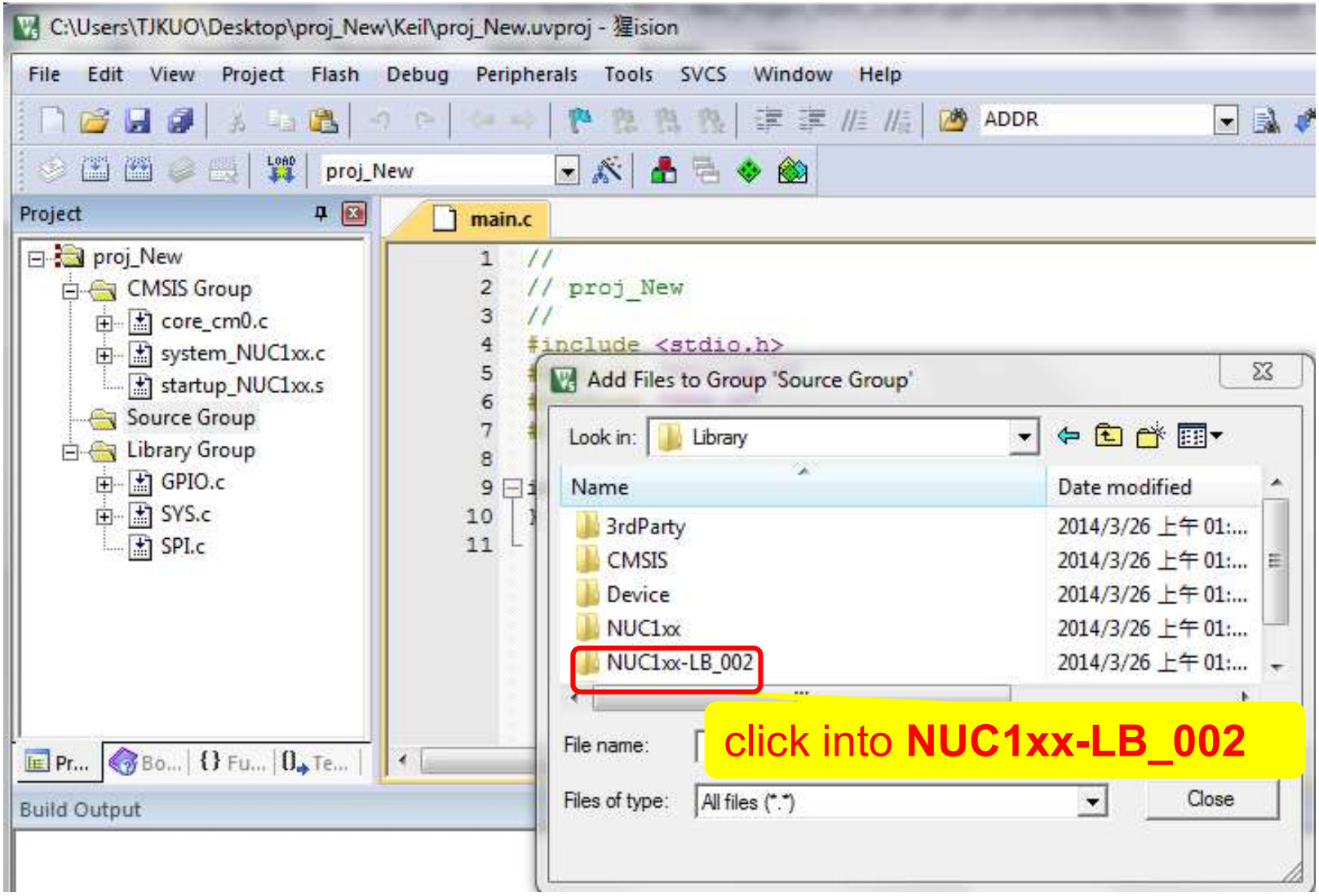
# Change to NUC140BSP_EDU directory



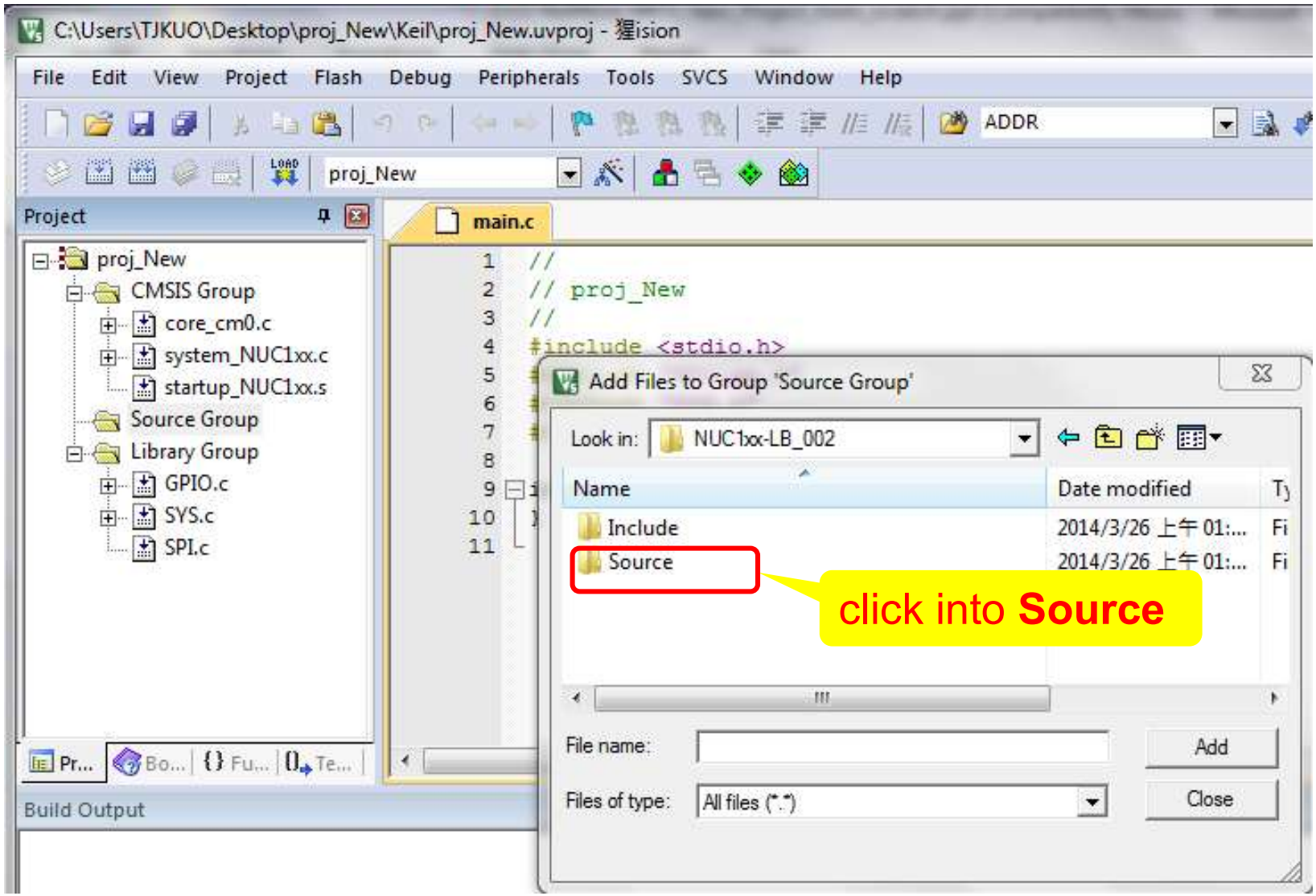
# Change to **Library** directory



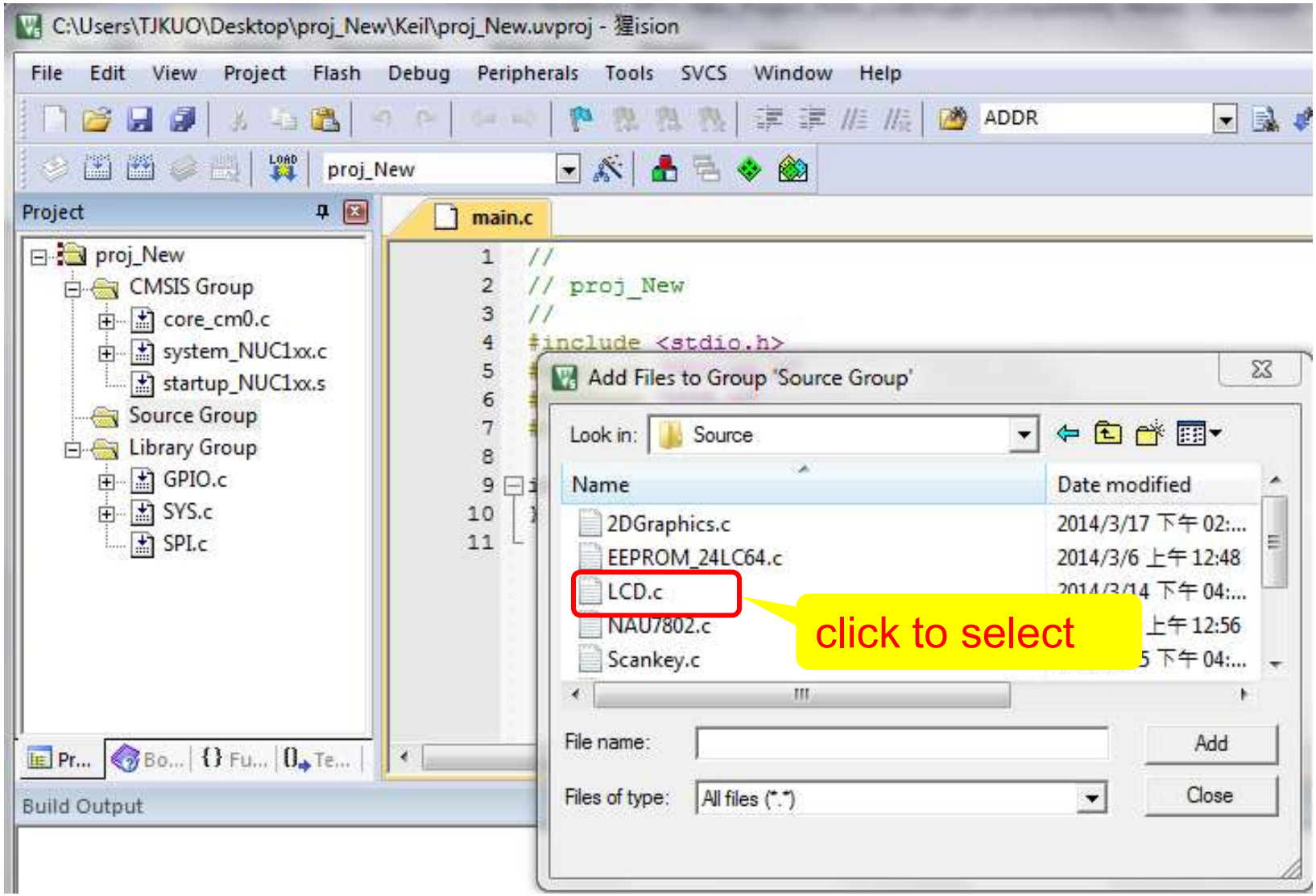
# Change to **NUC1xx-LB_002** directory



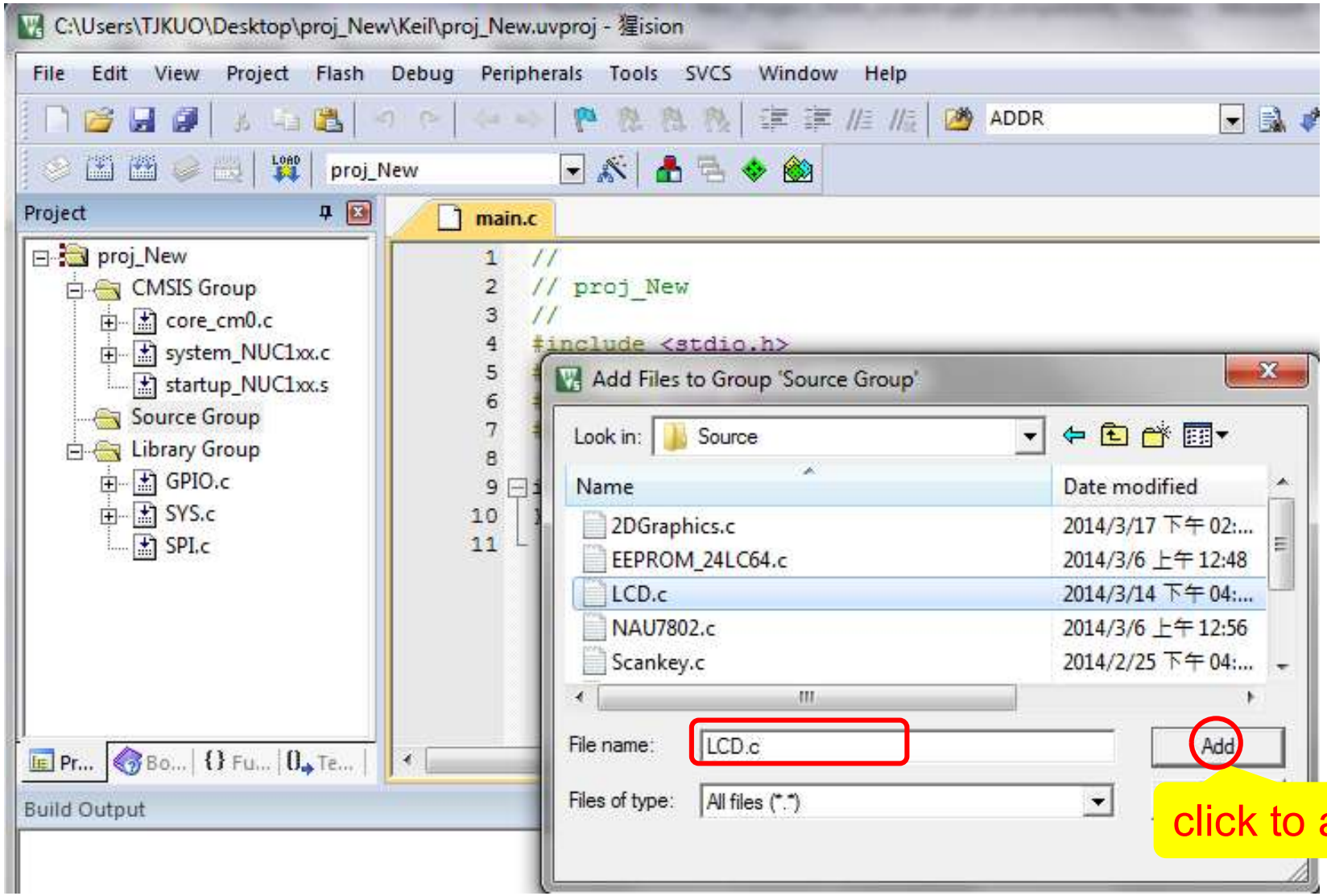
# Change to **Source** directory



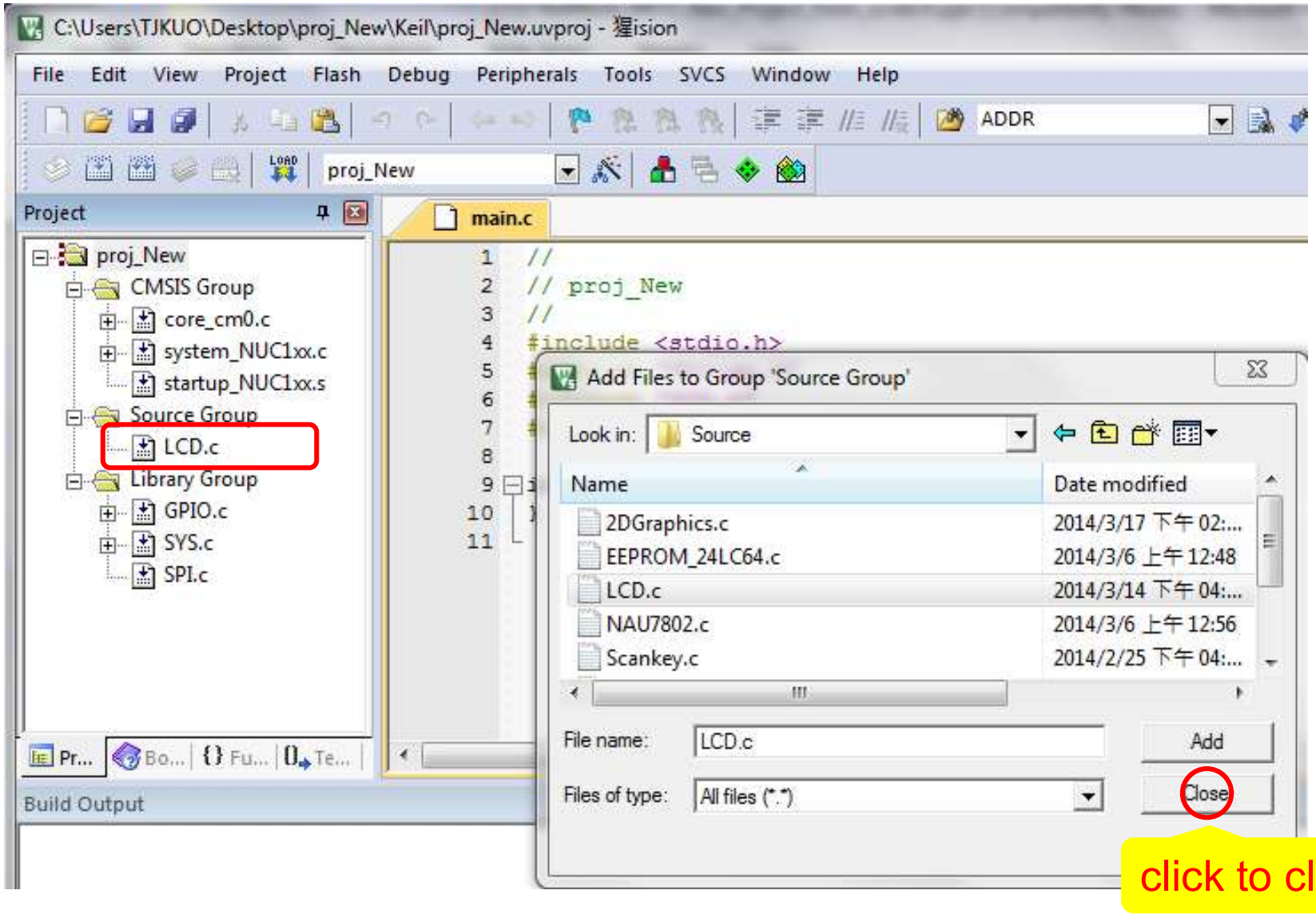
# Select **Peripheral Driver** to add



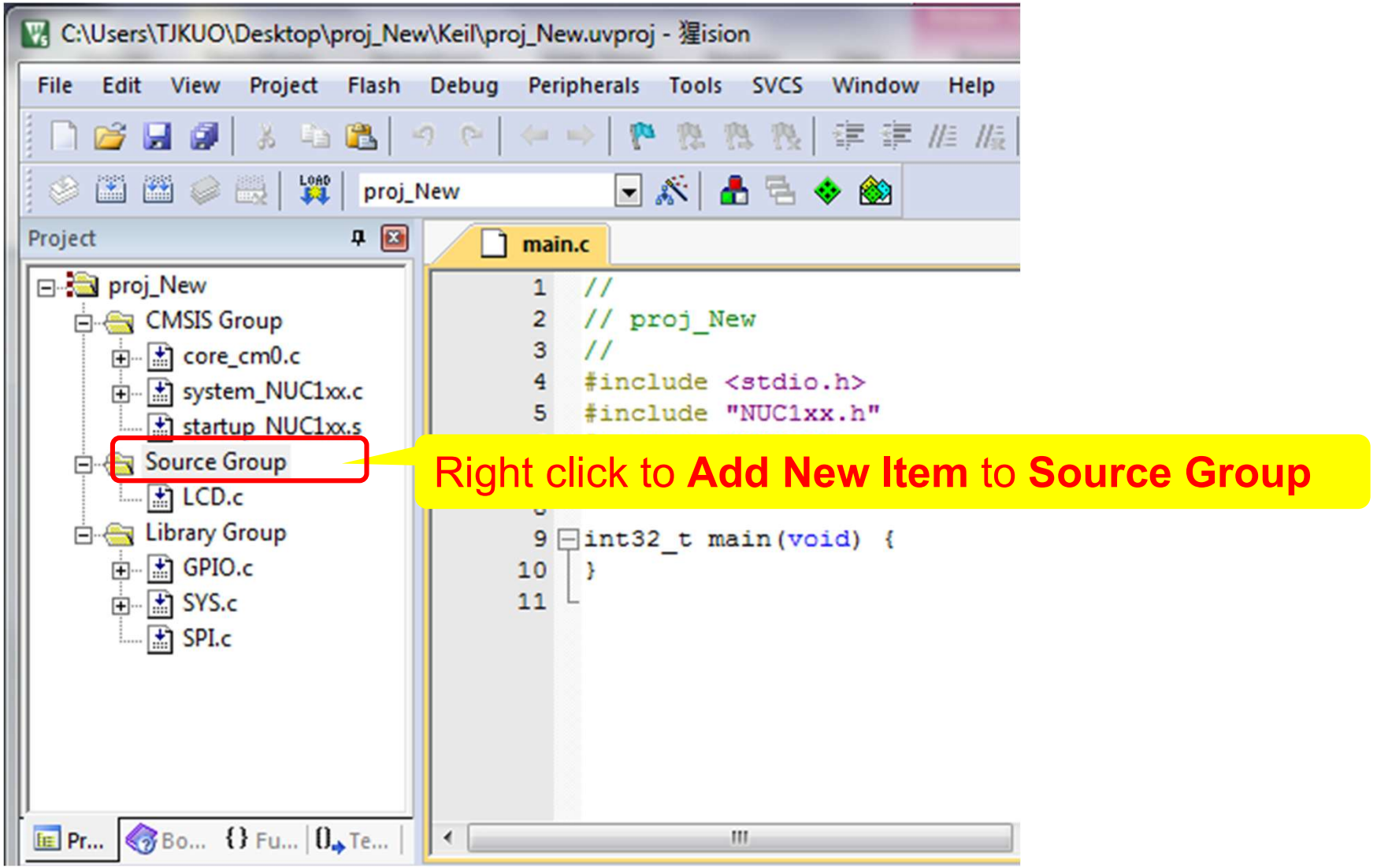
# Add **LCD Driver** to **Source Group**



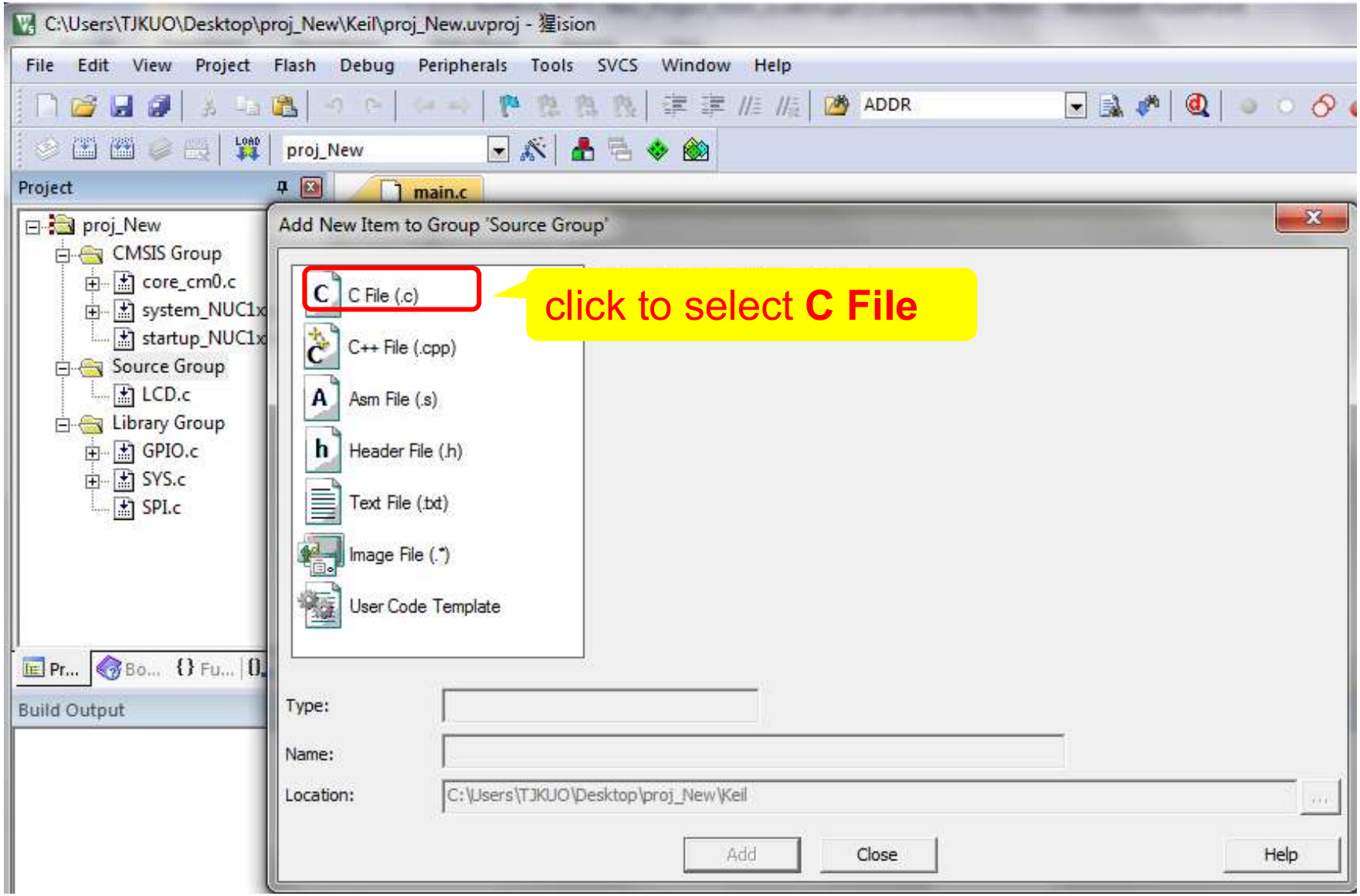
# Close Add Files



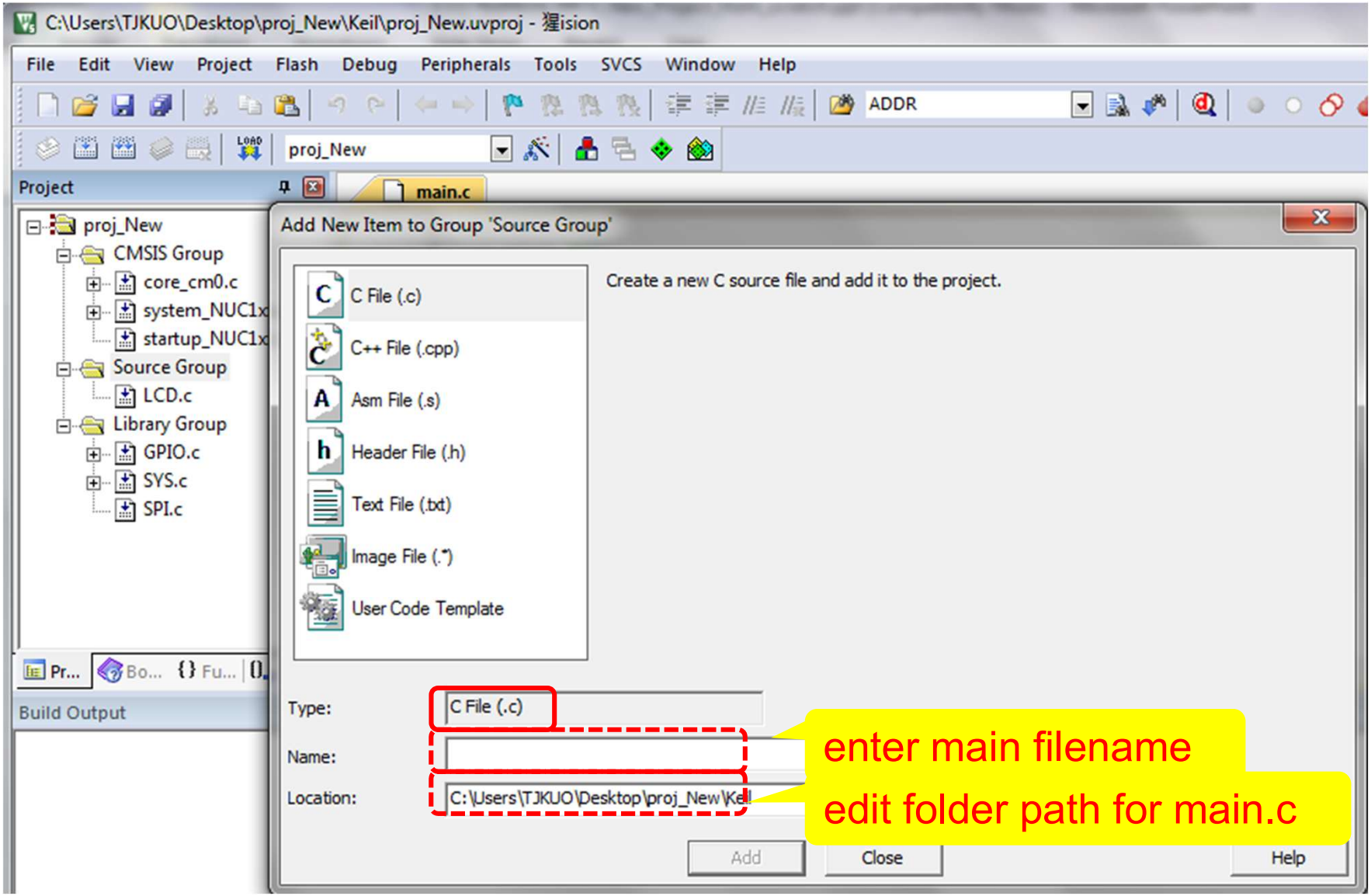
# To add **New Item** to **Source Group**



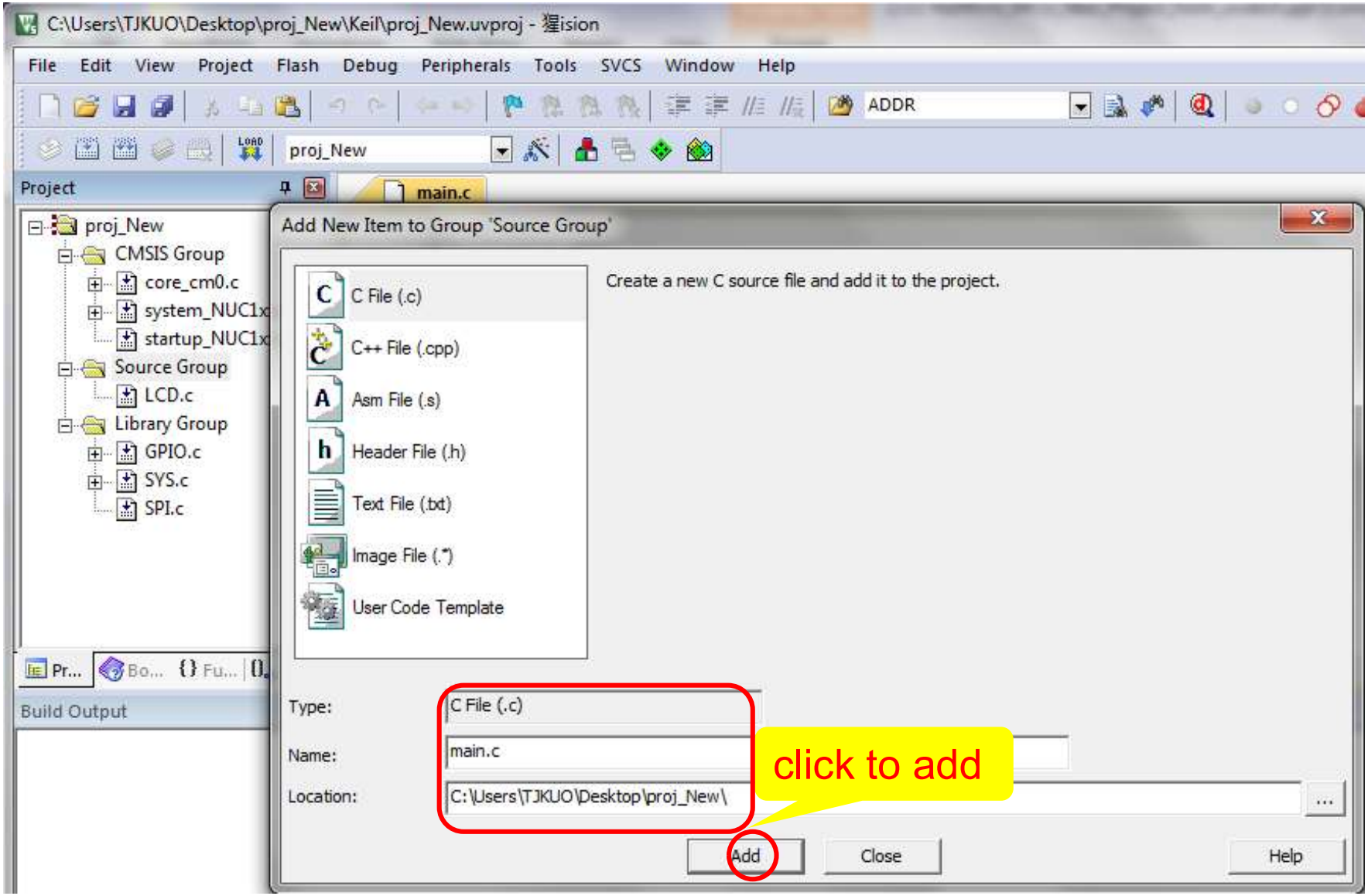
# To add **New Item** to **Source Group**



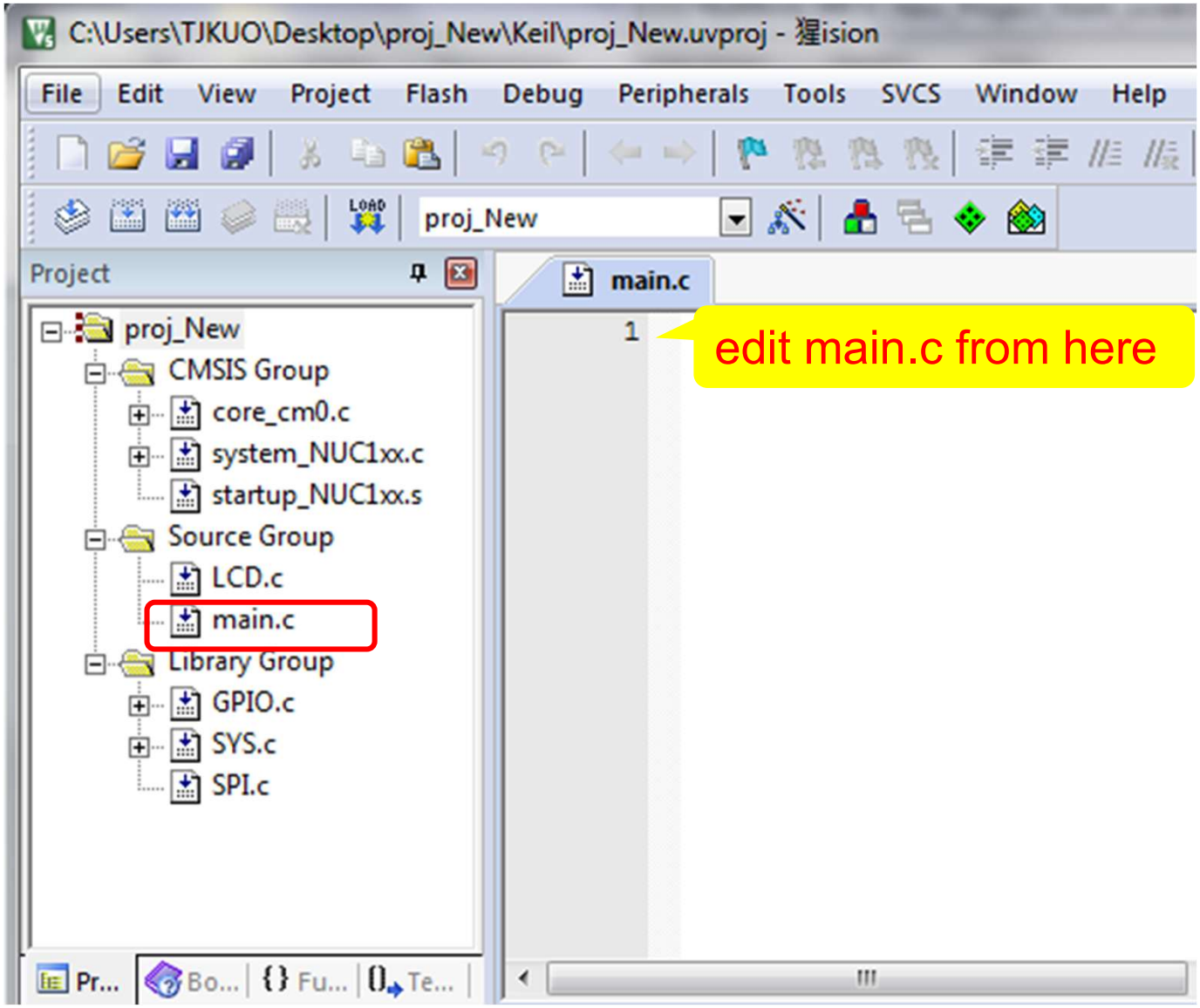
# edit main filename and its directory path



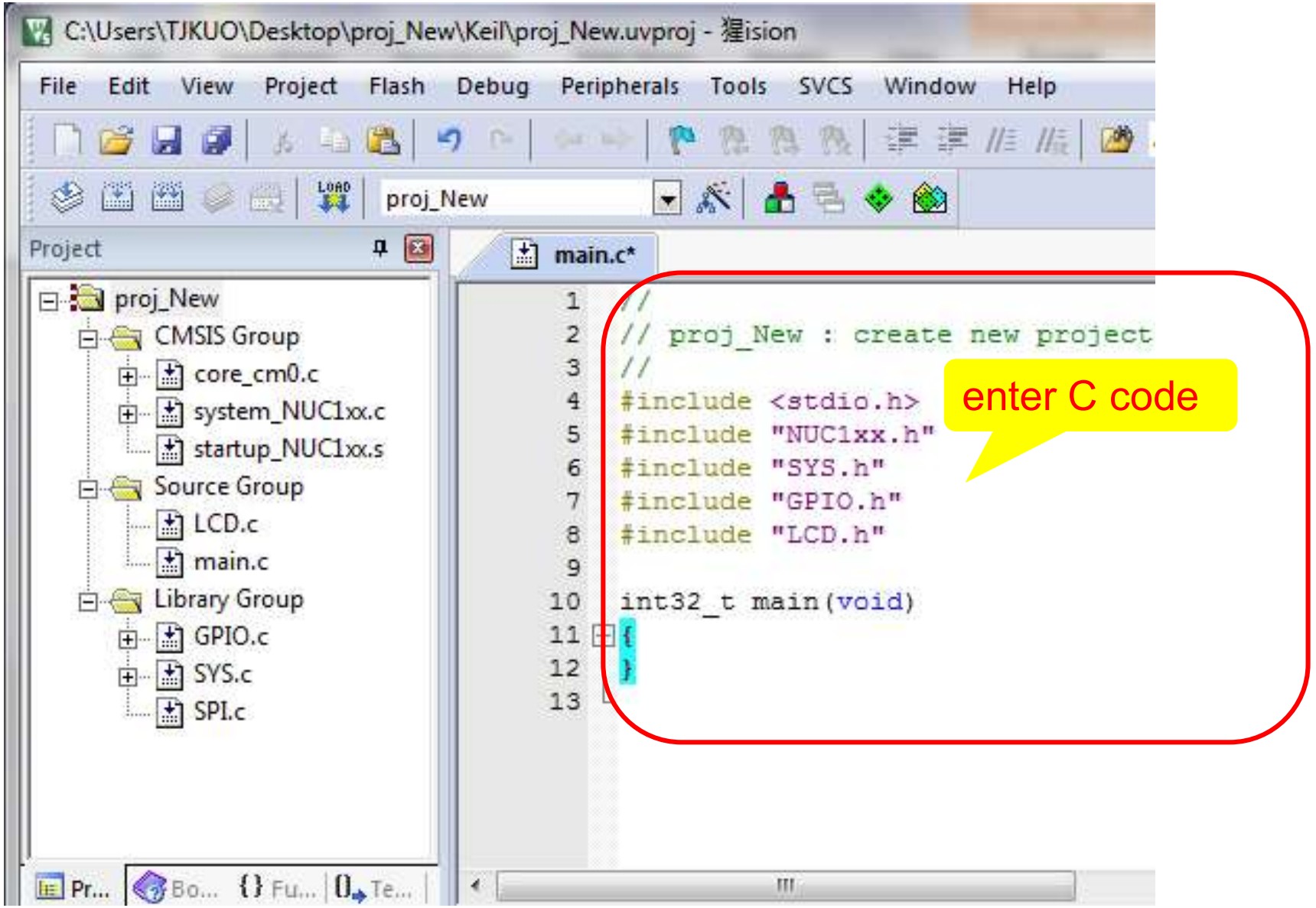
# Add **main.c** to **Source Group**



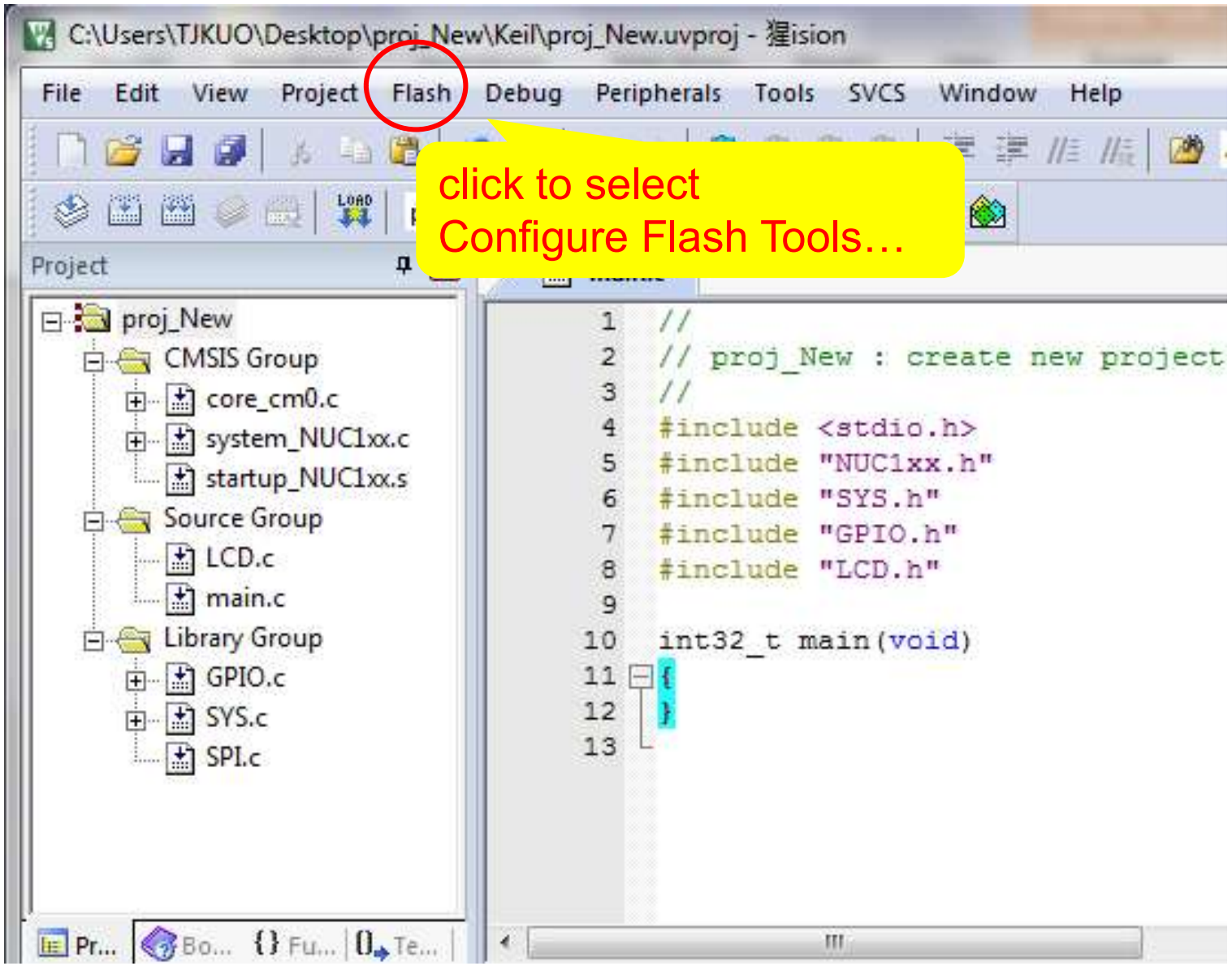
# edit **main.c**


C:\Users\TJKUO\Desktop\proj_New\Keil\proj_New.uvproj - 燈ision
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
proj_New
Project
proj_New
CMSIS Group
core_cm0.c
system_NUC1xx.c
startup_NUC1xx.s
Source Group
LCD.c
main.c
Library Group
GPIO.c
SYS.c
SPI.c
Pr... Bo... {} Fu... {}. Te...
main.c
1
edit main.c from here

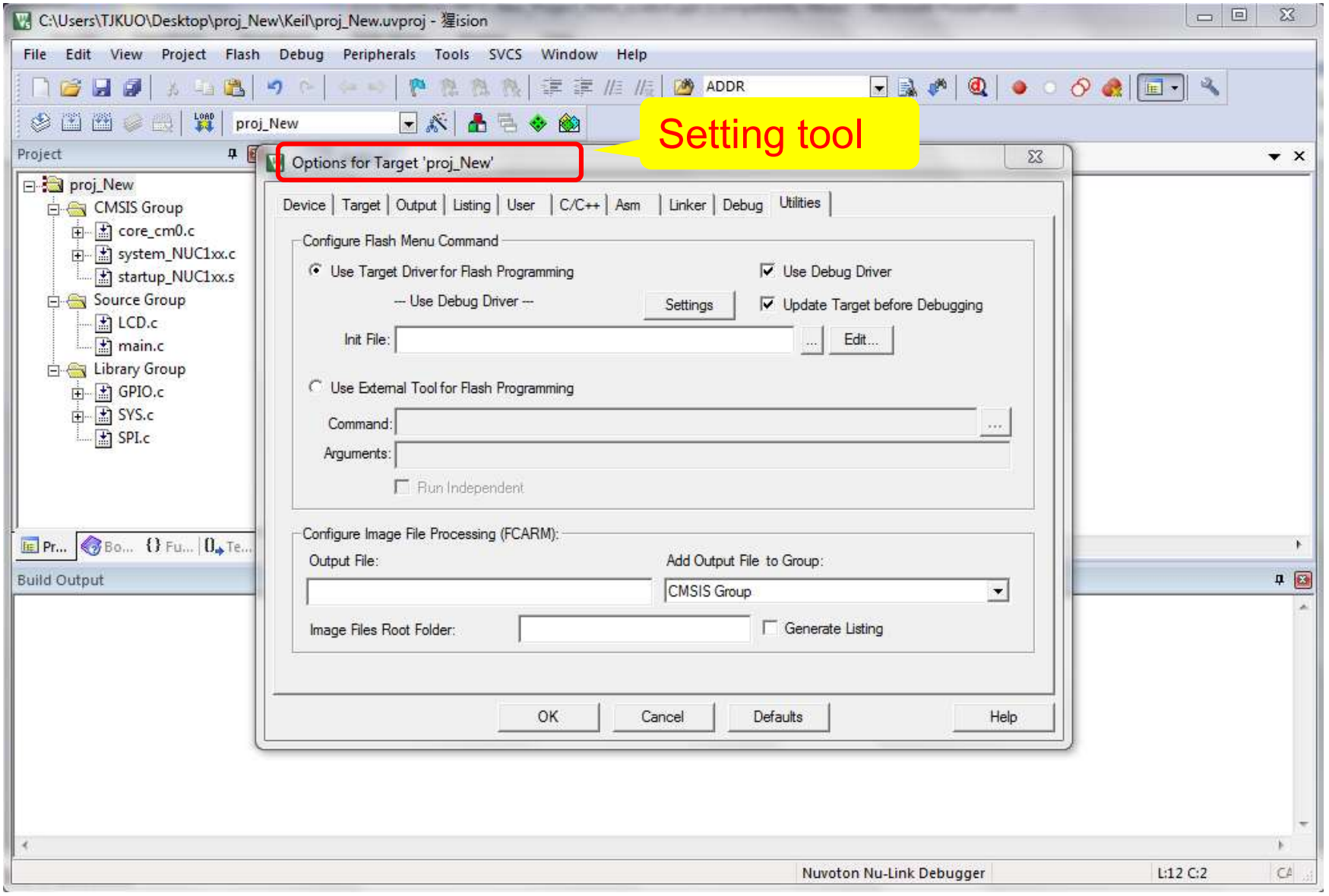
# edit **main.c** in C



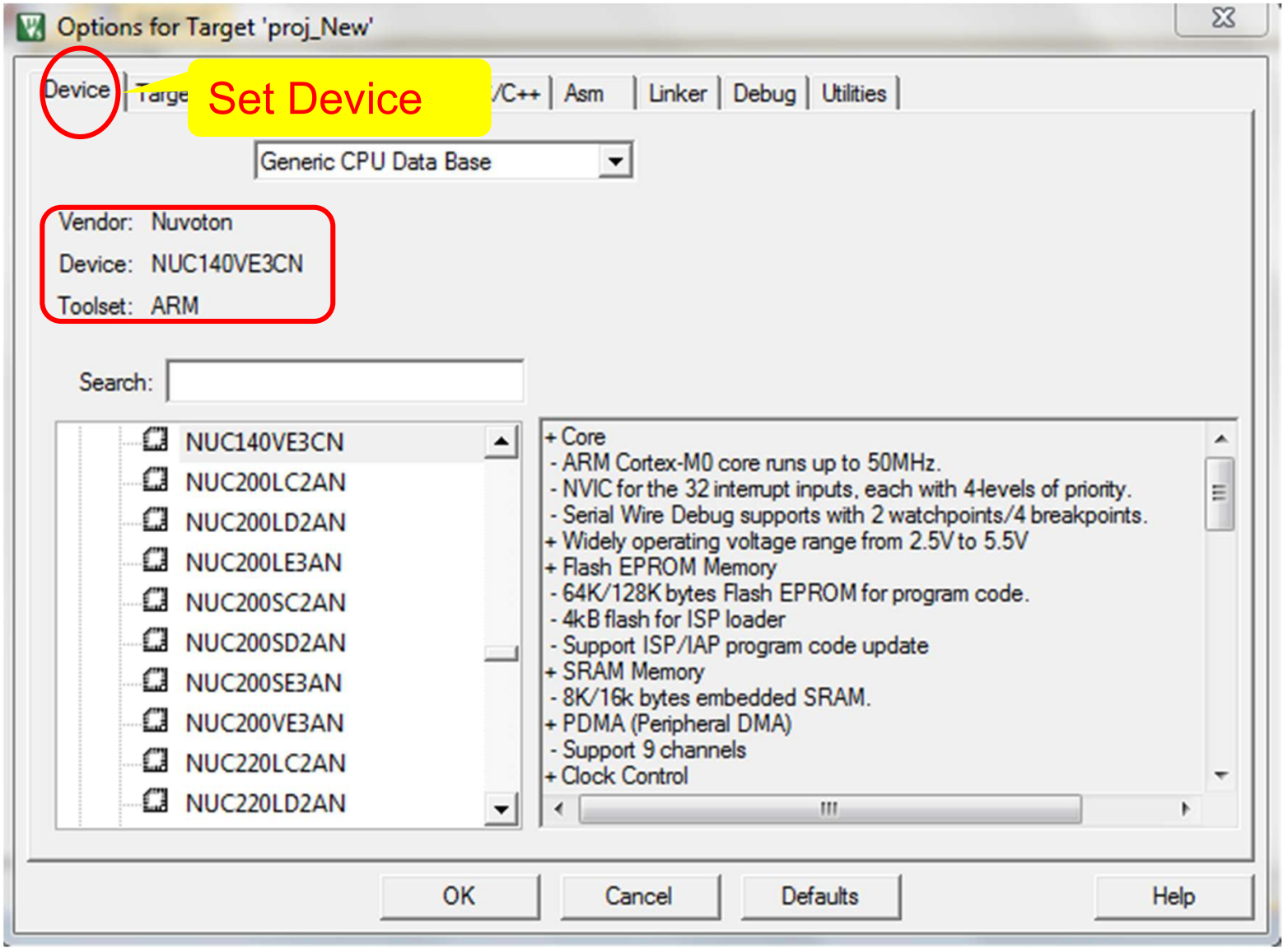
# edit **main.c** in C



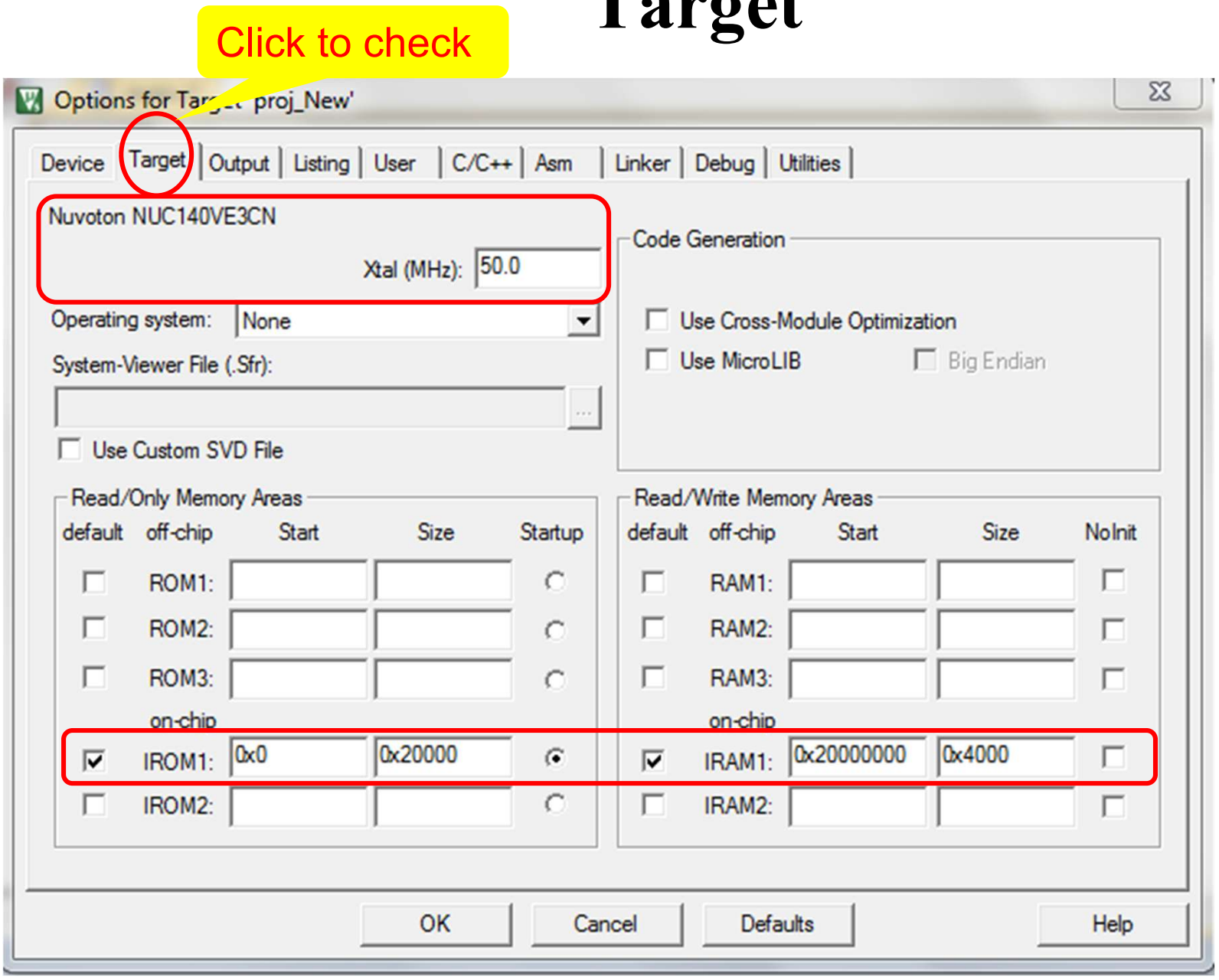
# Set Configuration of Keil MDK



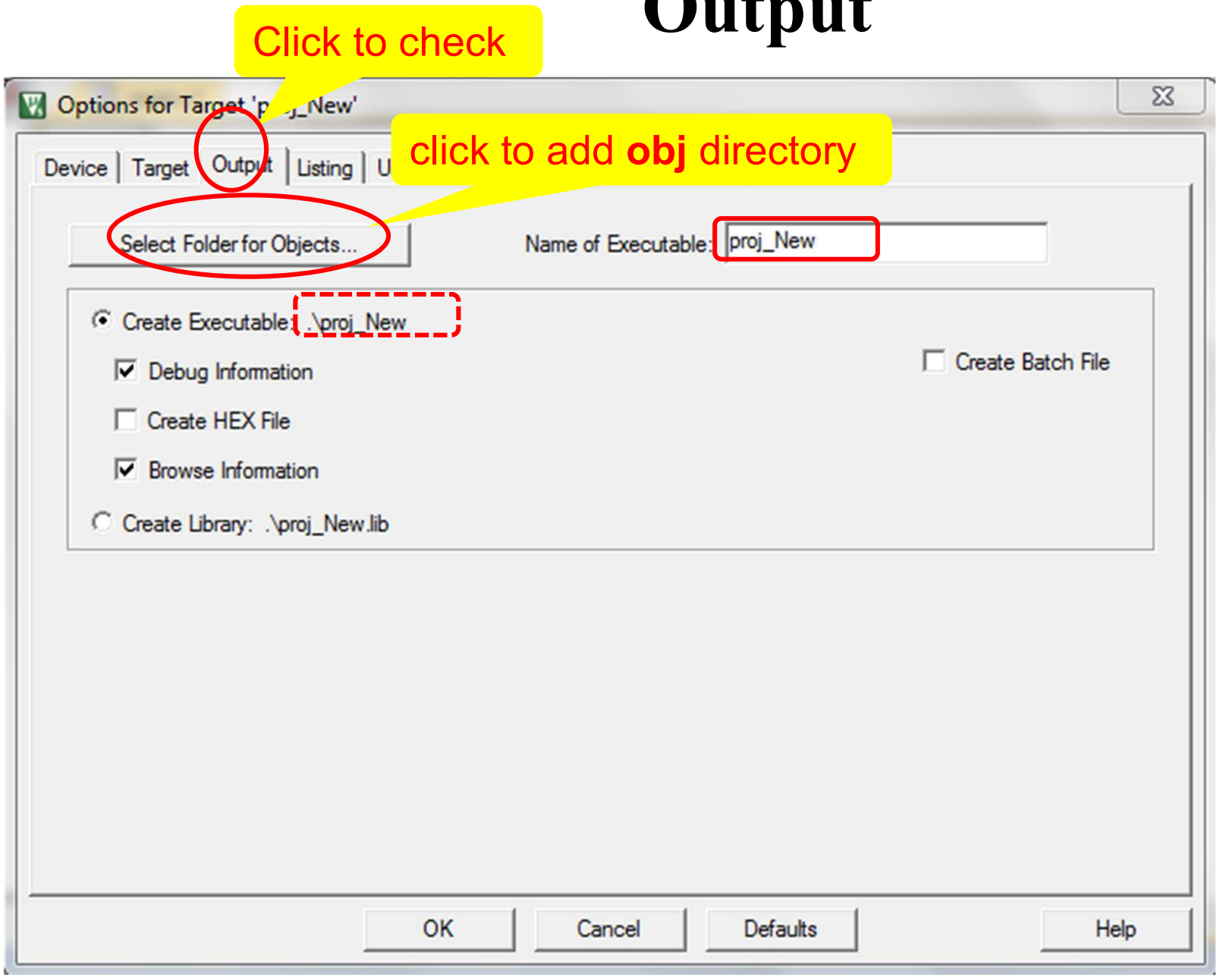
# Device


Options for Target 'proj_New'
Device
Set Device
Asm
Linker
Debug
Utilities
Generic CPU Data Base
Vendor: Nuvoton
Device: NUC140VE3CN
Toolset: ARM
Search:
NUC140VE3CN
NUC200LC2AN
NUC200LD2AN
NUC200LE3AN
NUC200SC2AN
NUC200SD2AN
NUC200SE3AN
NUC200VE3AN
NUC220LC2AN
NUC220LD2AN
+ Core
- ARM Cortex-M0 core runs up to 50MHz.
- NVIC for the 32 interrupt inputs, each with 4-levels of priority.
- Serial Wire Debug supports with 2 watchpoints/4 breakpoints.
+ Widely operating voltage range from 2.5V to 5.5V
+ Flash EPROM Memory
- 64K/128K bytes Flash EPROM for program code.
- 4kB flash for ISP loader
- Support ISP/IAP program code update
+ SRAM Memory
- 8K/16k bytes embedded SRAM.
+ PDMA (Peripheral DMA)
- Support 9 channels
+ Clock Control
OK
Cancel
Defaults
Help

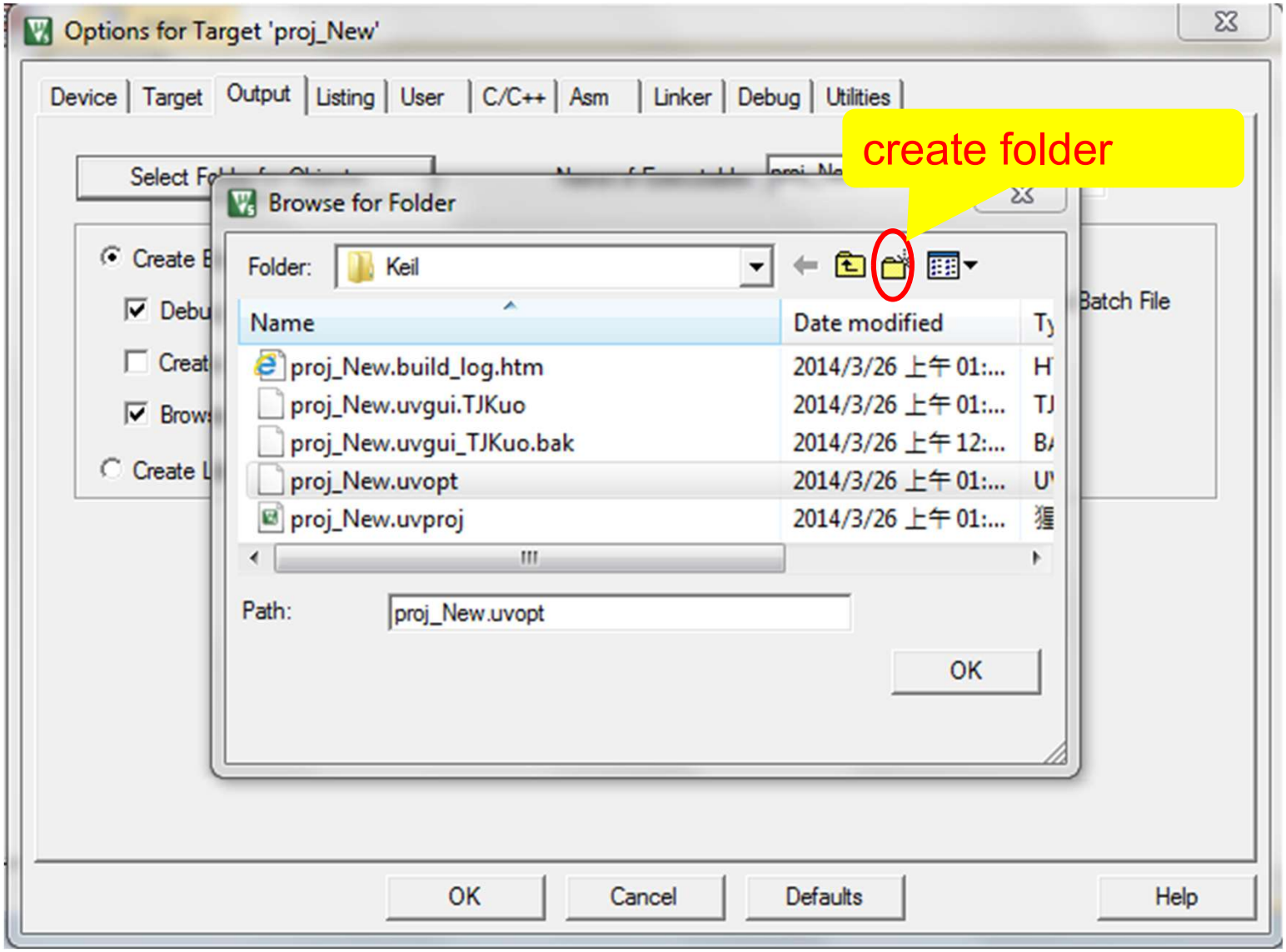
# Target


Click to check
Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Nuvoton NUC140VE3CN
Xtal (MHz): 50.0
Operating system: None
System-Viewer File (.Sfr):
Use Custom SVD File
Code Generation
Use Cross-Module Optimization
Use MicroLIB
Big Endian
Read/Only Memory Areas
default off-chip Start Size Startup
ROM1:
ROM2:
ROM3:
on-chip
IROM1: 0x0 0x20000
IROM2:
Read/Write Memory Areas
default off-chip Start Size NoInit
RAM1:
RAM2:
RAM3:
on-chip
IRAM1: 0x20000000 0x4000
IRAM2:
OK
Cancel
Defaults
Help

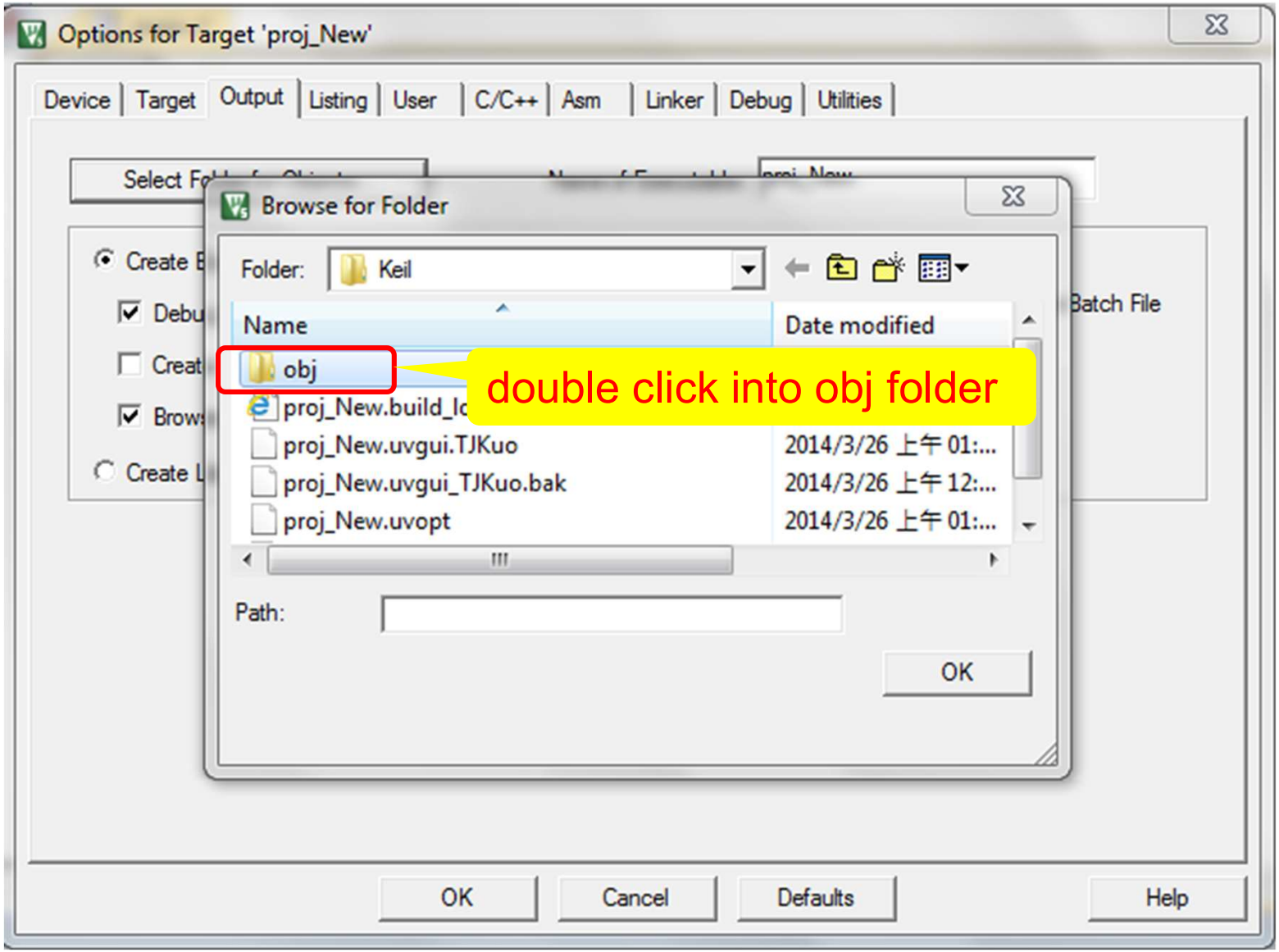
# Output


Click to check
click to add **obj** directory
Options for Target 'proj_New'
Device
Target
Output
Listing
U
Select Folder for Objects...
Name of Executable: proj_New
Create Executable: .\proj_New
Debug Information
Create Batch File
Create HEX File
Browse Information
Create Library: .\proj_New.lib
OK
Cancel
Defaults
Help

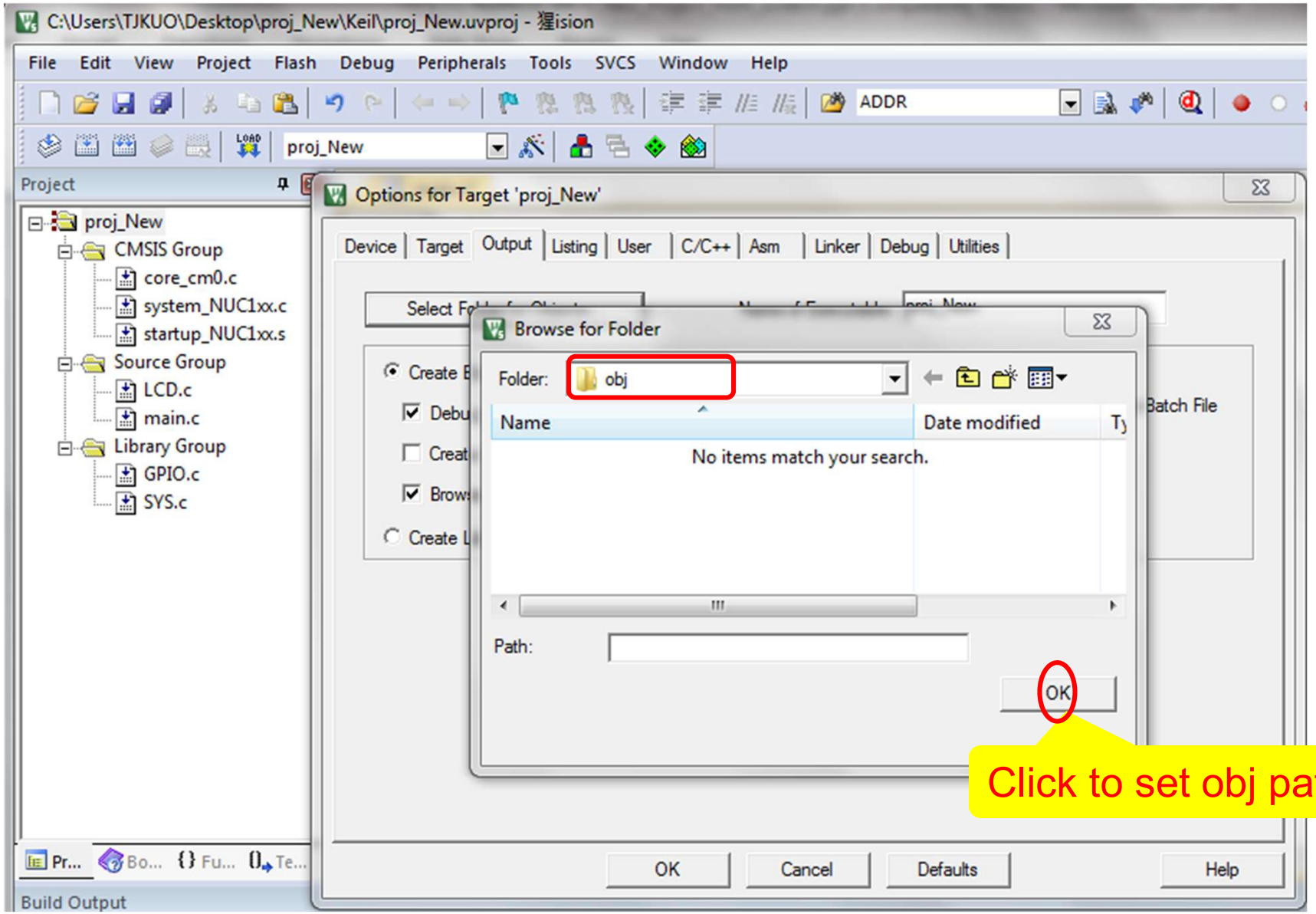
# **proj_New** directory



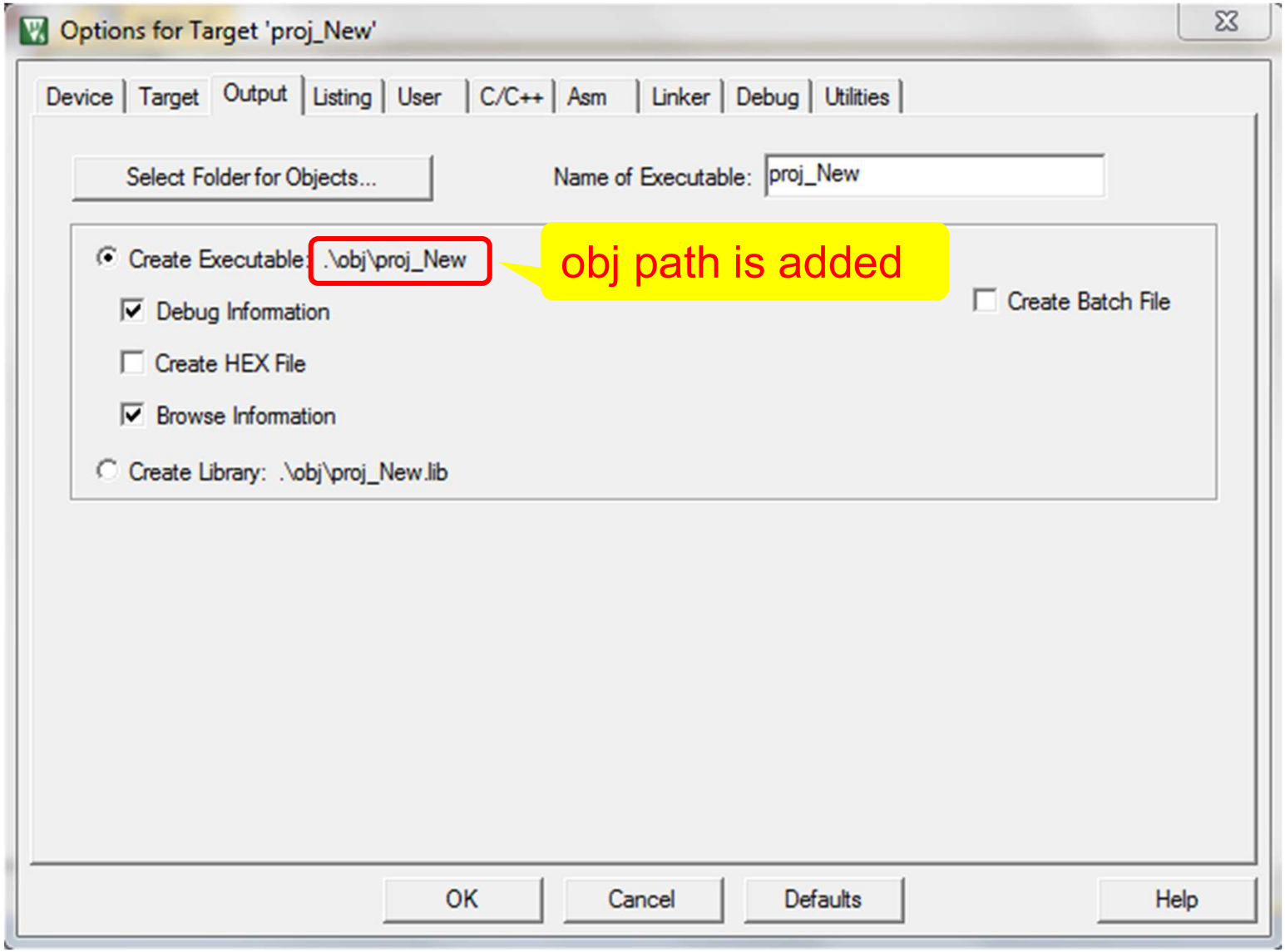
# **obj** folder created [1]



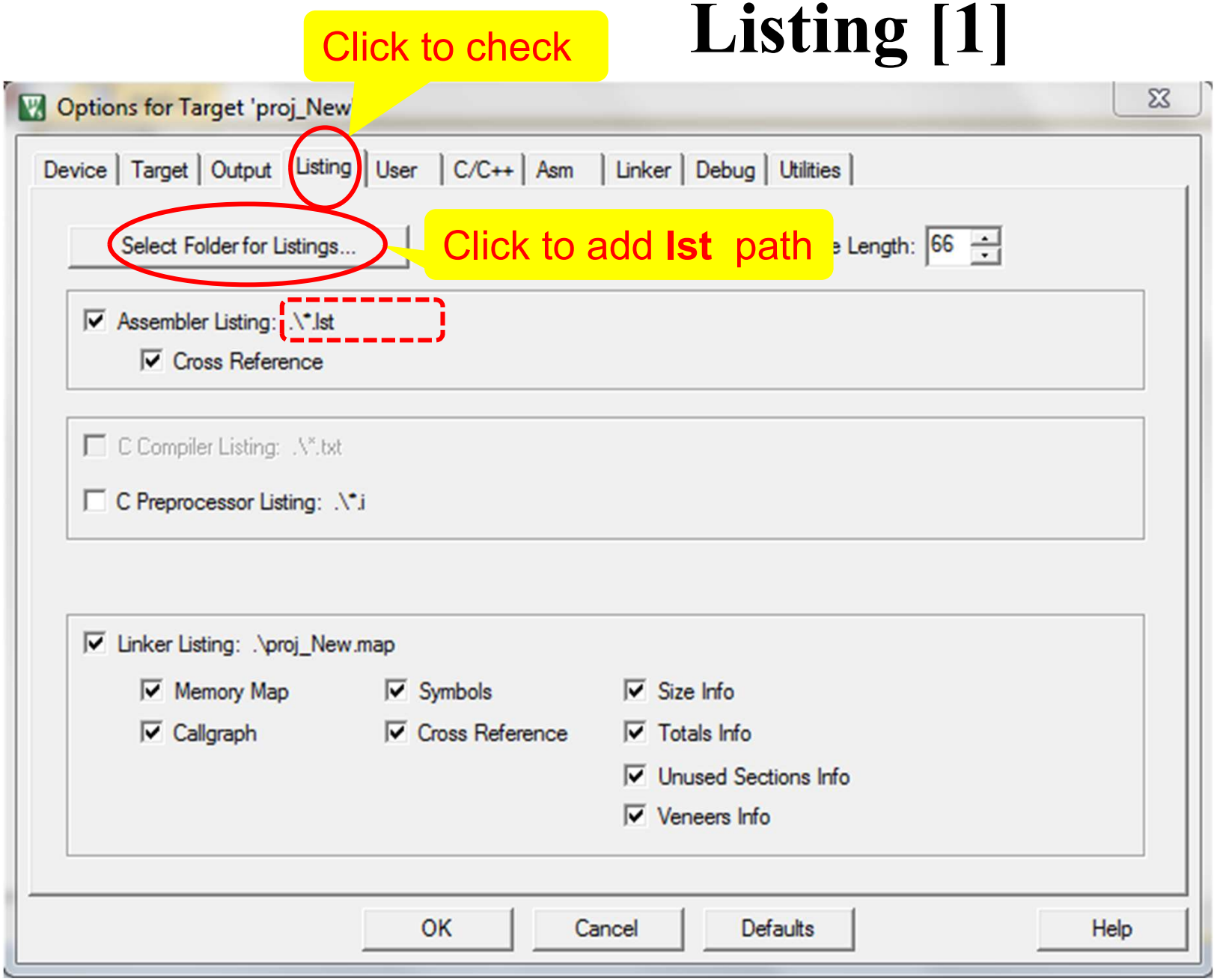
# **obj** folder created [2]



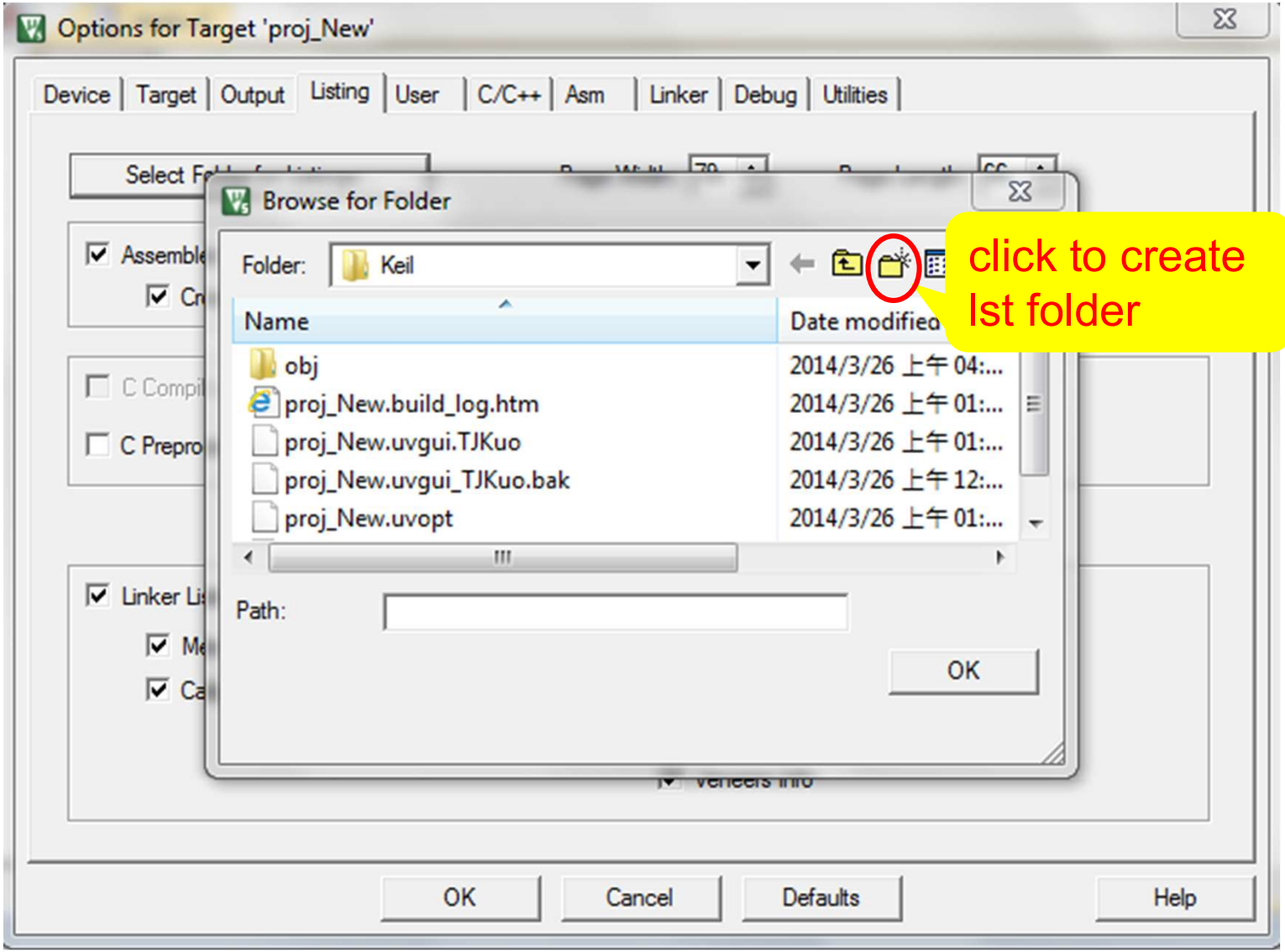
# Output


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Select Folder for Objects...
Name of Executable: proj_New
Create Executable: .\obj\proj_New
obj path is added
Debug Information
Create HEX File
Browse Information
Create Batch File
Create Library: .\obj\proj_New.lib
OK
Cancel
Defaults
Help

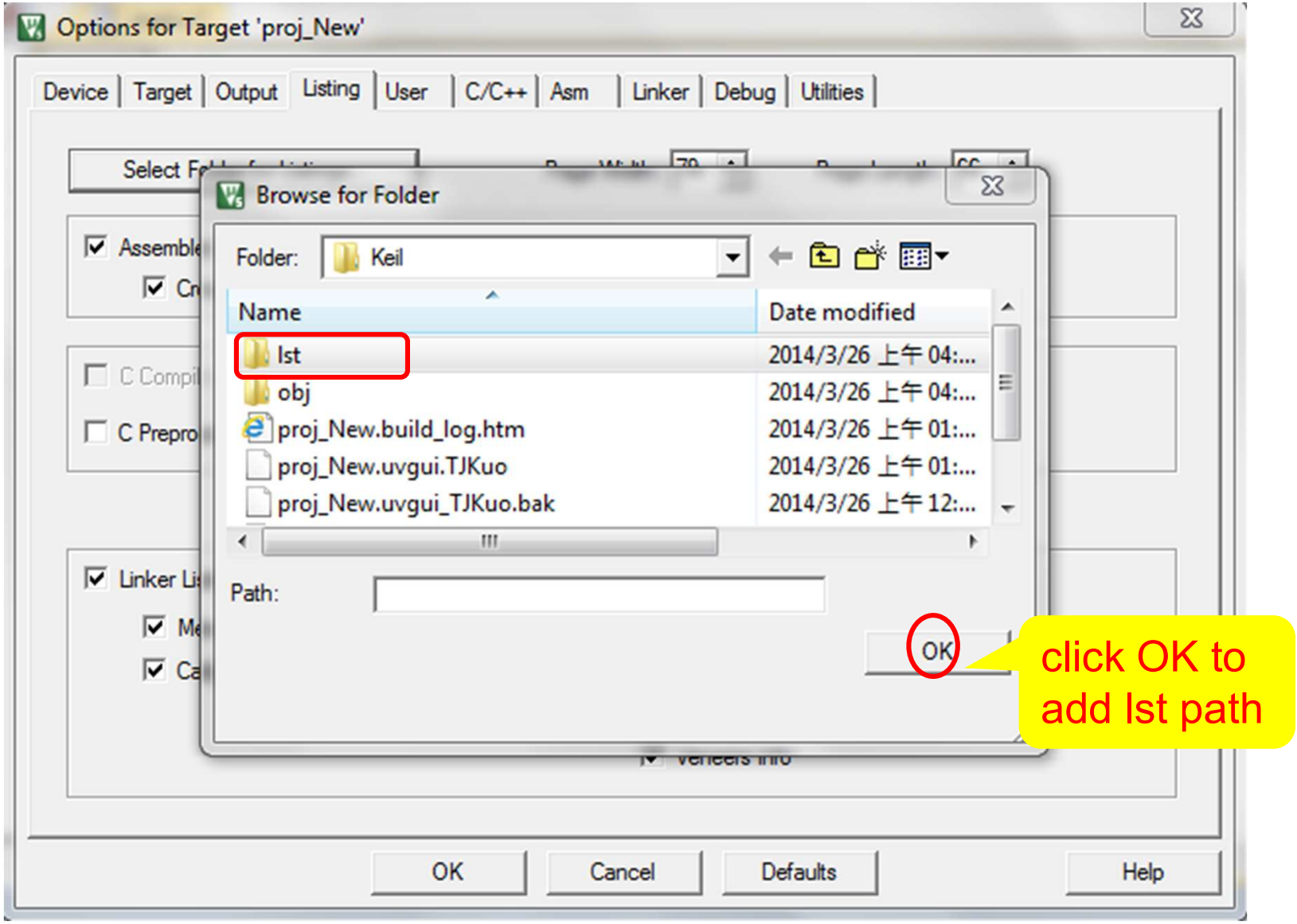
# Listing [1]


Click to check
Options for Target 'proj_New
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Select Folder for Listings...
Click to add lst path
Length:
66
Assembler Listing: .*.lst
Cross Reference
C Compiler Listing: .*.txt
C Preprocessor Listing: .*.i
Linker Listing: .\proj_New.map
Memory Map
Symbols
Size Info
Callgraph
Cross Reference
Totals Info
Unused Sections Info
Veneers Info
OK
Cancel
Defaults
Help

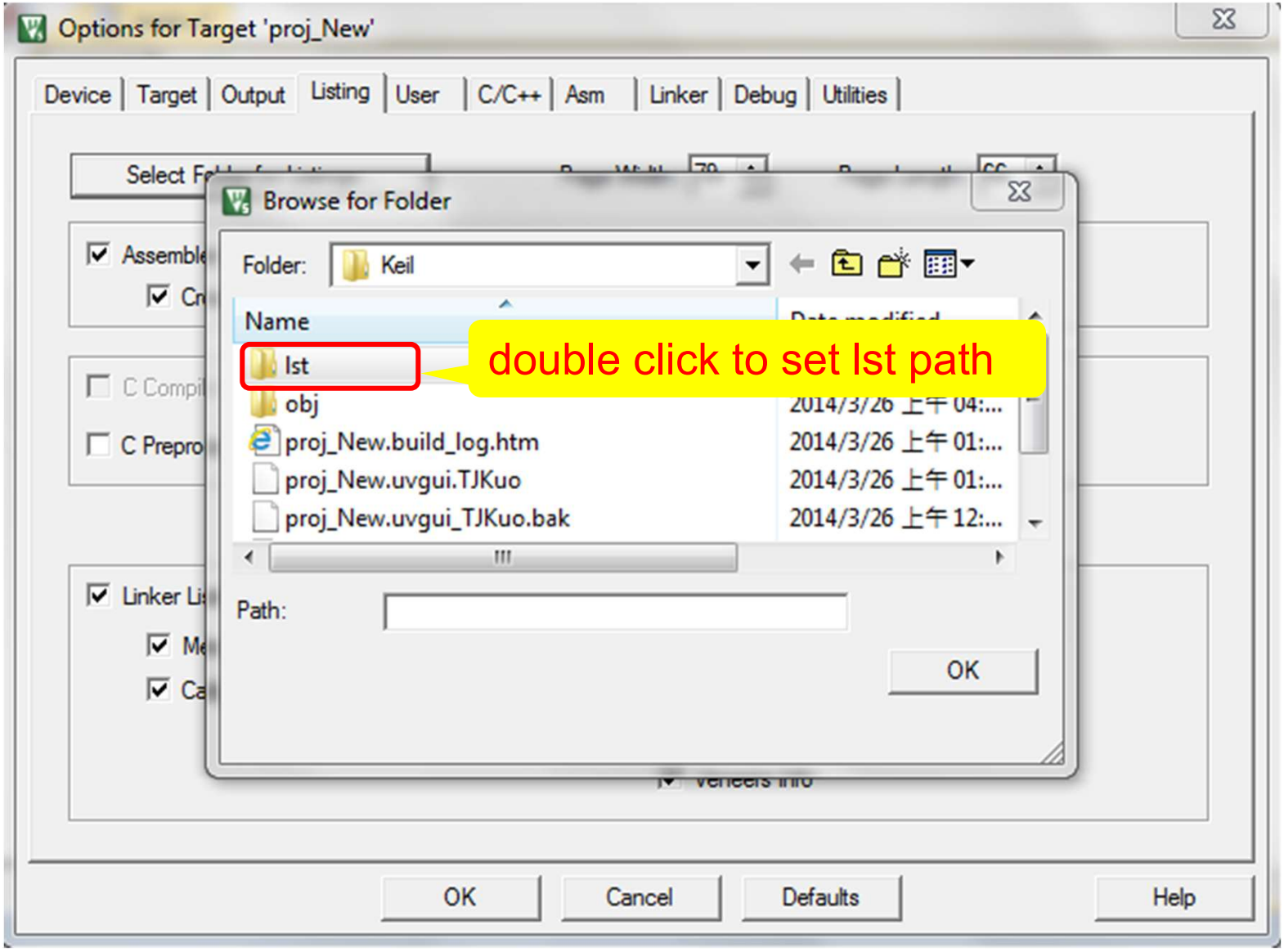
# Listing [2]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Browse for Folder
Folder:
Keil
click to create lst folder
Name
Date modified
obj
2014/3/26 上午 04:...
proj_New.build_log.htm
2014/3/26 上午 01:...
proj_New.uvgui.TJKuo
2014/3/26 上午 01:...
proj_New.uvgui_TJKuo.bak
2014/3/26 上午 12:...
proj_New.uvopt
2014/3/26 上午 01:...
Path:
OK
OK
Cancel
Defaults
Help

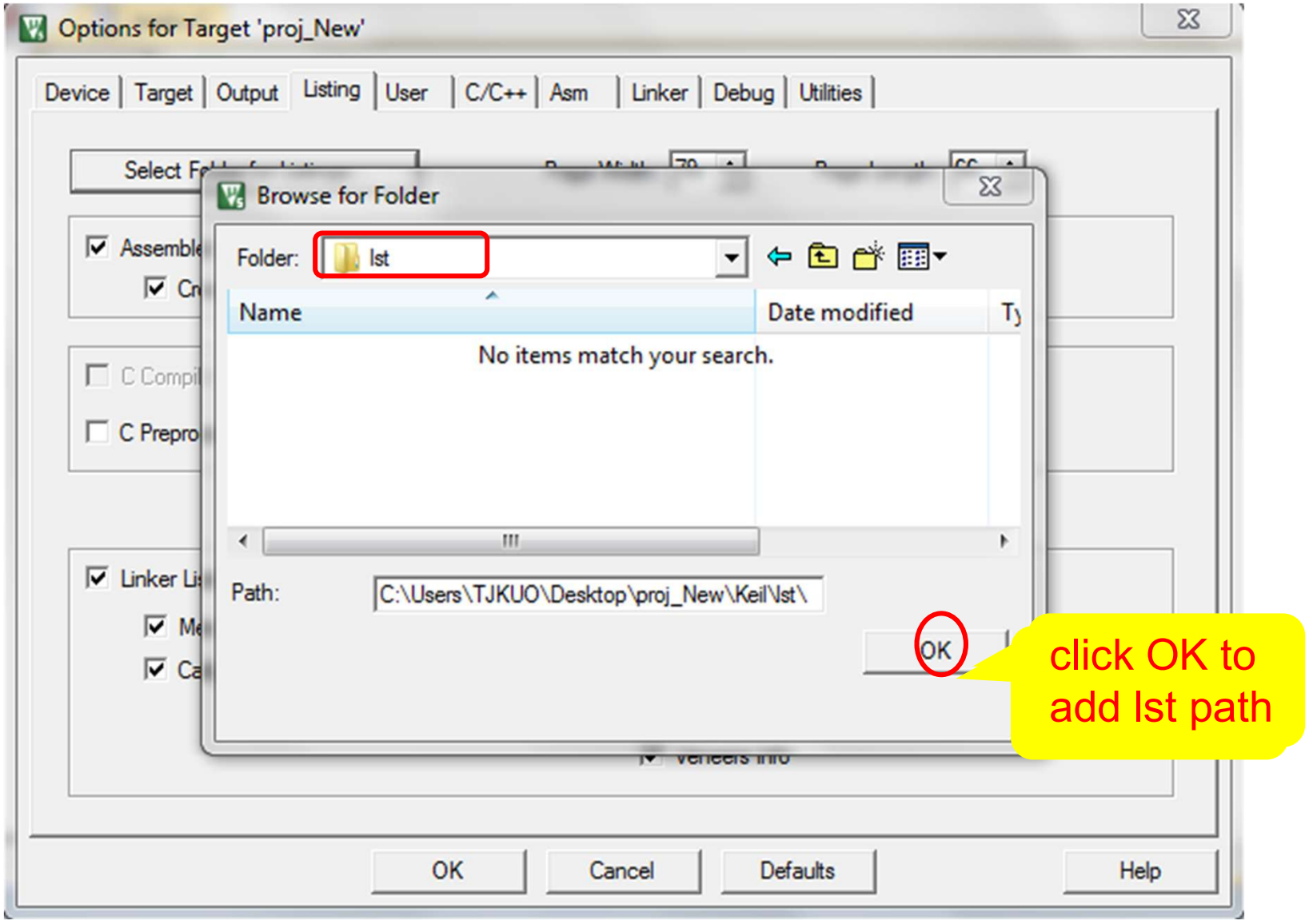
# Listing [3]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Browse for Folder
Folder:
Keil
Name
Date modified
lst
2014/3/26 上午 04:...
obj
2014/3/26 上午 04:...
proj_New.build_log.htm
2014/3/26 上午 01:...
proj_New.uvgui.TJKuo
2014/3/26 上午 01:...
proj_New.uvgui_TJKuo.bak
2014/3/26 上午 12:...
Path:
OK
click OK to add lst path
OK
Cancel
Defaults
Help

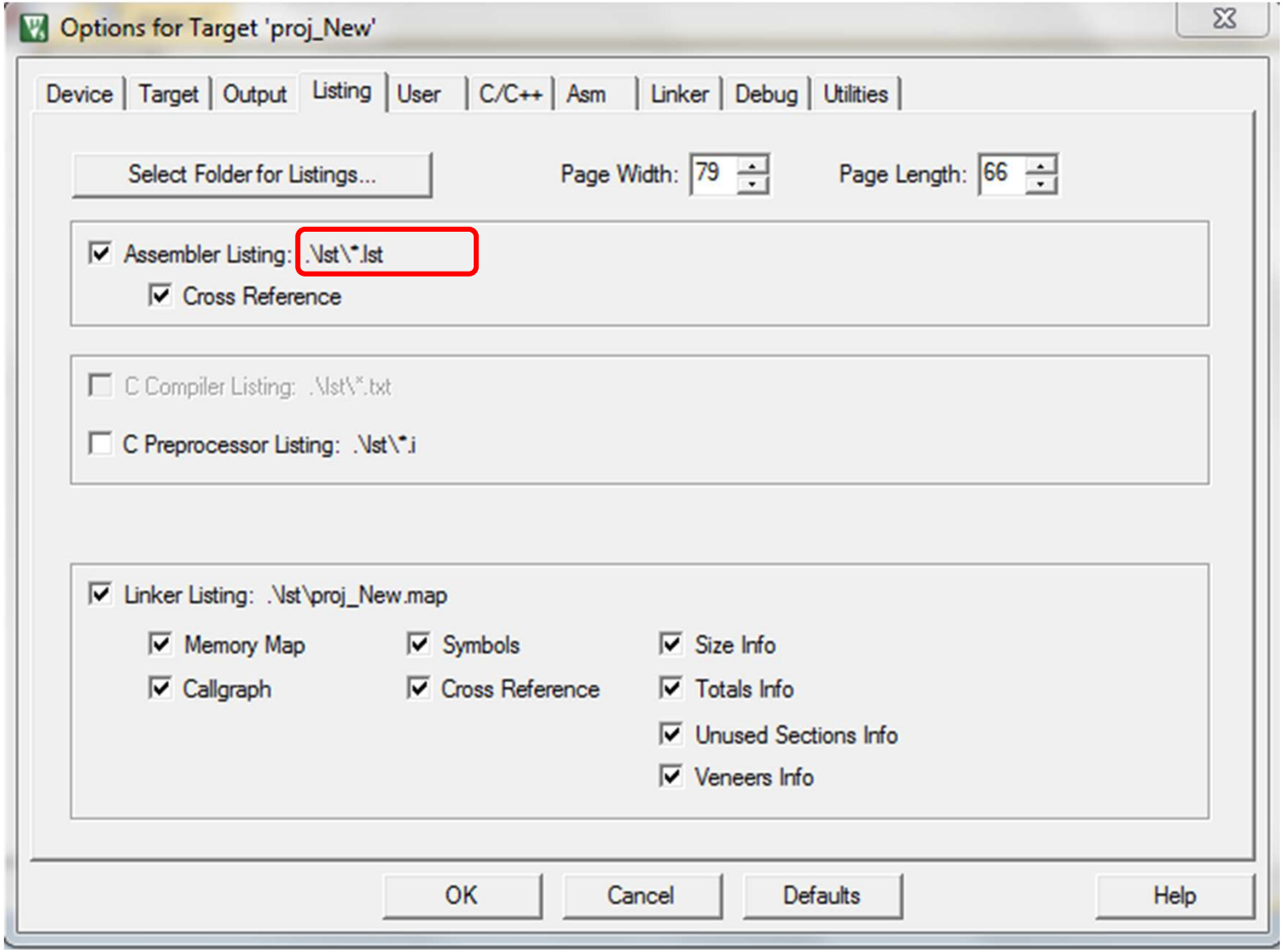
# Listing [4]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Browse for Folder
Folder:
Keil
Name
lst
obj
proj_New.build_log.htm
proj_New.uvgui.TJKuo
proj_New.uvgui_TJKuo.bak
2014/3/26 上午 01:...
2014/3/26 上午 01:...
2014/3/26 上午 12:...
double click to set lst path
Path:
OK
OK
Cancel
Defaults
Help

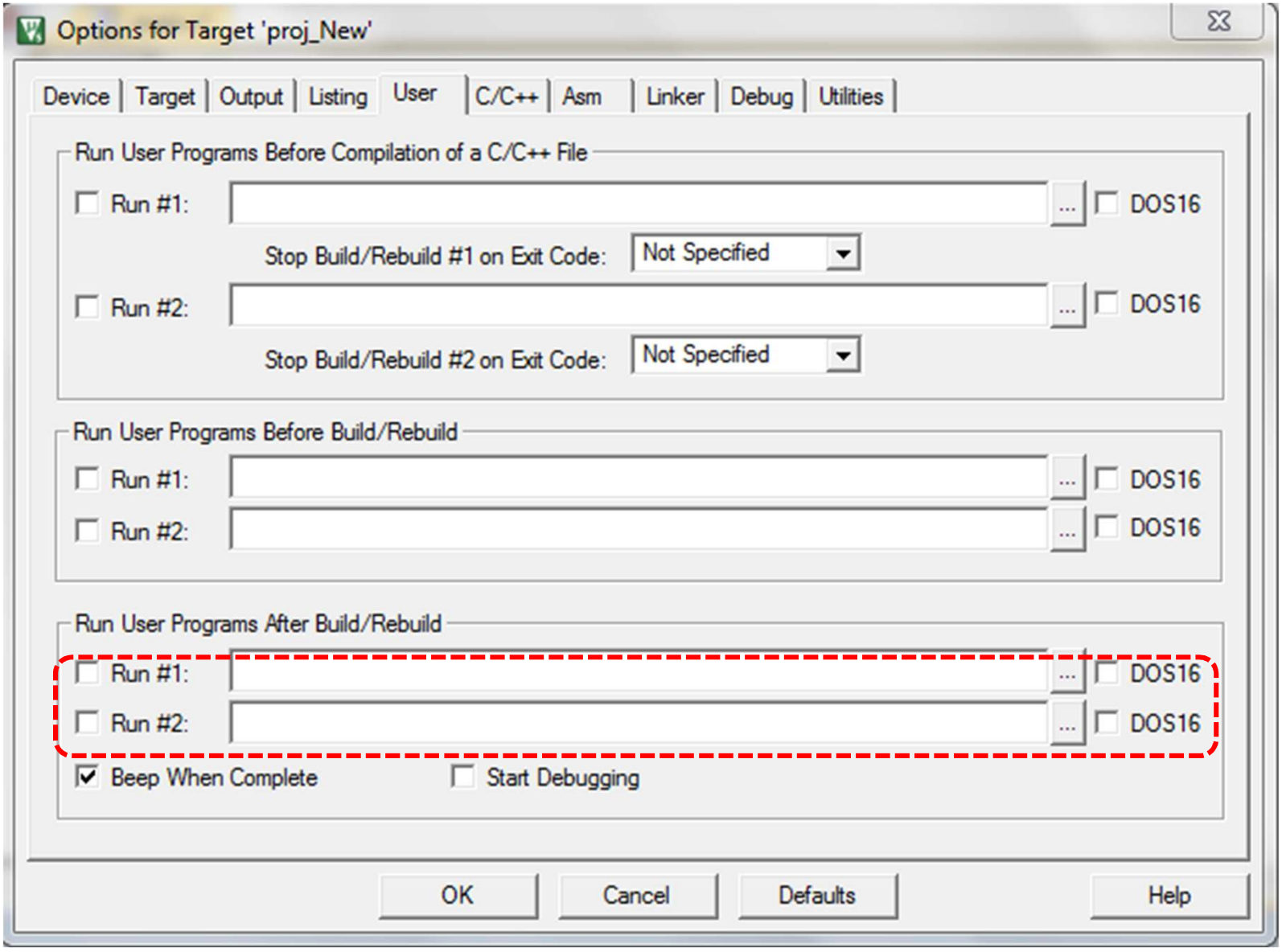
# Listing [5]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Browse for Folder
Folder:
lst
Name
Date modified
No items match your search.
Path:
C:\Users\TJKUO\Desktop\proj_New\Keil\lst\
OK
click OK to add lst path
OK
Cancel
Defaults
Help

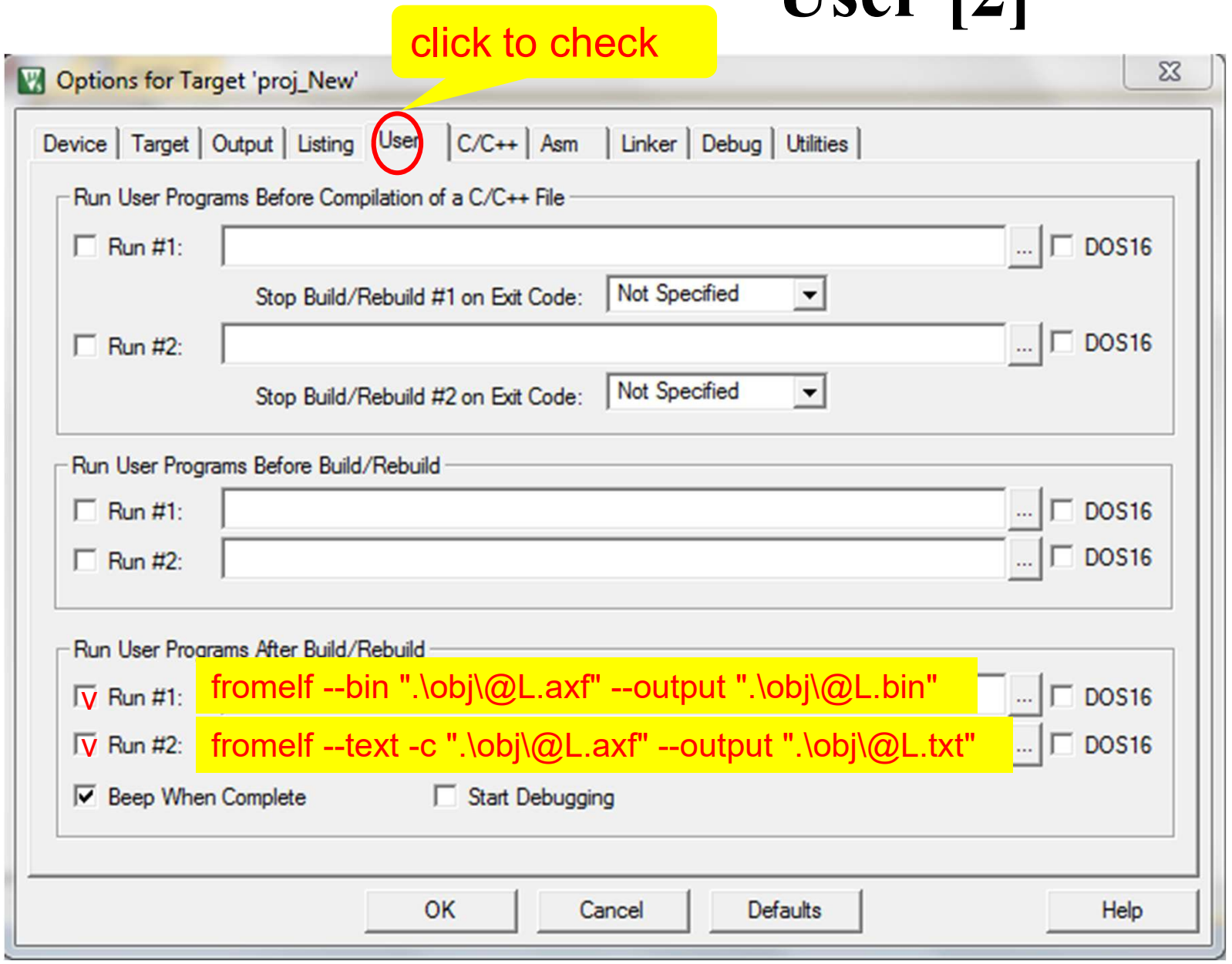
# Listing [6]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Select Folder for Listings...
Page Width: 79
Page Length: 66
Assembler Listing: .\lst*.lst
Cross Reference
C Compiler Listing: .\lst*.txt
C Preprocessor Listing: .\lst*.i
Linker Listing: .\lst\proj_New.map
Memory Map
Symbols
Size Info
Callgraph
Cross Reference
Totals Info
Unused Sections Info
Veneers Info
OK
Cancel
Defaults
Help

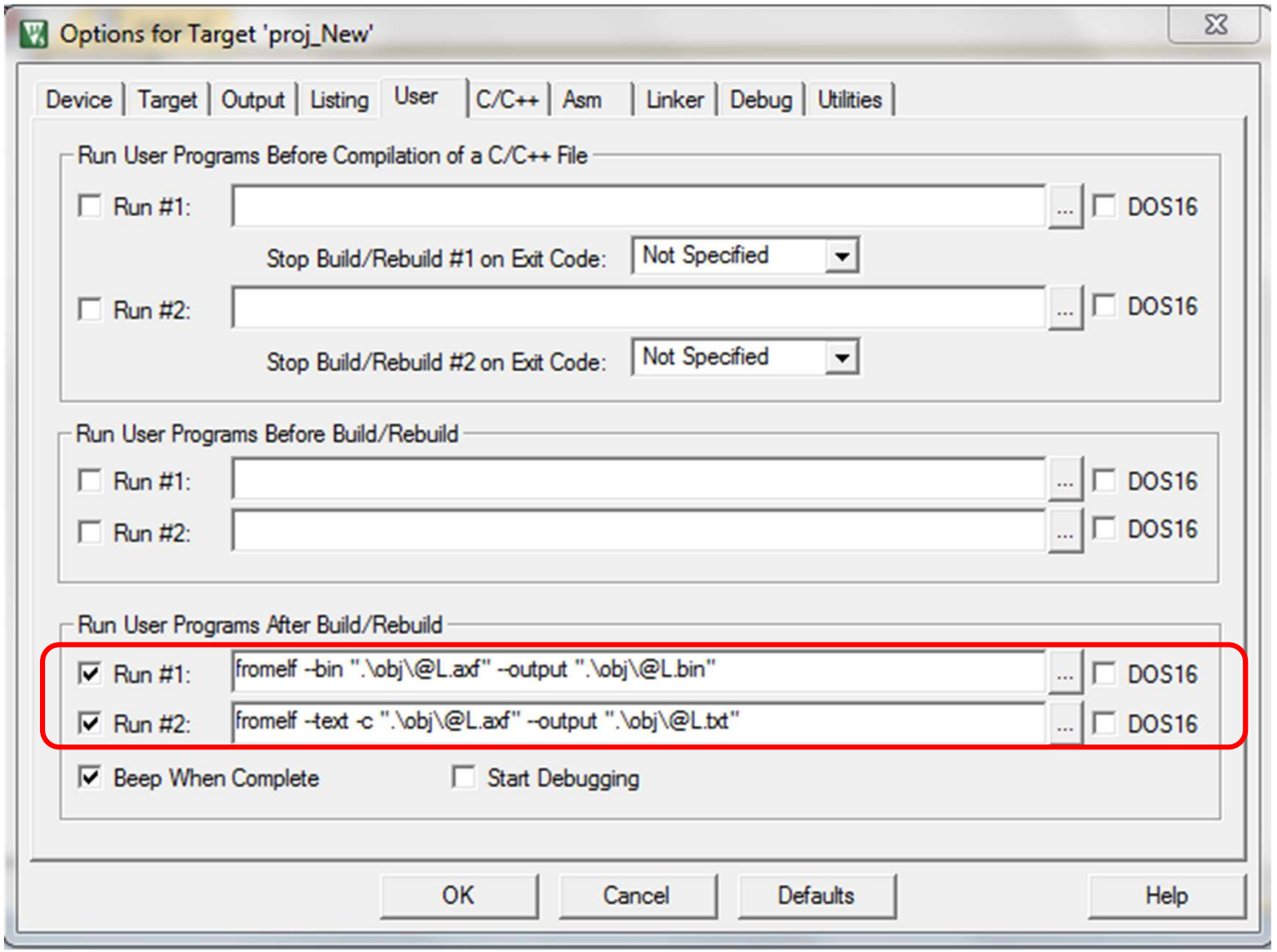
# User [1]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Run User Programs Before Compilation of a C/C++ File
Run #1:
DOS16
Stop Build/Rebuild #1 on Exit Code:
Not Specified
Run #2:
DOS16
Stop Build/Rebuild #2 on Exit Code:
Not Specified
Run User Programs Before Build/Rebuild
Run #1:
DOS16
Run #2:
DOS16
Run User Programs After Build/Rebuild
Run #1:
DOS16
Run #2:
DOS16
Beep When Complete
Start Debugging
OK
Cancel
Defaults
Help

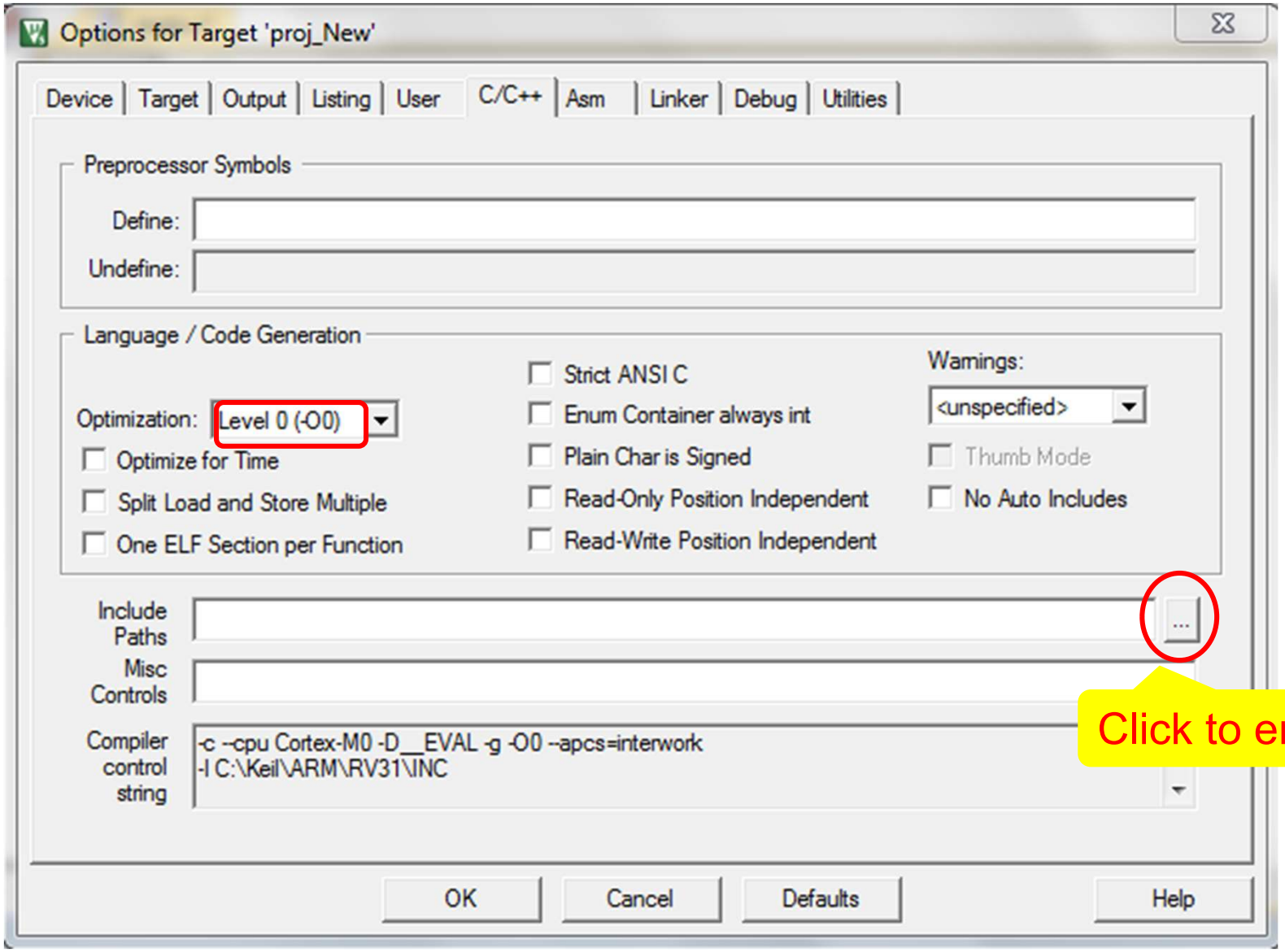
# User [2]


click to check
Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Run User Programs Before Compilation of a C/C++ File
Run #1:
DOS16
Stop Build/Rebuild #1 on Exit Code:
Not Specified
Run #2:
DOS16
Stop Build/Rebuild #2 on Exit Code:
Not Specified
Run User Programs Before Build/Rebuild
Run #1:
DOS16
Run #2:
DOS16
Run User Programs After Build/Rebuild
Run #1:
fromelf --bin ".\obj\@L.axf" --output ".\obj\@L.bin"
DOS16
Run #2:
fromelf --text -c ".\obj\@L.axf" --output ".\obj\@L.txt"
DOS16
Beep When Complete
Start Debugging
OK
Cancel
Defaults
Help

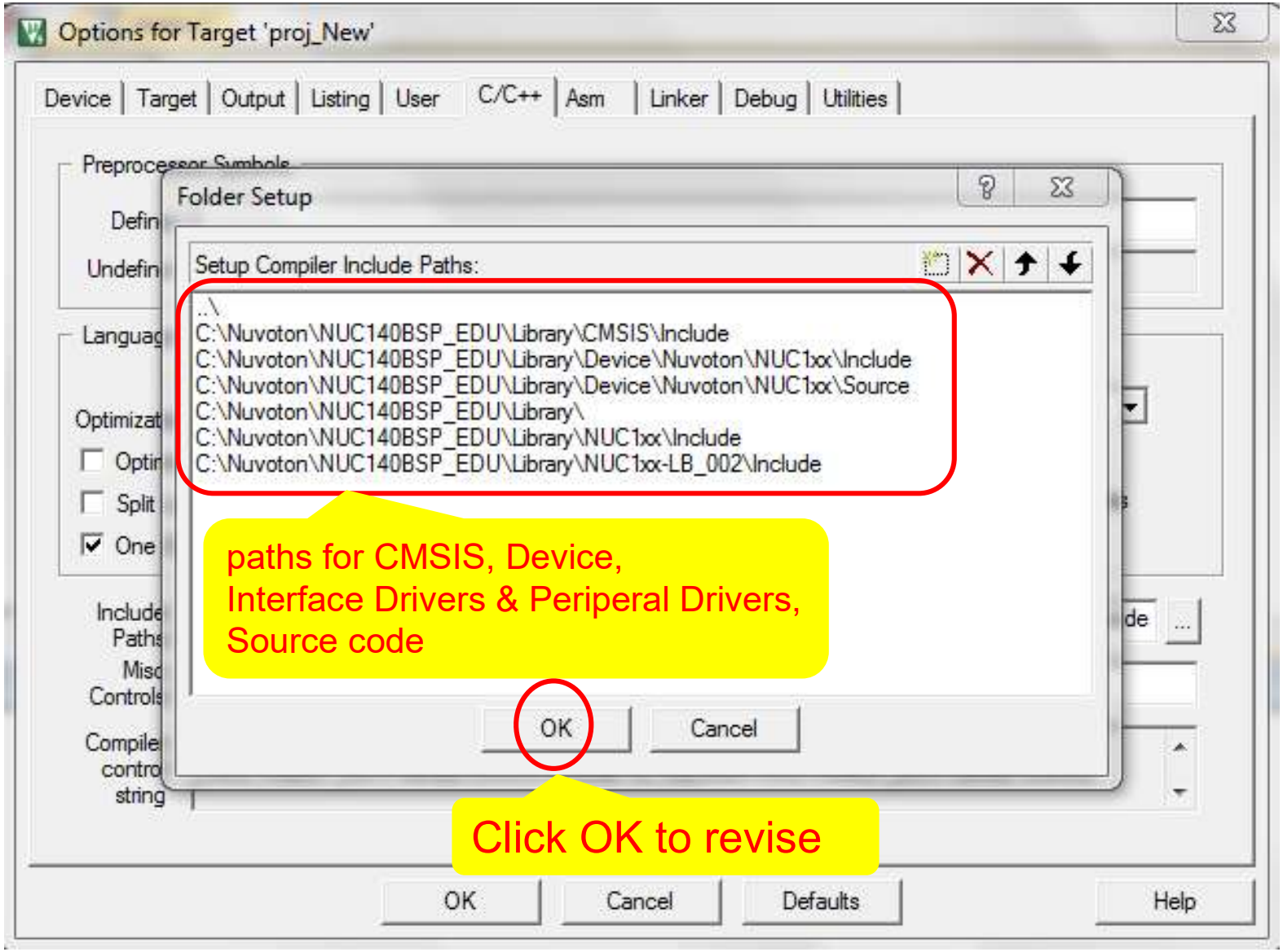
# User [3]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Run User Programs Before Compilation of a C/C++ File
Run #1:
DOS16
Stop Build/Rebuild #1 on Exit Code:
Not Specified
Run #2:
DOS16
Stop Build/Rebuild #2 on Exit Code:
Not Specified
Run User Programs Before Build/Rebuild
Run #1:
DOS16
Run #2:
DOS16
Run User Programs After Build/Rebuild
Run #1:
fromelf --bin ".\obj\@L.axf" --output ".\obj\@L.bin"
DOS16
Run #2:
fromelf --text -c ".\obj\@L.axf" --output ".\obj\@L.txt"
DOS16
Beep When Complete
Start Debugging
OK
Cancel
Defaults
Help

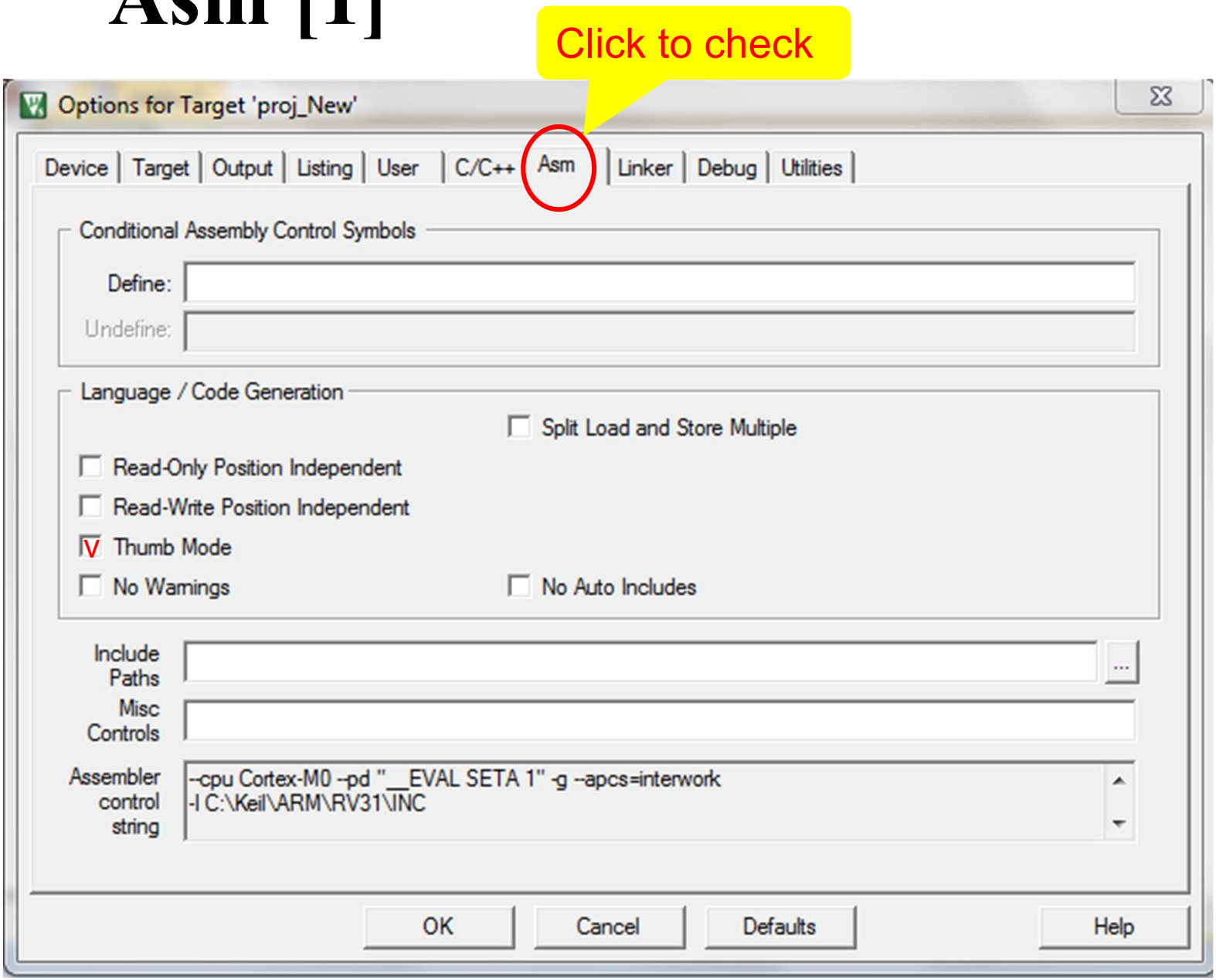
# C/C++ [1]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Preprocessor Symbols
Define:
Undefine:
Language / Code Generation
Optimization:
Level 0 (-O0)
Optimize for Time
Split Load and Store Multiple
One ELF Section per Function
Strict ANSI C
Enum Container always int
Plain Char is Signed
Read-Only Position Independent
Read-Write Position Independent
Warnings:
<unspecified>
Thumb Mode
No Auto Includes
Include Paths
Misc Controls
Compiler control string
-c --cpu Cortex-M0 -D__EVAL -g -O0 --apcs=interwork
-I C:\Keil\ARM\RV31\INC
Click to enter
OK
Cancel
Defaults
Help

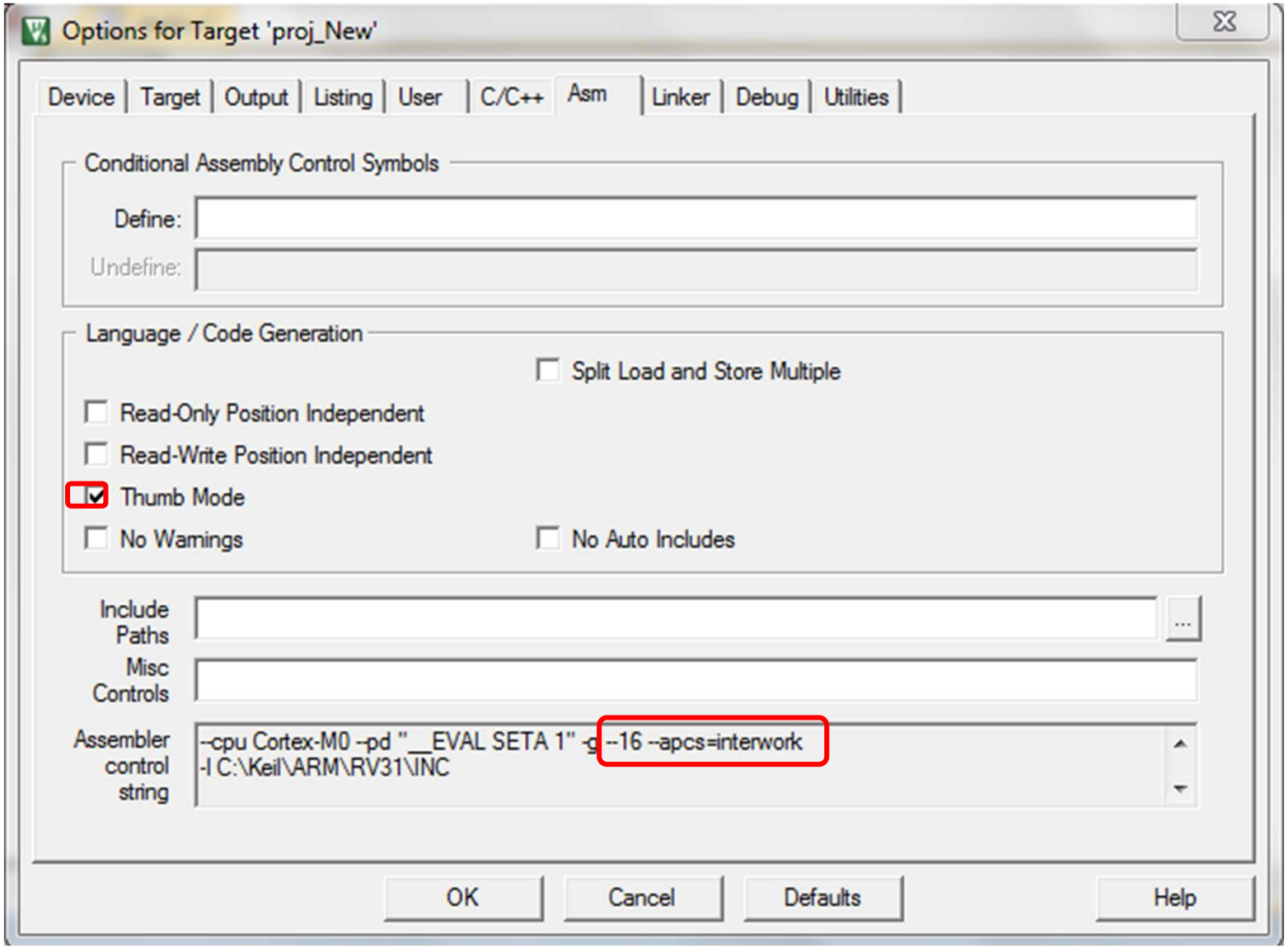
# C/C++ [2]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Folder Setup
Setup Compiler Include Paths:
..\
C:\Nuvoton\NUC140BSP_EDU\Library\CMSIS\Include
C:\Nuvoton\NUC140BSP_EDU\Library\Device\Nuvoton\NUC1xx\Include
C:\Nuvoton\NUC140BSP_EDU\Library\Device\Nuvoton\NUC1xx\Source
C:\Nuvoton\NUC140BSP_EDU\Library\
C:\Nuvoton\NUC140BSP_EDU\Library\NUC1xx\Include
C:\Nuvoton\NUC140BSP_EDU\Library\NUC1xx-LB_002\Include
paths for CMSIS, Device,
Interface Drivers & Periperal Drivers,
Source code
OK
Cancel
Click OK to revise
OK
Cancel
Defaults
Help

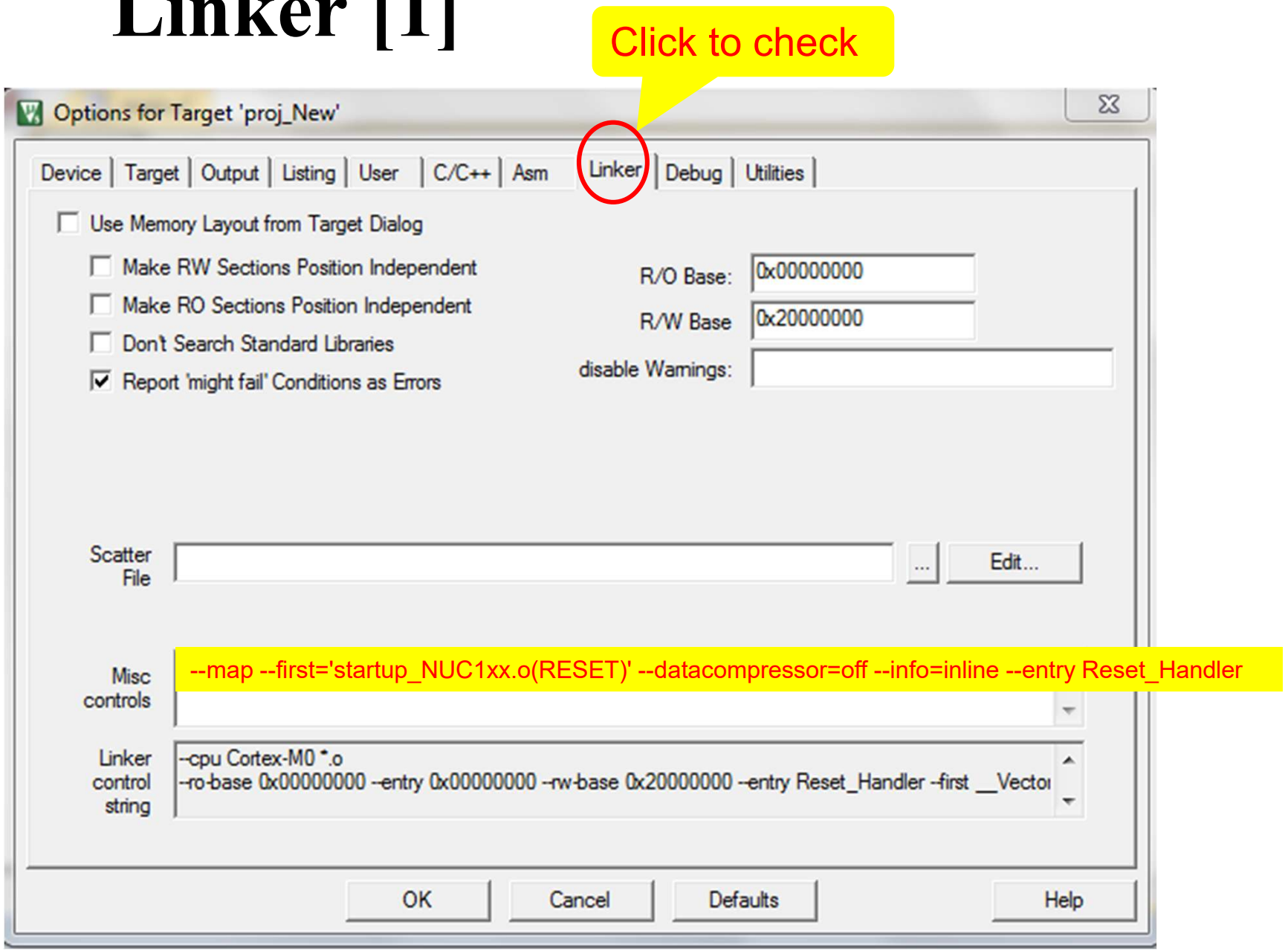
# Asm [1]


Click to check
Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Conditional Assembly Control Symbols
Define:
Undefine:
Language / Code Generation
Split Load and Store Multiple
Read-Only Position Independent
Read-Write Position Independent
Thumb Mode
No Warnings
No Auto Includes
Include Paths
Misc Controls
Assembler control string
--cpu Cortex-M0 --pd "__EVAL SETA 1" -g --apcs=interwork
-I C:\Keil\ARM\RV31\INC
OK
Cancel
Defaults
Help

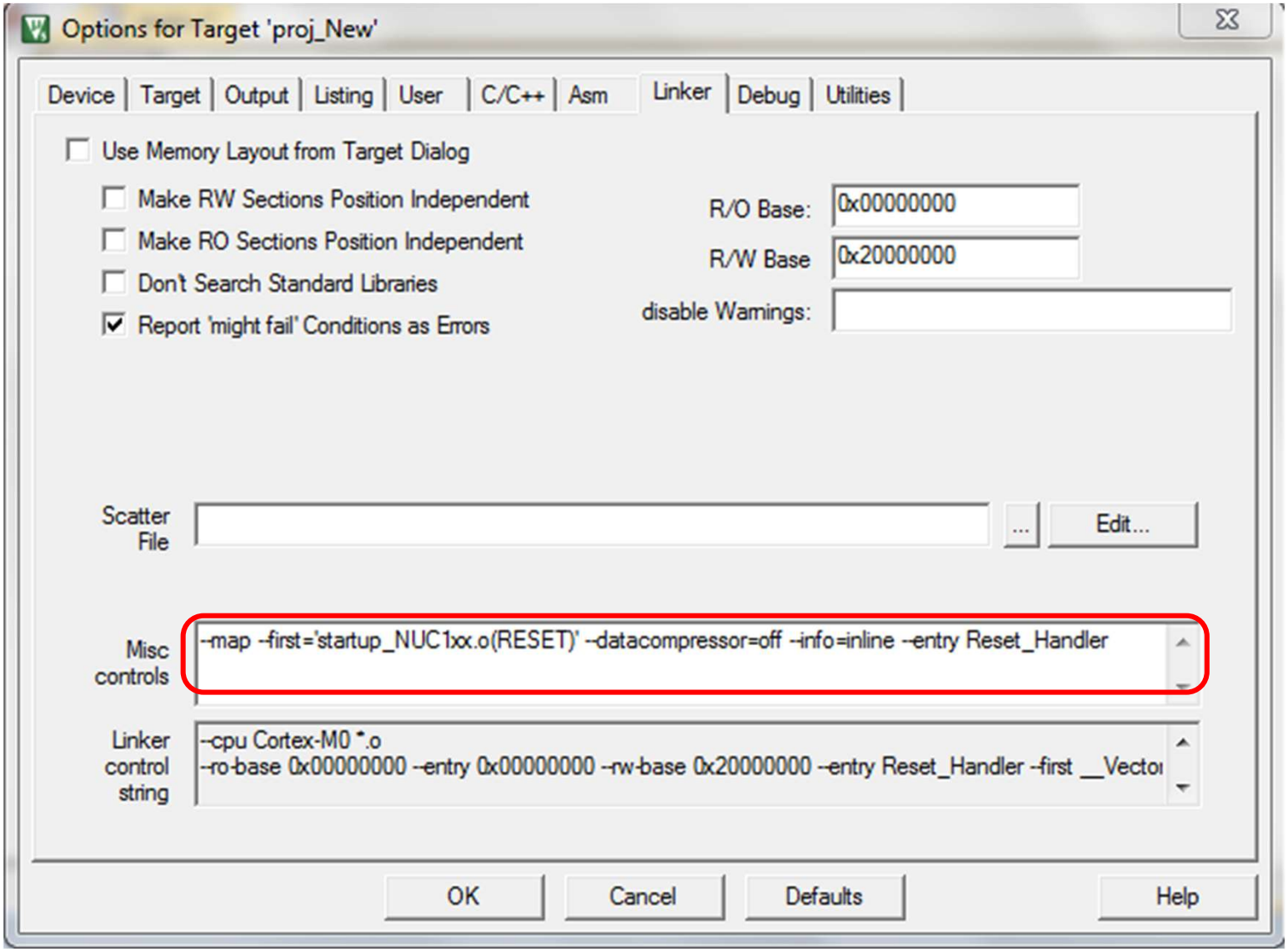
# Asm [2]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Conditional Assembly Control Symbols
Define:
Undefine:
Language / Code Generation
Split Load and Store Multiple
Read-Only Position Independent
Read-Write Position Independent
Thumb Mode
No Warnings
No Auto Includes
Include Paths
Misc Controls
Assembler control string
--cpu Cortex-M0 --pd "__EVAL SETA 1" -g --16 --apcs=interwork
-I C:\Keil\ARM\RV31\INC
OK
Cancel
Defaults
Help

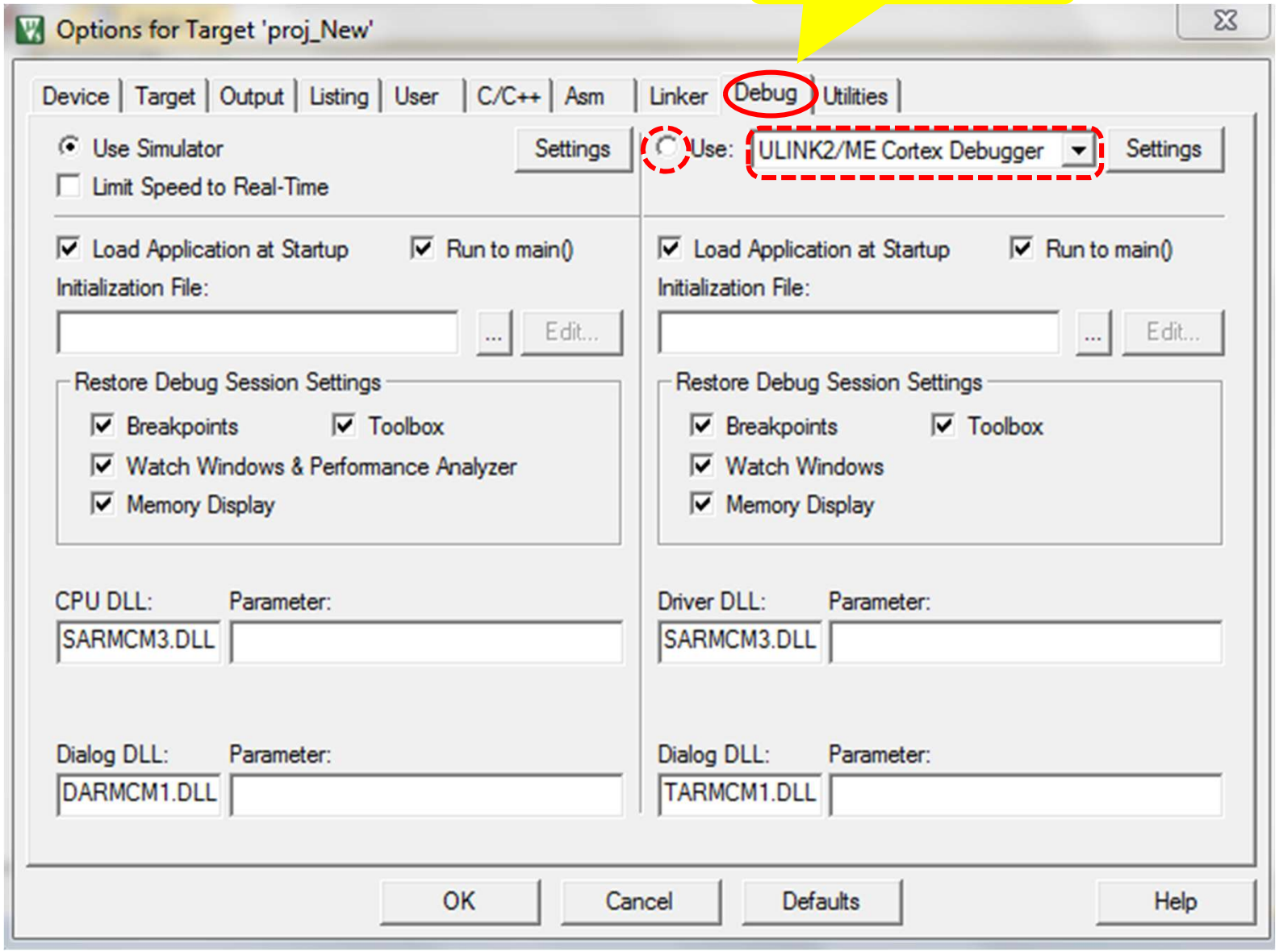
# Linker [1]


Click to check
Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Use Memory Layout from Target Dialog
Make RW Sections Position Independent
Make RO Sections Position Independent
Don't Search Standard Libraries
Report 'might fail' Conditions as Errors
R/O Base:
0x00000000
R/W Base
0x20000000
disable Warnings:
Scatter File
...
Edit...
Misc controls
--map --first='startup_NUC1xx.o(RESET)' --datacompressor=off --info=inline --entry Reset_Handler
Linker control string
--cpu Cortex-M0 *.o
--ro-base 0x00000000 --entry 0x00000000 --rw-base 0x20000000 --entry Reset_Handler --first __Vecto
OK
Cancel
Defaults
Help

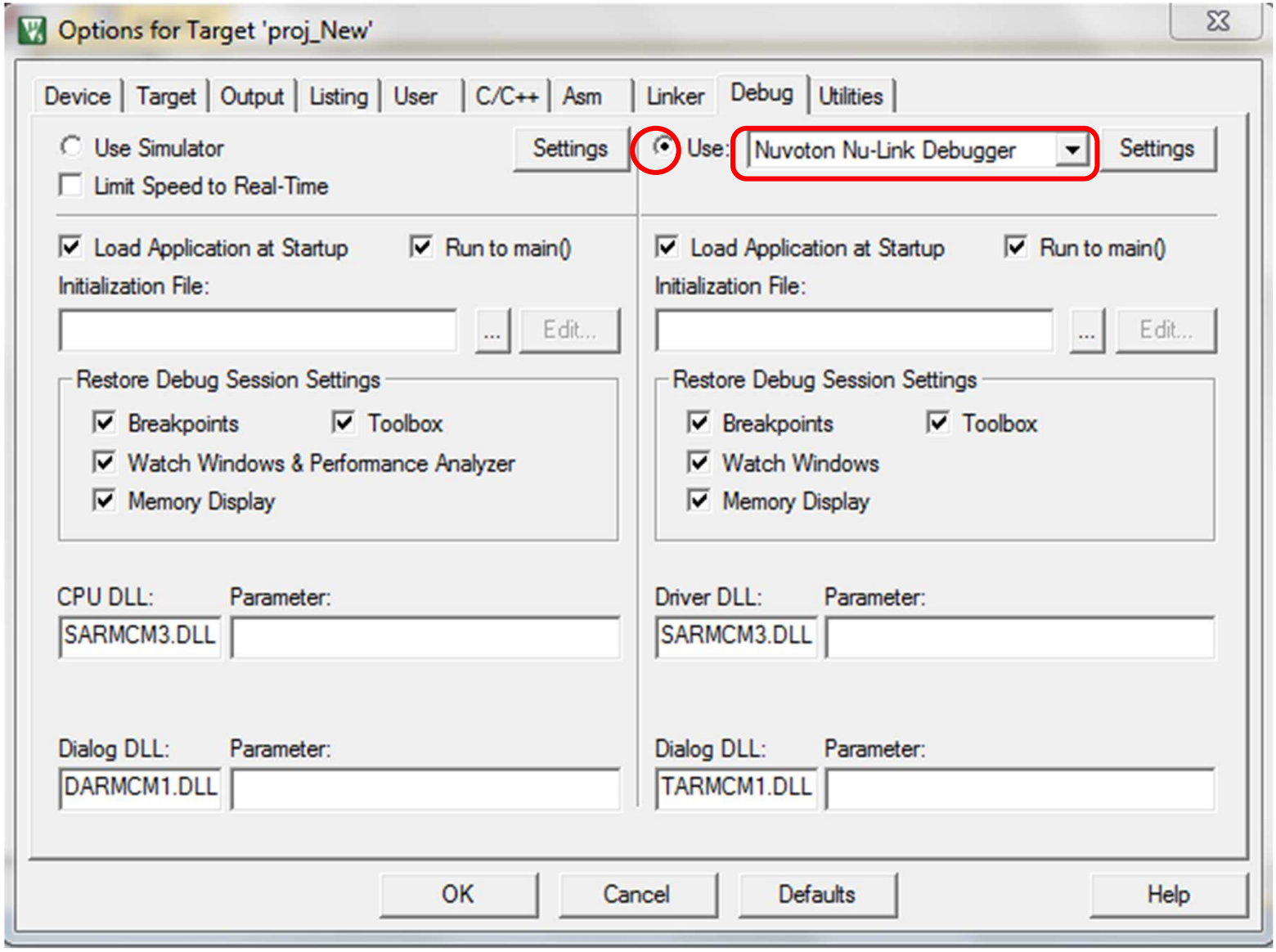
# Linker [2]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Use Memory Layout from Target Dialog
Make RW Sections Position Independent
Make RO Sections Position Independent
Don't Search Standard Libraries
Report 'might fail' Conditions as Errors
R/O Base:
0x00000000
R/W Base
0x20000000
disable Warnings:
Scatter File
...
Edit...
Misc controls
--map --first='startup_NUC1xx.o(RESET)' --datacompressor=off --info=inline --entry Reset_Handler
Linker control string
--cpu Cortex-M0 *.o
--ro-base 0x00000000 --entry 0x00000000 --rw-base 0x20000000 --entry Reset_Handler --first __Vecto
OK
Cancel
Defaults
Help

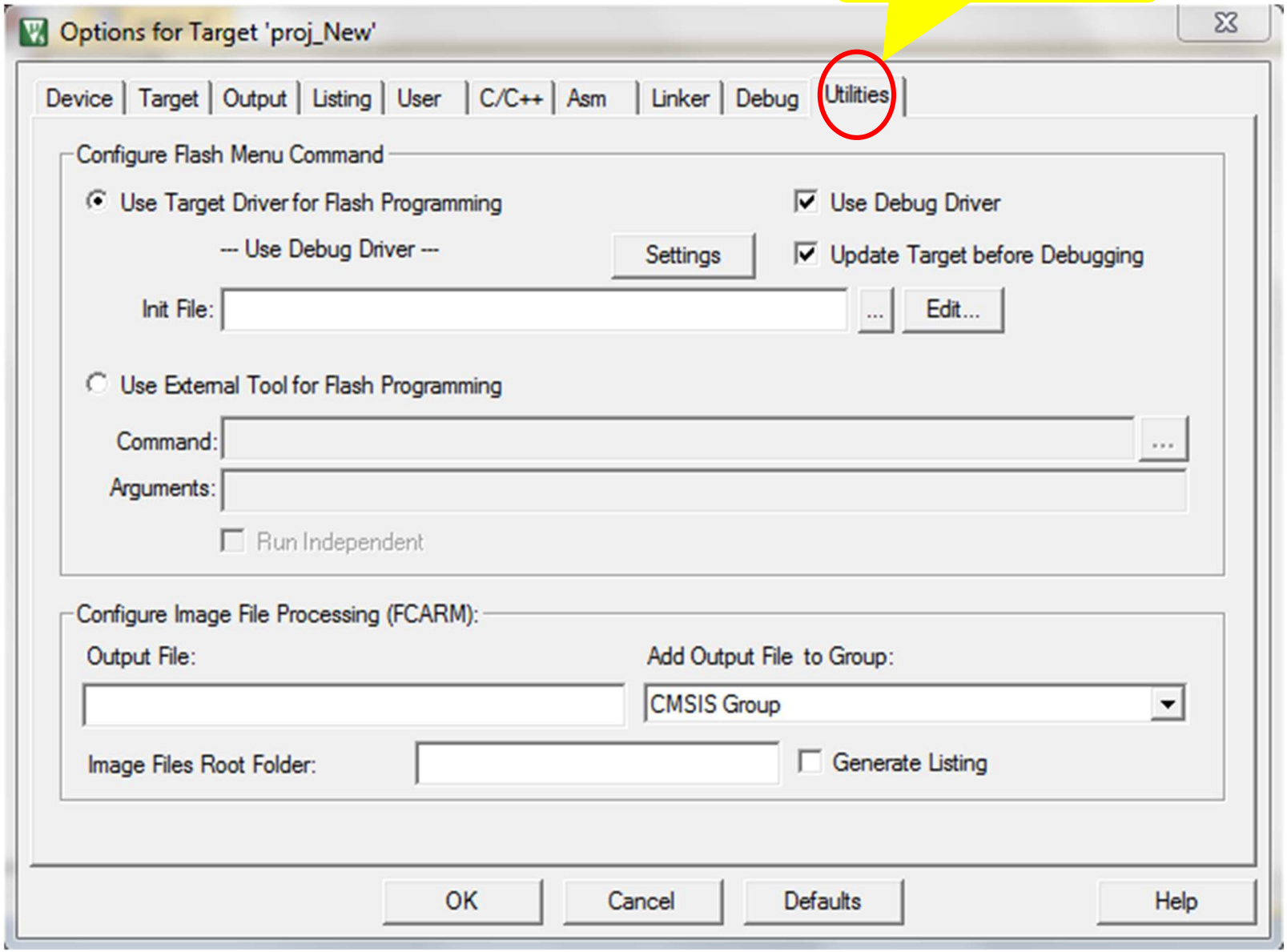
# Debug [1]


Click to check
Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Use Simulator
Settings
Use:
ULINK2/ME Cortex Debugger
Settings
Limit Speed to Real-Time
Load Application at Startup
Run to main()
Initialization File:
Edit...
Restore Debug Session Settings
Breakpoints
Toolbox
Watch Windows & Performance Analyzer
Memory Display
Load Application at Startup
Run to main()
Initialization File:
Edit...
Restore Debug Session Settings
Breakpoints
Toolbox
Watch Windows
Memory Display
CPU DLL:
Parameter:
SARMCM3.DLL
Driver DLL:
Parameter:
SARMCM3.DLL
Dialog DLL:
Parameter:
DARMCM1.DLL
Dialog DLL:
Parameter:
TARMCM1.DLL
OK
Cancel
Defaults
Help

# Debug [2]


Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Use Simulator
Settings
Use:
Nuvoton Nu-Link Debugger
Settings
Limit Speed to Real-Time
Load Application at Startup
Run to main()
Initialization File:
Edit...
Restore Debug Session Settings
Breakpoints
Toolbox
Watch Windows & Performance Analyzer
Memory Display
Load Application at Startup
Run to main()
Initialization File:
Edit...
Restore Debug Session Settings
Breakpoints
Toolbox
Watch Windows
Memory Display
CPU DLL:
Parameter:
SARMCM3.DLL
Driver DLL:
Parameter:
SARMCM3.DLL
Dialog DLL:
Parameter:
DARMCM1.DLL
Dialog DLL:
Parameter:
TARMCM1.DLL
OK
Cancel
Defaults
Help

# Utilities


Click to check
Options for Target 'proj_New'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Configure Flash Menu Command
Use Target Driver for Flash Programming
Use Debug Driver
--- Use Debug Driver ---
Settings
Update Target before Debugging
Init File:
...
Edit...
Use External Tool for Flash Programming
Command:
...
Arguments:
Run Independent
Configure Image File Processing (FCARM):
Output File:
Add Output File to Group:
CMSIS Group
Image Files Root Folder:
Generate Listing
OK
Cancel
Defaults
Help

# Build Project (Compile)

